सत्यमेव जयते

प्रत्नकीर्तिमपावृणु

REPORT ON EXCAVATION AT RATNAGIRI, JAJPUR, ODISHA

2024-2025

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Glimpses of Ratnagiri Excavation 2024 -2025

Chapter - I

Introduction

Ratnagiri archaeological site (Lat. 20º38’32’’ N., Long. 86º 20’10” E, MSL 54mt) is located within the village Ratnagiri in tehsil Bari, of Jajpur district, Odisha and about 100 km away from state capital Bhubaneshwar. The archaeological site is over an isolated outcrop of khondalite formation of the Asia hill range abruptly rising amid a wide expanse of an alluvial plain. The River Kelua a tributary of Brahmani is flowing close to the site in the western direction. Two major rivers Brahmani and Birupa flow either side of the site. The important Buddhist establishment close to the sites are of Udayagiri at a distance of about 8 km and Lalitagiri is about 15 km. The site is connected well with motorable road from the capital city Bhubaneswar as well as from Chandikhol and district headquarter Jajpur. The site was first reported in Bengal District Gazetteers by Manmohan Chakravarti in1906 and afterward in 1928 and the site was initiated by Debala Mitra of Archaeological Survey of India in 1958 to 1961.

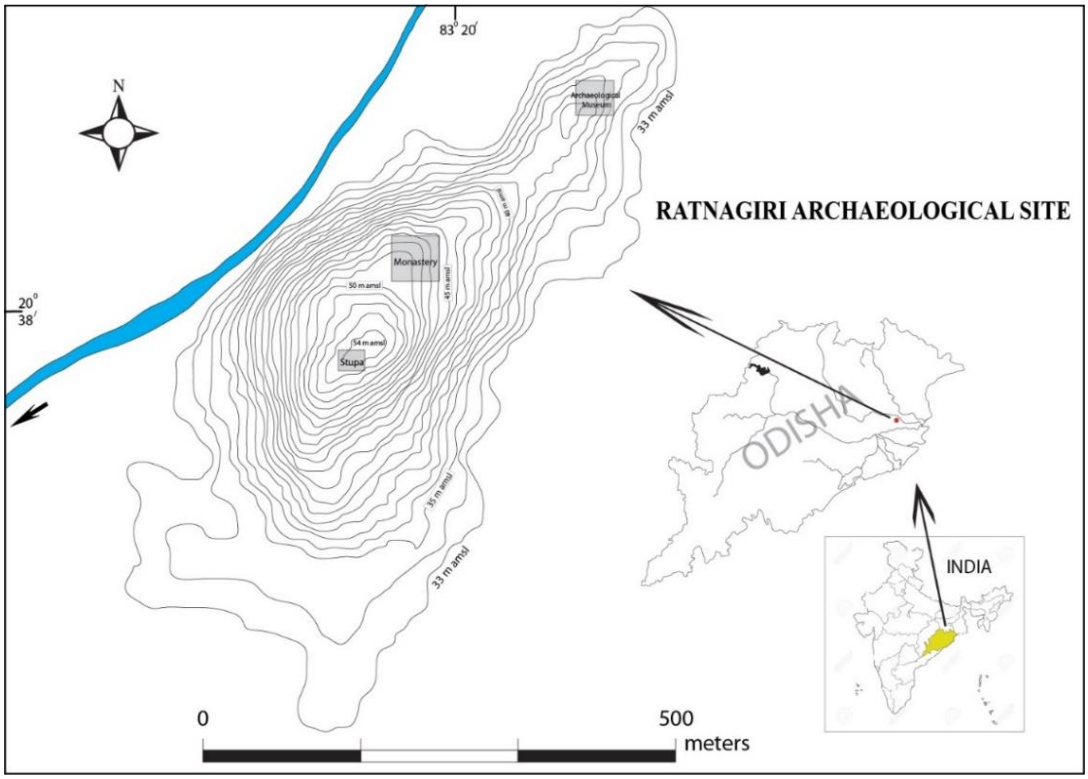


Fig. 2. Location of the site with contour

Brief History about the Site

The monastic establishment received patronage from different well known royal dynasties such as Śailodbhava's during 7th century CE, Bhauma-Kara's in 8th to 10th century CE when the site witnessed its zenith. The Bhauma King Śivakaradeva-I (756 to c 795 CE) was credited for sending an envoy to the court of the Chinese emperor Te-Tsong to present a Buddhist manuscript Gandavyuha to the emperor. After Bhauma the Somavaṃśi's ruled over this region in the last part of the 11 the century CE, king Karṇadeva donated the village Kona in Uttara Tosali to Rani Karpurasri who was reside and responsible for the maintenance of the Ratnagiri monastery as mentioned in the copper plate charter recovered from the village Ratnagiri, presently housed in Ratnagiri site Museum.

The copper plate charter belongs to the last Somavamsi/Keshari king named Paramamahesvara Paramabhattaraka Maharajadhiraja Somakulatilaka Trikalingadhipati Karnadeva. It mentioned about a grant of rent-free village named Kona with the Brahma-Atthavisa Khanda of Uttara Toshali to a woman named Karpurasri who hailed from Salomapura-Mahavihar in the twelfth day of the bright fortnight of the Asvina in his sixth regnal year. The name Karpurasri associated with the legend and in the copper plate record is same.

Ratnagiri means the hill of jewels on Buddhism and for the local people the site is rich in term of unique sculptural members and sounds monumental architecture in comparison to other Buddhist establishments of the locality. The name of the site mentioned in a terracotta sealing reads as '*Śrī-Ratnagiri-Mahāvihāriy ārya-bhikṣu-saṅghasya*'. There is a strong legend regarding establishment of the site by queen Karpurasri. The legend has its historical significance as one of the copper plates charter retrieved from the village Ratnagiri mentioned about the patronage of the site by queen Karpurasri. Karpurasri belongs to a family of professional dancer which was evident from the charter, because this charter was issued in the name of her mother and grandmother instead of his father and grandfather. Later she became the queen of king Karnadeva and in the later life she settled in Ratnagiri as a devotee or may be as a nun. The village granted in favour of her was meant to meet her personal expenses and to provide her to make contribution to the Buddhist settlement at Ratnagiri.

The account of Lama Tāranāth of Tibet (1608 century CE) shed some light on the site by ascribing the affiliation of Gupta king Narasiṃhagupta Bālāditya (4th-5th

century CE), who has been identified as *Buddhapakṣa* by the Buddhist scholar Nalinaksha Dutta, who built a vihāra named Ratnagiri on a crest of hill near the sea in Odisha that accommodated 500 monks. This incident also indirectly corroborates with the presence of an inscribed stone slab which bears an inscription having *eastern variety of* Gupta Brahmi legend *reads as Pratitya-samutpāda-suira* stylistically dated to 5^{th}-6^{th} cent CE and the recent findings of a terracotta sealing dated to 4^{th}-5^{th} century CE of Gupta Brahmi Script with legend Ye dharmā hetu-prabhāvah, hetuṁ teṣāṁ tathāgato, hy avadat teṣāṁca, ya nirodha evaṁ-vadī, mahāśramaṇaḥ. Various Tibetan records also gave data about the flourishing religious trends of Ratnagiri towards the end of the 10^{th} century CE. One Tibetan source composed during 1476 CE to 1478 CE mentioned that Acharya Tsi-lu-pa (Cheluka) had learnt Kalachakrayana, an offshoot of Vajrayana in the vihara of Ratnagiri (Rin-chen-rito). Another Tibetan work by Mkhas-Grub -Thams chad says that the great preceptor Chilu Pandita of Orbisan (Odisha) studied the completed Tripitaka in Ratnagiri vihar, Vikaramasila, and Nalanda. One of the Tibetan sources named "Pag-sam-jon-Zang" mentioned that Acharya Bitoba who obtained the kalachakrayana from Sambhala, was instructed by Vajrapani to go through magic to Sambhala where the kalachakrayana would be found. He went there and brought it to Ratnagiri and explained the doctrine to Bhiksu Abadhutipa, Buddhisri & Naropa (an eminent tantric sage who was immortalised in the Tibetan works.)

Another two inscribed stela of 12^{th} century CE found from the south eastern corner of the Monastery-I records the gift of the paddy to the Samgha (monastery) of Ratnagiri. The first one contains the gift of paddy by Jananda of Kayasthakula, who was a native of Nalanda, while the second one contains the grant of paddy by an offspring of a dead person in the motive to keep the gate of hell closed for that dead person. It also furthers records about some grant of paddy by Mandgalaraj, over a village called Chiro-Siddhapura for salvation and about a reprimand for attempt to resumption of the grant in the last stanza. The present excavation revealed a stone inscription written in Sanskrit language and Nāgari characters of 9th/10th century CE states that Karachanda was a noble soul, who is in this age of Kali where the virtue is rare (kali-yuga-durlabha-sucarikaḥ), still performs deeds of dharma. Further it records that he renovated the stupa enshrined the sacred relics (dhātu-garbha-stūpa) which was dilapidated. He also erected a beautiful monastery, a stepped well and a pavilion near

the site. All these achievements stand like a bridge across the vast ocean of worldly sufferings (saṁsāra-durggata-mahārṇava-setu-kalpaḥ)

Legend and Tradition

The site retains few living tradition till date related to the Mahakal, the presiding deity of the Mahakal temple, which was built in the 12th -13th century CE on the remains of the earlier Buddhist structures.

The native people of Ratnagiri celebrated Changada *jatra*/mela (festival) on this site in the day of *Maha Vishuba Sankranti* or the first day of the month of Baishaka. There are two main purposes behind celebrating this festival. According to the people of the Ratnagiri, it is the birth anniversary of the presiding deity of the Mahakal while another reason for celebrating this festival is related to fertility, in which the childless or barren women observes this festival in order to become a mother by eating the offering Mango to the god. The word 'Changda' in colloquial language meaning 'a container or a basket made out of Bamboo' in which seven mangoes were given as an offering to the presiding deity Mahakal during the festival. One of the rituals of this festival started before seven days of the Maha Vishuba Sankranti, in which the *Baghuti* (person who belongs to a fisherman community) worshipped the '*Ghanta'*(bell) of the Mahakal deity. Then in the day of the festival, the *changada* is prepared by the priest of the temple, which contains one effigy made on the *Basanga* leaf (*justicia adhatoda L.*) of the presiding deity Mahakal and seven mangoes along with other offerings. Then the Changada has been taken to the *Changda ganda* of the Kelua River for the *Buda* (religious bath). After completion of this bath, the devotees returned to the temple, during this time the mangoes were distributed among all *guharia*/alia (a person whose wish to be fulfilled by God). All social classes of Ratnagiri village participated in this festival. A carnival is also prepared for three days during this festival.

Fig 1: Location of the excavated site Ratnagiri in yellow mark, Jajpur, Odisha

The Site and Its Environment

Ratnagiri archaeological site is situated over an isolated outcrop of the Khondalite formation of the Asia Hill Range, abruptly rising amid a wide expanse of an alluvial plain. The River Kelua a tributary of Brahmani, is flowing close to the site in the western direction. The rivers Brahmani and Birupa are two major rivers on both sides of the site. Once, Ratnagiri was a flourishing centre of Buddhist learning and spirituality between 4^{th}- 12^{th} centuries CE, and an integral part of Udaygiri & Lalitgiri Buddhist establishment. The hillock contains the ruins on the uneven top of the rugged hills. The site is surrounded by deltaic alluvial soil and best for paddy and cereal cultivation. During archaeological field survey in and around the landscape, it has been noticed that human settlements began in the area around 1^{st} -2^{nd} Century BCE. The fertile landscape and Khondalite outcrops which are suitable place for habitation attract the settlers to settle in. The region experiences a tropical monsoon besides summer and winter. The summer is from March to June is hot & humid, with temperatures ranging from 30°c to 45°c. The monsoon persists between June to September, the average annual rainfall is around 1500-1600 mm. Winter season starts from November and continues till February with mild & pleasant temperature between 10°c- 25°c. Humidity

remains high year-round, particularly during the monsoon season. The best time to visit the site Ratnagiri is from November to February.

The religious establishment has grown up at the site due to favourable weather, perfect landscape for religious practices and as evidence from the epigraphical records, the establishment had well supported by royal patronage.

Present Flora & Fauna

The hillock where the site settled features a mix of natural vegetation and agricultural landscape. Common types of flora found here are deciduous trees like, Neem (*Azadiracuta Indica)* and Banyan (*Ficus benghalensis*), Kadam (*Burflower)*, Katha Champa (*Magnolia-champaca*), Mango (Mangifera indica). Palm tree /tal (*Borassus flabellifer*), Gauva (*Psidium guajava*), Bael tree (*Aegle marmelos*), Bada Chakhunda (*Senna occidentalis*), Curry leaves (*Murraya koenigii*), Amla/Indian gooseberry (*Phyllanthus emblica*), Jhaun/She-Oak (*Casuarina*), Arjuna Tree (*Terminalia arjuna*), Coconut Tree (*Cocos nucifera*), Serisa/Lebbeck (*Albizia lebbeck*), Baula/Spanish cherry (*Mimusops elengi*), Radhachura/Peacock flower (*Caesalpinia pulcherrima*), Sugandhraj/Cape Jasmine (*Gardenia Jasminoides*), Bhringraj/False daisy (*Eclipta prostrata*), Vishalyakarani (*Tridax procumbens*), Anchu (*Morinda coreia*), Three-seed mercury (*Acalypha rhomboidei*), Karanja/pongame oiltree (*Millettia pinnata*), Shisham/Indian rosewood (*Dalergia sissoo*), Simili/Red-silk cotton (*Bombax ceiba*), Bottlebrush (*Callistemon citrinus*), Maloo creeper (*Buahinia vahlii*), Mahakal lata/ Bitter snake gourd vine or Great fringed flower vine (*Trichosanthes tricuspidate*), Date palm (*Phoenix dactylifera*), Seetaphal/Custard apple (*Annona squamosa*), Banana Tree (*Musa acuminatamusa bablisiana*), Waterchest nut (*Trapa natans L.),* Bamboo (*Bambusa vulgaris)*, shrubs & grasses, various wild grasses & medicinal plants grow naturally in the surrounding uncultivated areas, cultivated plants like paddy, sugarcane, pulses and oilseeds dominated the agricultural field around the site.

Due to human activity and agricultural expansion, large wild animals are extinct day by day in the region, but the species still exist in the area are jackals (*Canis aureus*), Hyena (*Hyaen striate*), Mongoose (*Urva edwardsii)*, Indian palm civet (*Paradoxurus hermaphroditus*), House cats (*Felis catus*), small rodents (*Mus musculus*), Mole (*Melanocytic nevus),* Bat (*Pipistrellus pipistrellus)*, Indian palm Squirrel (*Funambulus palmarum*), Monkey (*Macaca mulatta*), Langur (*Presbytis entellus*), besides the

domesticated animals seen around the site: Cow (*Bos indicus*), Buffalo (*Bubalus bubalis*), Ox (*Bos taurus*), Bull (*Bos sp.*),Goat (*Capra Aegagrus hircus*), Dog (*Canis familiaris*). Reptiles noticed in the site: Garden Lizard (*Calotes versicolor*), Monitor lizard (*Varanus),* House wall lizard (*Hemidactylus frenatus),* Indian rat snake (*Ptyas mucosa),* Cobra (*Naja Naja),* Centipede *(Scolopendra subspinipes).*

Birds like, Rock pigeon (*Columbia Nivia*), Black-hooded Oriole (*Oriolus xanthornus*), Spotted owl (*Athene brama*), Black Drongo (*Dicrurus macrocercus*), Rufous Treepie (*Dendrocitta vagabunda*), Red-vented Bulbul (*Pycnontous cafer*), Black cape Crow (*Corvus capensis*), Indian pied Starling(*Gracupica contra*), Sparrow (*Passer domesticus*), Common Myna (*Acridotheres tristis*), Red-wattled lapwing (*Vanellus indicus*), Jerdon's Bushlark (*Plocealauda affinis*), Jungle babbler (*Argya striata*), Common Hawk-Cuckoo (*Hierococcynx varius*), Herons (*Ardea cinerea)*, Crow Pheasant (*Greater coucal*), Verditer flycachter (*Eumyias thalassinus)*, Coppersmith barbet (*Megalaima haemacephala)*, Brown-headed barbet (*Psilopogon zeylanicus),* Spotted Dove (*Spilopelia chinensis),* Oriental magpie Robin (*Copsychus saularis),* Indian robin (*Copsychus fulicatus),* Common Tailorbird (*Orthotomus sutorius),* Shikra (*Tachyspiza badia),* Black Kite (*Milvus migrans),* Indian scops Owl (*Otus bakkamoena),* Barn swallow (*Hirundo rustica),* Asian Koel (*Eudynamys scolopaceus),* Alexandrine Parakeet (*Psittacula eupatria),* Rose-ringed Parakeet (*Psittacula kramei),* Indian Roller (*Coracias benghalensis),* Asian green Bee-eater (*Merops orientalis),* White- throated Kingfisher (*Halcyon smyenensis),* Black-rumped Flameback Woodpecker (*Dinopium benghalense),* purple Sunbird (*Cinnyris asiaticus) are found in the site.* Arthropods like Scorpion (*Deccanometrus bengalensis*), Termites (*Odontotermes horns*), locust (*Cyrtacanthacris succinata*), Honey bee (*Apis mellifera*), Dragonfly (*Anax indicus*), Cricket (*Acheta domestica*), Ant (*Solenopsis invicta*), Dung beetle (*Scarabaeus sacer*), Common tree frog (*Polypedates maculatus*), Common frog (*Bufo bufo*) are also seen.

Demography

Population of the village Ratnagiri is approximately 3,500 at present, live in 870 houses. The villagers and nearby areas are mostly Hindus. A minority of Muslim population is inhabited in Aliabad situated at the distance of 2.6 km from the site. Recently, two families belong to *kandara* caste are converted into Buddhism, one

kumbhara family converted into Christianity in the village Ratnagiri. The major occupation of the people is based on agriculture irrespective of caste. The major castes living in the area are of *khandayat, brahmin, gauda, harizana, kandara, pana, teli, kachara, kumbhara*.

Thus, it can be assumed that the flourishing belief system on Buddhism in the area was wiped out with the passing of time and only the ruins were left behind in the form of legacy and heritage.

Archaeological environment

The area surrounding the Ratnagiri archaeological site is rich with archaeological landscape, deeply associated with Buddhism. Beyond the main monastic complex of Ratnagiri, the wider environment is dotted with numerous ancient remains that collectively shed light on the religious, cultural & socio-economic importance of the region between 1^{st}-2^{nd} centuries CE to 12^{th} -13^{th} century CE.

The major archaeological sites close to Ratnagiri are Lalitagiri, nearly 15km south-west, and Udayagiri is at 8km west, have resemblance with the material culture and shows their cultural and religious integrity from 7^{th} century onward till the declination of the site. The other sites Radhanagar and Kaima situated approx. 20 km away from the site are major urban and religious centre of early historic period. The Kolangiri site lies 2 km north-west of Ratnagiri was known for its Buddhist remains as well as habitational structures contemporary or little earlier to the site under study. A hillock in the village Parbat Panasua nearly 1 km south-west of the site has an image of Vajrasattva worshiped as Shree Shree Bana Madha. The deity venerated as a village deity since time immemorial without having knowledge of its Buddhist affinities by the villagers.

Besides above, archaeological remains in the form of sculpture are found in the nearby villages either worshiped as village deity or as Brahmanical deity. The recent excavation at Parabhadi, 12 km from Ratnagiri exposed a huge Buddhist stupa dated to 7^{th} -8^{th} Century CE. At the same hillock there are rock cut chambers locally known as Hatikhal at Bandreswar not only shows about the Buddhist potentiality of the area but also shows various art and architectural forms within the whole landscape of the Asia hill range.

Archaeologically, the region also reveals the remains of ancient archaeological features such as water reservoir, rock cut caves, rock cut sculptures, masonry brick and stone built structures etc. affiliated to Buddhism. Besides, several notable temples & religious sites affiliated to Brahmanism noticed in the region which reflects the region's rich spiritual heritage. It suggests a syncretic religious environment where Buddhist & Brahmanical practices coexisted.

In essence, the archaeological environment around Ratnagiri is not just a backdrop to the main site – it is an active part of its history. The surrounding landscape, with its scattered ruins & explored mounds, holds immense potential for further discoveries, offering a broader & more nuanced understanding of Buddhist life, architecture, and society in early medieval Odisha.

Previous Work

The site was reported in Bengal District Gazetteers by Manmohan Chakravarti in 1906 and the next mention of Ratnagiri comes from the accounts of Ramaprasad Chanda, who was the Superintendent of the Archaeological Section at the Indian Museum. He visited the site during his exploration of Odisha in 1927–28. In 1927, Birendra Nath Ray, Secretary of the Orissa Historical Society, also visited the site. At his request, in 1928, Haran Chandra Chakladar, a lecturer at the University of Calcutta, along with Nirmal Kumar Bose, visited Ratnagiri. That same year, their findings were published in an article of *The Modern Review* titled "A Great Site of Mahayana Buddhism in Orissa." The Somavamsi Copper plate found from the Ratnagiri village referred to the site was published by Narayan Tripathy in 1930. R. P. Chanda later published a short report in 1930 titled *Explorations in Orissa*, which included a description of the sculptures found at Ratnagiri. One more important reference came from Devaprasad Ghosh, who wrote an article in *The Modern Review* comparing a large Buddha head and a Bhumisparsha Buddha statue from Ratnagiri to the Dhyani-Buddha images at Borobudur. In 1949 P. Acharya wrote a summary in *The Orissa Review*, suggesting that remains of the ancient Pushpagiri monastery described by the Chinese traveller Xuanzang could be combined remains found at Naltigiri (Lalitagiri), Udayagiri, and Ratnagiri. In 1954 S. C. Chandra, who was the Assistant Superintendent of the Archaeological Survey of India, wrote an article called '*Early Medieval Sculptures of Utkala (Orissa)'* in *The Orissa Historical Journal* (Volume III, No. 2). In

this article, he also discussed the sculptures found at Ratnagiri. However, a detailed and in-depth study on Ratnagiri was not published till 1987. There is a PhD dissertation by Nancy Hock, titled '*Buddhist Ideology and the Sculpture of Ratnagiri, seventh through thirteenth centuries*,' submitted to the University of California, Berkeley. Her research concludes that certain ritual texts, especially the *Manjushri-mula-kalpa* were used at Ratnagiri in connection with the sculptures found. The large number of Amoghapasha images found at the site also indicates that the worship of Bodhisattva was popular at Ratnagiri. M.J. Sindhu completed a PhD in 2014 from the Ancient Indian History, Culture and Archaeology Department of Utkal University on "Iconography of Buddhist sculptures at Ratnagiri" based on the excavated findings of Debala Mitra.

The first excavation at the site was initiated by Debala Mitra of Archaeological Survey of India in 1958 to 1961. These excavations uncovered the impressive remains of an important Buddhist site, known as Ratnagiri-maha-vihara. The site includes a large Maha stupa, two square-shaped monasteries (Monastery No. 1 and No. 2), a single-winged monastery, many small stupas, a large number of votive stupas, eight temples, and a large number of sculptures mostly made of stone with some in bronze. According to Mitra, main part of the site was established around the 5th century CE and continued to develop till the 12th century CE. The most important among the artifacts recovered earlier was the terracotta seal with legend 'Śrī-Ratnagiri-Mahāvihāriy ārya-bhikṣu-saṅghasya'.

Chapter II

Excavation- 2024-2025

The present excavation report has been based on the spade work initiated at the site from 1st Dec 2024 after getting permission from the directorate, New Delhi. Prior to this the major excavation was conducted by D. Mitra in 1958-61, but most of the things have not been addressed in that excavation. The scientific spade work has taken in the current field season with the aim and objective as given bellow.

Aim and objective

- The basic aim of the excavation to unearth and preserve partially visible structures and sculptures damaged due to human vandalism.
- To find out if any shrine complex/Chaitya complex existed in the unexcavated area.
- To know the ceramic assemblage of the site which has not been addressed in the earlier excavations.
- To find out any material culture of south-east Asian origin or beyond.
- To understand the cultural framework of colossal Buddha head from the site.

Excavation

The spadework has been conducted in trench number YA1 Qd $^{3\ \&\ 4}$; YB1, YC1, YD1, YE1, XE1, XD1, XC1, XB1, XA1, Qd $^{1,\ 2,\ 3,\ 4}$; & XA2 Qd $^{2\&\ 3,}$ having 928.4983 sq. m area in the current field season. In the majority of the trench, three stratigraphical layers have been revealed. In the XA^{2,} a maximum of four cultural layers have been exposed within a limited area due to constraints of space and multi-layer structural activities. By considering the availability of undisturbed space measuring 1m × lm has been taken for further excavation to know whether there is earlier cultural deposit/sequence below layer 3. The closing depth of the trial pit within the trench from the ground level is 285 cm, and from the top to bottom, four cultural layers have been observed. The site has various nature of deposits and cultural remains in different location, due to the uneven surface of the hilltop and the nature of the settlement. Space allotment played a crucial role within the hilly landscape. The settlements has its beginning from the early time of the Common Era and continued till the 12th-13th century CE and beyond. The early phase of material culture encountered in the eastern

part of the hillock, close to the recent findings of the Chaitya complex while the late period activities are noticed in the western part of the site. The available and suitable space in the hill has been divided into three major sectors as observed from the recent excavation findings i.e., Stupa complex, Chaitya complex and the Monastic complex. The area in the eastern direction of the chaitya complex has earlier and more cultural deposits.

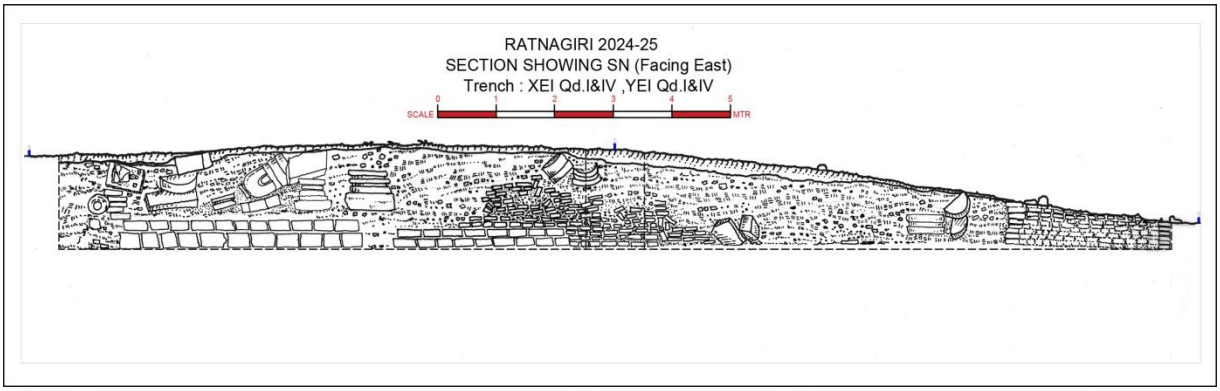
Stratigraphy

Layer 1 is light grey in colour, semi-compact in nature, mixed with brickbats, stone fragments, potsherds etc. The ceramic assemblage was dominated by greyware with less amount of redware. It is worthwhile to mention here that the nature of the cultural deposit is varying in nature in different locations of the site. The layer 1 in trench XE1 and XD1 is of later time period in comparison to the layer 1 in the XA1, XB1 trenches. The basic difference is noticed in the ceramic assemblage, structural activities and epigraphical records found incised in the architectural members.

Layer 2 is reddish grey in colour semi compact in nature and encountered with brick and stone structures, votive stupas, evidence of brick jelly flooring and comparatively less amount of ceramic assemblages noticed in this layer.

Layer 3 is similar in nature in comparison to the previous layer where the red ware pottery increased as well as different ritualistic pottery out number, mainly noticed in quality and shape.

Layer 4. The limited excavation in the XA2 trench revealed a succeeding cultural deposit of light brown colour, loose in nature and contained brickbats, stone chips, fine variety of red ware, red slipped ware and finger impressed bricks fragments. Red ware has been a dominating ceramic assemblage followed by grey ware in this layer. The same area has yielded a fragment of sprinkler, incense burner in red ware and an inscribed potsherd in grey ware which can be assigned to 4th-5th century CE are noteworthy.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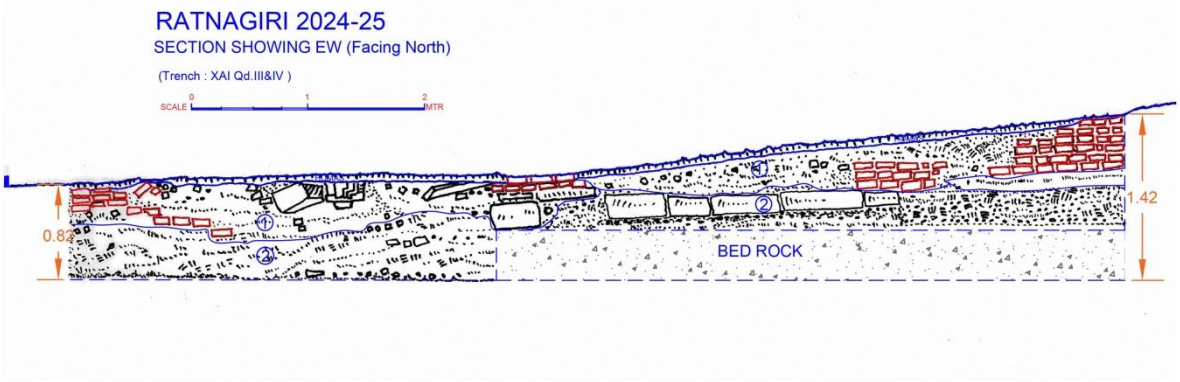
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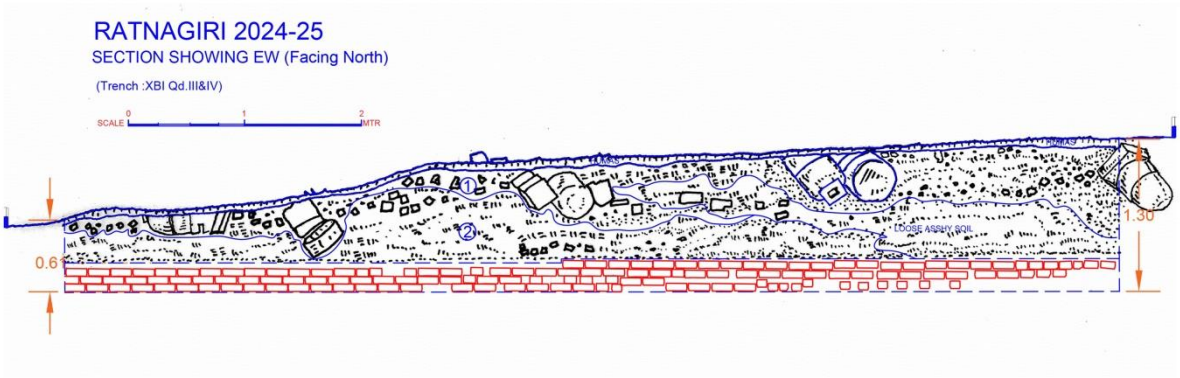
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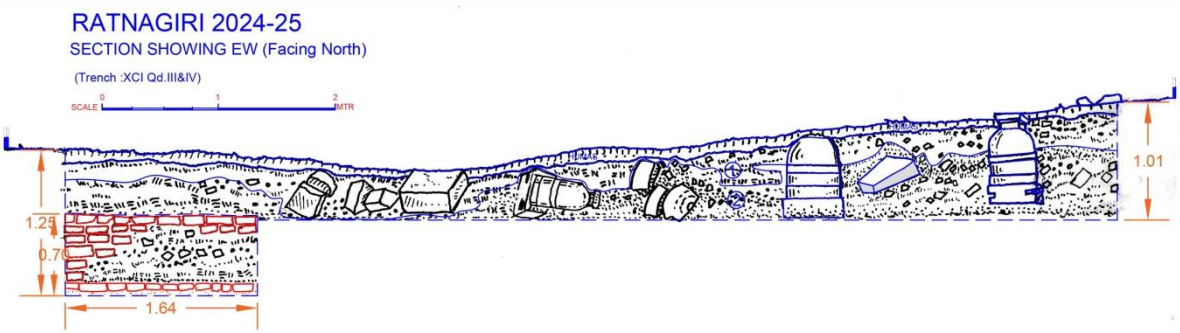
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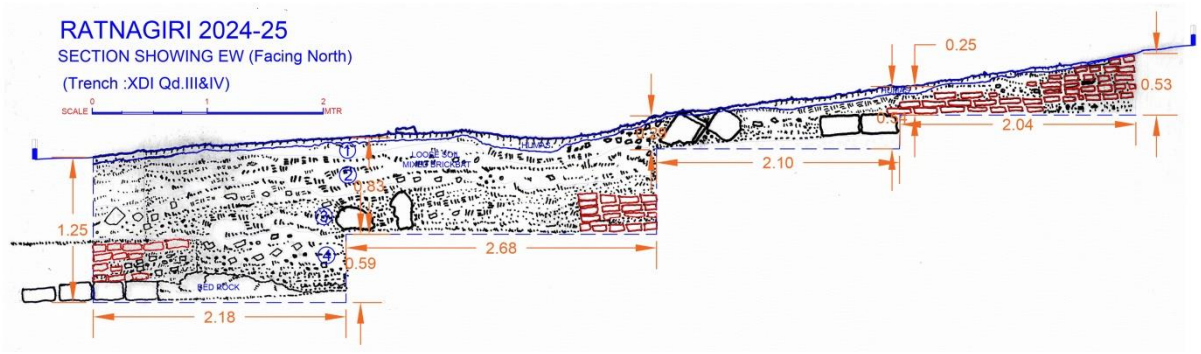
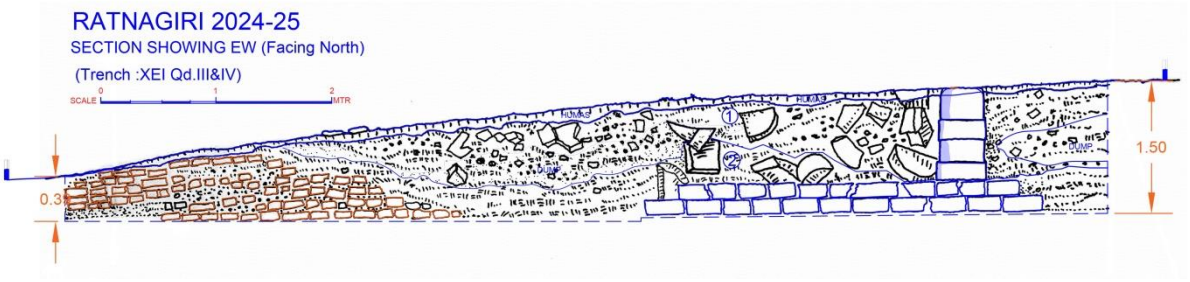




Section facing North, Trench YE3

General view of the excavated trenches

Chronology of the site

The land of Odisha was not unknown to Buddhism as the people were already acquainted with this pantheon since the time of Gautama Buddha. The early Buddhist canonical literature tells us the story of two merchants *Tapussa* and *Bhalluka* from *Ukkala* (Utkal) who were among the first disciples of Buddha and are supposedly credited for the expansion of Buddhism. The first archaeological evidence of this religious tradition is major rock edict and rock-cut elephant of Dhauli near Bhubaneswar dated to 3rd century BCE. The existence of other sites like Lalitagiri (2nd century BCE) and Langudi (2nd to 1st century BCE) exemplify the popularity of Buddhism in this region. Though, it was followed in different forms but people were coming under the same umbrella.

The chronological time frame of the Ratnagiri site has been determined on the basis of excavated material culture. The present excavation has yielded structures, numbers of inscriptions, pottery, architectural and sculptural remains which speaks about their chronological frame starting from early 4th century CE to 12th-13th century CE. The palaeographical evidences in terracotta plaques, stone tablets, inscribed potsherds and inscriptions on the surface of stupas retrieved from the excavation has given relative dates and chronology to the site.

Period-I (4th to 7th century CE)

Period-I of the site can be assigned to 4th-7th century CE on the basis of paleographical evidence such as the terracotta plaque retrieved from the excavations having legends in Brahmi script, Sanskrit language and can be stylistically dated to 4th - 5th century CE that reads as *Ye dharma hetu prabhavā hetu..stesāṁ tathāgato hyavadata tesāṁ cha..yo nirōdha evamvādi mahāśramaṇa.* The excavation conducted by Debala Mitra in 1958-61 also documented an inscribed on stone slab bears an inscription having eastern variety of Gupta Brahmi legend deciphered as '*Pratitya-samutpāda-suira*' stylistically dated to 5th-6th century CE. Besides, the sculptures of Gana / Yaksha retrieved from the present excavation stylistically dated to post Gupta period and can be dated to 7th century CE.

Period- II (7th to 10th century CE)

The recent excavation revealed a Chaitya Complex and multiple votive stupas which housed images of Vajrayana sect in their niches. Besides this, three colossal heads of Buddha along with stone tablets bearing *Siddhamatrika* script in Sanskrit language have also been documented from the same strata can be stylistically dated, 7th-8th century CE. Besides, other inscriptions and sculptures can be assigned to 9th-10th century CE. During the above mentioned period the site was under the control of Bhaumakara dynasty who donated many grants for the construction of monumental architecture at the site.

Period -III (10th to 12th century CE)

This period can be assigned to 10th -12th century CE on the basis of the findings of monolithic votive stupas designed as a replica of mature phase of Kalingan temple architecture dated to 10th -12th century CE (*vimana* having a curvilinear super structure attached with a pyramidal *pidha jagamohana*). The frontal porch of the curvilinear stupa, designed as pyramidal *pidha* housing a Buddhist deity Tara. Besides, the stone votive tablets having Nagari inscription can be assigned to the same time bracket. The fusion of art and architecture noticed in sculptures and stupa architecture during this period. The political suzerainty has also changed during this period. Another significant dated finding of the period was the copper plate charter of Karnadeva of the Somavamsi dynasty recovered from the site can be dated to early 12th century CE, presently housed in the Ratnagiri museum.

Period- IV (12th - 13th century CE)

In the present excavation, the material remains associated with Mahakal Temple has been yielded from the upper layer of trenches XE1, YE and YF1. The findings were of semi-oxidized, medium fabric dark grey ware, architectural fragments of earlier stupa used as the platform of the temple. The epigraphical evidence of 13th century and onwards has not found from the site and the sculptures kept inside the Mahakal Temple including the presiding deity cannot be assigned beyond 13th century CE. The desertification of the site may occur in between 12th -13th century CE as per the tangible evidence retrieved from the site.

Desertification of the site

In the post 12th century CE, the rise of Shaivism and Vaishnbisim under the Ganga and changes in the religious scenario leads to lack of patronage towards Buddhism by the royal dynasties were major cause for the decline of Buddhism in the state. The root can be also observed during 9th century CE when Buddha was considered as 9th avatar of lord Vishnu.

The glorious chapter of Buddhism under Bhaumakara's started collapsing during the last part of Somavansi ruling dynasty who were Shaivite as well as worshiper of Shakticult as instance from the revival of Virajakhetra/Jajpur, the capital city as one of the Shaktipithas. The situation became worse during the reign of Gangas because of their staunch support to Vaishnavism. According to *Madalapanji*, a 12th-14th century temple chronicle of Jagannath Puri, Gangas were believed to have supressed Buddhism. It recounts the tale of bringing more than six hundred monks to death by the ruler Anangabhima Deva III, when they failed to answer his spiritual inquiries satisfactorily. Another such event is mentioned in the same text about Madana Mahadeva, brother of Anangabhima Deva III, persecuted Buddhists (Ref. K C Panigrahi, *History of Orissa: Hindu period*, pp. 312).

The text *Chaitanya Bhagabata* composed by Isvara Dasa towards the end of 16th century CE mentioned the conflict between Buddhists and king of Keshari (Somavanshi) dynasty which led to assassination of large number of Buddhist monks (Ref. Isvara Das, *Shri Chaitanya Bhagabata*, compiler, Rai Bahadur Artaballava Mohanty, published by Utkal University, page no. 323)

During 15th century CE, the coming of Chaitanya Mahaprabhu (1486-1534 CE) the patronization of Vaishnavism was encouraged. *Chaitanya Charitamruta* composed by Krishnadas Kaviraj Goswami (1528-1620) also described about the suppression of Buddhism (Ref. Shyam Sunder Tripathy. 1988 *Buddhism and other religious cults of south-east India*, Delhi, pp. 188).

The construction of a temple for Mahakal, a fierce form of Lord Shiva shows shifting of ideology from Buddhist to Brahmanical dominance in the region. It was constructed during 13th century CE over Buddhist structural remains. This act of replacement was not only merely architectural but also ideological, reflecting the decline of Buddhist pantheon. By the 14th-15th century CE, the remnants of Buddhist philosophy, especially during the time Chaitanya, were further diminished. The demolition of Buddhist remains by non-Buddhist is partially true because most of the sculptures are found intact and in situ during excavation. One of the best examples is deity of *Shri Banamadhava* at the site of Parbata Panasua near Ratanagiri which is an image of a Buddhist deity Vajrasattva.

It has been observed from the material culture retrieved from the excavation that the site has a long span about 1000 years of history and slowly declined with the passing of time, changing of religious and political scenario of the state.

In-situ Sculptural Remains

Chapter - III

Archaeological Findings

The excavation revealed hundreds of art objects of various materials, shapes, and sizes, both secular and religious in nature. Out of which 304 art objects are documented as antiquities, which define the nature and time scale of the site. The inscribed stone tablets, sculptures, inscribed votive stops, terracotta plaques, terracotta beads, terracotta animal figurines, stone and terracotta polisher/pestle, saddle-quern, dabber, discs, sling balls, bangles, spool, etc., are key findings of the spadework. The details of all the findings are as follows-

Epigraphical Records From The Ratnagiri Excavation

The excavation retrieved epigraphical records in the form of stone tablets of both square and rectangular shapes, terracotta plaques (seals-sealings), potsherds, inscribed surfaces of architectural members, base/body of votive stupas, and sculptures with Buddhist sacred hymns known as Dhārani. The scripts are Brahmi (4th century CE), Siddhamatrika/ Kutila (8th-9th century CE), and Nagari, which is dated between the 10th and 12th century CE. In the previous excavation, instances of inscriptions have also been noticed in copper plates and metal objects. Two inscribed Stelas are noteworthy as their texts are in the Gaudiya /proto-Bengali variant script of the 12th century CE, with the language in Sanskrit.

History of Dhāraṇī

Buddhist Dhāraṇīs have been found at many Buddhist sites of India, which received less scholarly attention earlier. Although attempts to identify the textual references and their purposes have been made in the recent issue of the Bulletin of School of Oriental and African Studies (vol 77 (1) 2014, Ronald M. Davidson), the works of T. E. Donaldson, Iconography of the Buddhist Sculpture of Orissa (2001), M. Benestie (1981), Contribution a l'etude du stūpa bouddhique indien: Les stūpas mineurs de Bodh-Gaya et Ratnagiri, 2 vols. Paris, N. Hock, (1987) Buddhist Ideology and the Sculpture of Ratnagiri, 7th through 13th century. Ph. D Dissertation. Berkley: University of California and Dhāraṇīs from the Buddhist Sites of Orissa by Umakanta Mishra in Pratnatattva 2016.

It is known from texts like Sādhanamāla and Nispannayogavali that the Manjusri Namsangiti Tantra is a kind of Buddhist Tantra. In Tibetan, it is also known as 'Jampal-Tsenjo', which recites names of Manjusri and is considered one of the advanced teachings of Sakyamuni Buddha. It has seven principal Mandalas and one of them is 'Dharmadhātuvāgiśvara Mandala' which describes another set of twelve interesting deities called the Dhāraṇī. Another theory is that Dhāraṇīs are formulas which are repeated by the Buddhists for various purposes. Dhāraṇīs i.e., 'dhārayati anaya iti' which means something is sustained were used as protective spells from 5th century CE onwards (R. Davidson, 2009) Studies in Dhāraṇī literature I, Journal of Indian Philos). Vāsubandhu in the text 'Bodhisattvabhụmi' categorised four types of Dhāraṇīs - Dharma Dhāraṇī, Artha Dhāraṇī, Mantra Dhāraṇī and the Bodhisattvakṣāntilabdha Dhāraṇī (M. Winternitz, 1983, History of Indian Literature, 2 vols., Delhi: Motilal Banarsidass).

The inscribed Dharani retrieved in the present excavation are mostly dealt with Dharma. The details are given below:

S. No. 01, ACC No. 272

The tablet, measuring 32 cm in length, 24 cm in height, and 8 cm in thickness, is carved from khondalite stone. It is inscribed on both sides with 11 lines on each. The tablet is broken or chipped off at one corner. The inscription written in Sanskrit language using the Kutila or early Nāgari script contains a Dhāraṇī mantra. Based on palaeographic analysis, the artefact is dated to the 8th century CE.

First Side

dhiṣṭhita dhātu-garbhe svāhā | samay-ādhiṣṭhite svāhā | sa-
rvva tathāgata hṛdaya-dhātu mudre svāhā | su-pratiṣṭhita stu-
pe svāhā | tathāgat-ādhiṣṭhite huru 2 hūṁ hūṁ svāhā |
sarvva tathāgato ṣṇāṣa dhamma-dhātu mudre | sarvva ta-
thāgata dhātu vibhūṣitā dhiṣṭhite hūṁ hūṁ svāhā | ti-
lavimva-nāma-dhāraṇī stupa madhye prakṣeptavyāḥ sā-
dhiṣṭhito bhavet || ye dharmmo hetu prabhavā hetuṁ
teṣāṁ tathāgato hyavraja datt=eṣāṁ ca yo nirodhaḥ eva-
mvādī mahāśāmantaḥ | namo vuddhāya || namo dharmmāya | na-
ma samghāya | tady-eṣāṁ ñ=ca dve viśuddhe | śodhaniti śodhani-
tān | viśodhani citta viśodhani | mā varṇṇa viśodhani |

Second Side

kṣaṇa parikṣīṇe | puṣye supuṇya jali
caraṇe | sarvva-pāpa vimocani | hara 2 sarva
caraṇāni | ḍada 2 sarva karmācaraṇāni | ḍadde
padmākṣī | padma viśāla | ḍa ḍa ḍa ḍa ḍa | maṇḍa 2
maṇḍala svāhā | nam-asmai ddhātu 2 tārā sarva ta-
thāgatānāma | vande vara-traya śuddhaṁ | akṣobhya
thāpitā bhava | hagravīnāy-āvalokiteśu |
tathāgatadmaṣī 3 sūkṣme 2 śānte 2 supte 2
nirākule | yaśovati | tejovati | sa-
rvva vuddh-ādhiṣṭh-ādhiṣṭhito bhu-modita svāhā |
kāṁtī dhāraṇī samāptaḥ || 0 ||

S. No. 02, ACC No. 273

An inscribed tablet made of khondalite stone, measuring 21 cm in length, 12 cm in width and 9 cm in thickness, was unearthed. The tablet bears a six-line inscription on one side, which is partially eroded and chipped out. The inscription is in the Sanskrit language, written in the Nāgari script and contains a Dhāraṇī mantra. Based on script and stylistic features, the tablet can be dated to the 11th –12th century CE.

S. No. 03, ACC No. 274

Inscribed Dhāraṇī mantra tablet made of khondalite stone has been uncovered at a depth of 96 cm from the excavation. The tablet measures 23 cm in length, 21.5 cm in width, and 15 cm in thickness. It is partially damaged/eroded, broken along one edge, and chipped at the upper portion. The inscription is in Sanskrit language and written in the Nagari script, tentatively dated to the 11th – 12th century CE. The inscription is written on three sides of the tablet and comprises a total of 16 lines.

S. No. 04, ACC No. 271

The sculpture of Dhyani Buddha Mahavairochana have been recovered at a depth of 1.12 meters. The sculpture is carved from khondalite stone, the sculpture measures 49 cm in length, 33 cm in width, and 27 cm in thickness. It is in good condition. Inscriptions are present on all sides of the sculpture, written in Sanskrit language using the Nagari script. The text is deciphered as a Dhāraṇī mantra. The sculpture is dated to the 11th –12th century CE.

S. No. 05, ACC No. 108

The inscribed tablet made of khondalite stone has been recovered, measuring 15 cm in length, 11 cm in width, and 2.5 cm in thickness. The tablet is broken and bears an inscription in Sanskrit, written in the Nagari script, comprises three lines and has been identified as a Dhāraṇī mantra. The object is dated to the 11th –12th century CE.

S. No. 06, ACC No. 277

A votive stupa carved from khondalite stone has been recovered contains an inscription of three lines engraved on the medhi portion. The stupa is in good condition and bears an inscription in Sanskrit, likely written in the Nagari script. The inscription has been identified as a Dhāraṇī mantra. The object is dated to the 11th–12th century CE.

S. No. 07, ACC No. 256

A sculpture of Buddha in bhūmi sparśa mudrā has been unearthed; the sculpture has carved with khondalite stone, measures 52 cm in length, 33 cm in width, and 15 cm in thickness. The sculpture is in eroded condition contains an inscription in Sanskrit, in the Nagari script, is present on the back side of the sculpture. It comprises two lines and has been identified as a Dhāraṇī mantra dated to the 11th –12th century CE.

S. No. 08, ACC No. 279

An inscribed Dhāraṇī mantra tablet dated to the 10th –12th century CE, carved from khondalite stone has been discovered, measuring 19 cm in length, 28 cm in width, and 9 cm in thickness in inner side, with outer dimensions of 23 cm in length, 33 cm in width, and 9 cm in thickness. The tablet is in good condition. The inscription is in Sanskrit language, using the Nagari script and consists of 10 lines engraved on one side of the tablet.

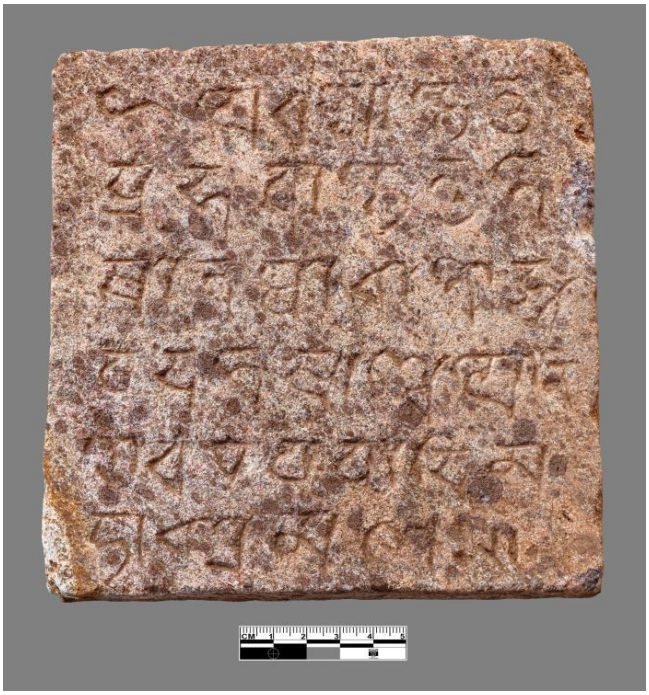
S. No. 09, ACC No. 280

A *medhi* portion of a votive stupa carved from khondalite stone has been unearthed, measuring 55 cm in length, 8 cm in thickness, and 69 cm in diameter. The object is in eroded condition. An inscription in Sanskrit language using Nagari script, is present on half of the *medhi* portion and consists of two lines. The surface also shows two nail holes and is chipped at the flat side. The inscription has been identified as a *Dhāraṇī* mantra dated to the 11th –12th century CE.

S. No. 10, ACC No. 281

A votive stupa made of khondalite stone has been unearthed, is in good condition, and bears an inscription in Sanskrit language using Nagari script. Five lines of the inscription with *Dhāraṇī* mantra are engraved on the *vedi* portion of the stupa. The object is dated to the 11th –12th century CE.

S. No. 11, ACC No. 282



An inscribed tablet carved from khondalite stone has been recovered, measuring 16 cm in length, 16 cm in width, and 4 cm in thickness. The tablet is in good condition, chipped on one side, written in Sanskrit language by using Nagari script, and consists of six lines engraved on the front side of the tablet. The inscription has been identified as a Dhāraṇī mantra dated to the 11th –12th century CE.

It includes the verse:

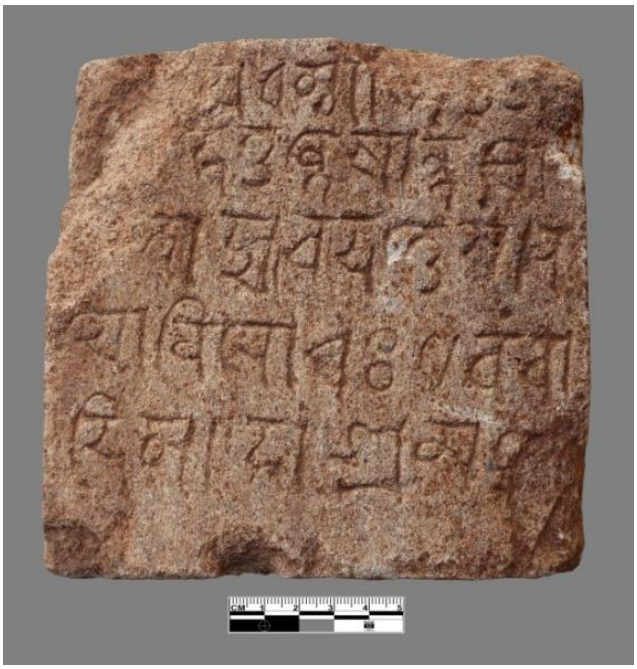
Ye dharmā hetu-prabhāvah hetuṁ teṣāṁ tathāgato hy avadat teṣāṁ ca ya nirodha evaṁ-vadī mahāśramaṇaḥ

S. No. 12, ACC No. 283

An inscribed tablet made of khondalite stone in a good condition, has been unearthed, measuring 16 cm in length, 16 cm in width, and 4 cm in thickness. It bears an inscription in Sanskrit language written in the Nagari script, consisting of six lines engraved on the front side. The inscription has been identified as a *Dhāraṇī* mantra, dated to the 11th–12th century CE. The text includes the verse:

Ye dharmā hetu-prabhavāh
hetuṁ teṣāṁ tathāgato
hy avadat teṣāṁ ca ya nirodha
evaṁ-vadī mahāśramaṇaḥ

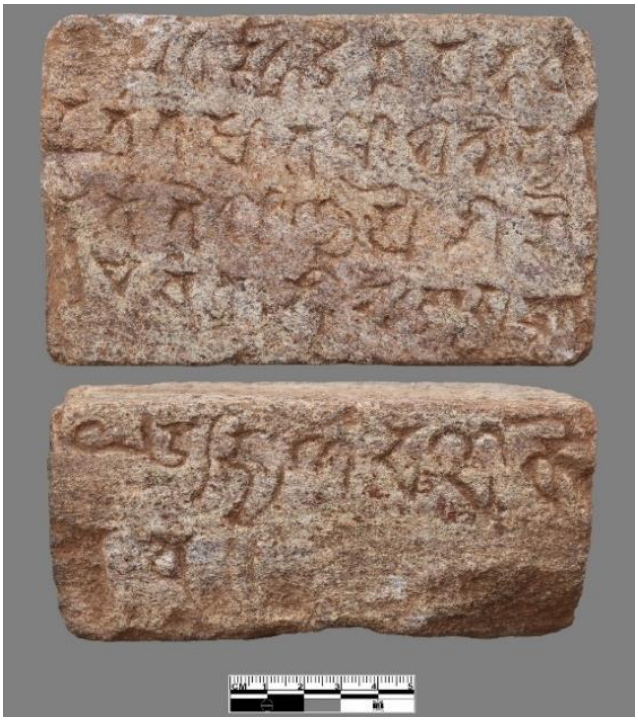
S. No. 13, ACC No. 284



A *Dhāraṇī* mantra inscribed tablet carved from khondalite stone has been discovered, with a measurement of 15 cm in length, 15 cm in height, and 4.5 cm in thickness. The tablet have a well preserved inscription in Sanskrit, written in the Nagari script. The object is dated to the 11th –12th century CE. The tablet is chipped on the upper and back sides. The inscription has five lines of inscriptions on the front side and contains the following verse:

Ye dharmā hetu-prabhavāh
hetuṁ teṣāṁ tathāgato
hy avadat teṣāṁ
ca ya nirodha evaṁ-vadī
mahāśramaṇaḥ

S. No. 14, ACC No. 285



The inscribed tablet made of khondalite stone has been recovered, measuring 13.5 cm in length, 8.5 cm in width, and 7 cm in thickness. Although the stone is broken, the inscription remains in good condition written in Sanskrit, using the Nagari script, and consists of six lines inscribed on the front and bottom sides of the tablet. The inscription has been identified as a *Dhāraṇī mantra*. The tablet is chipped on the back side, dated to the 11th –12th century CE. The inscribed text reads:

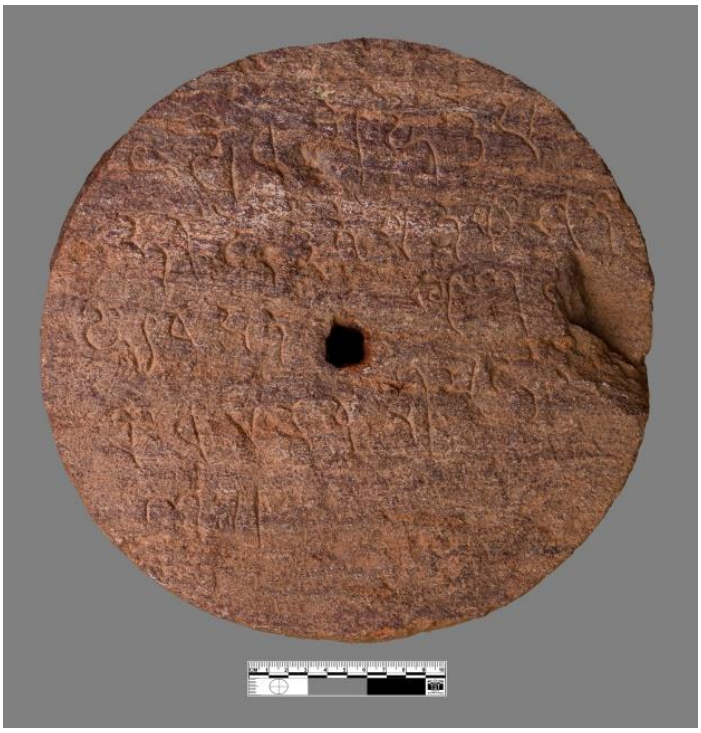
ये धर्मा हेतु

प्रभवा हेतुं तेषां

तथागतो ह्यवदत्तेषां च

यो निरोध एवं वादी

महाश्रमणः।।



S. No. 15, ACC No. 286

The votive stupa carved from khondalite stone bears an inscription in Sanskrit language by using Nagari script. The inscription, consisting of five lines, is engraved on the base of the stupa and has been identified as a *Dhāraṇī mantra* dated to the 11th –12th century CE.

S. No. 16, ACC No. 287

The recovered votive stupa made of khondalite stone has been is in eroded condition. It bears an inscription in Sanskrit, in the Nagari script. A single lined circular inscription is present on the *medhi* portion of the stupa. The inscription has been identified as a *Dhāraṇī mantra* dated to the 11th –12th century CE.

S. No. 17, ACC No. 288

A fragment of khondalite stone block measuring 40 cm in length and 5 cm in height is in good condition, bears a two-line inscription on the surface. The inscription has been identified as a *Dhāraṇī mantra* dated to the 11th –12th century CE. The inscription is in Sanskrit language and written in Eastern Nagari script, and reads:

ये धर्मा हेतु प्रभवा हेतुं तेषां तथागतः ह्य[व]-
दत् तेषाञ्च यो नी(नि)रो[ध] एवं वादी महाश्रम[णः]
य ञ्चिण दण्डके- न च

S. No. 18, ACC No. 289

An eroded and broken on one side votive stupa made of khondalite stone has been recovered, bears a four-line inscription on the *medhi* portion, written in Sanskrit and likely using the Nagari script. The inscription has been identified as a *Dhāraṇī mantra* dated to the 11th –12th century CE.

S. No. 19, ACC No. 290

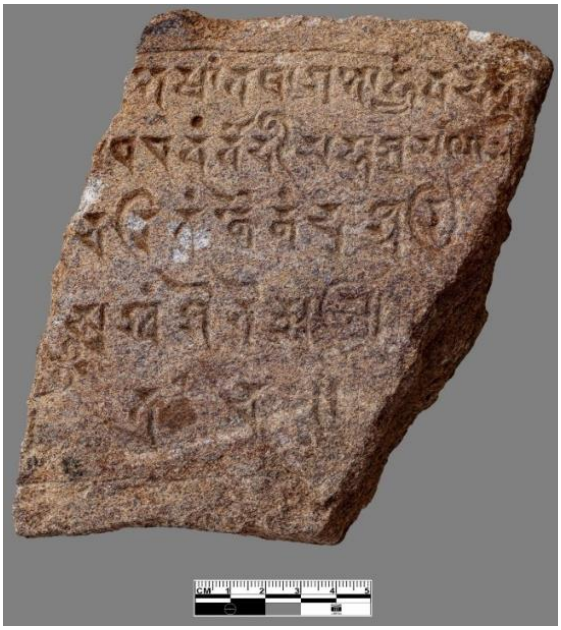
A fragment of a sculpture made of khondalite stone has been recovered, bears a two line inscription measures 20 cm in length and 5 cm in height. The inscription appears to be a *Dhāraṇī mantra* dated to the 10th -12th century CE. A flying Gandharva flanked at one corner of the inscription.

S. No. 20, ACC No. 291

The inscribed stone tablet made of khondalite stone has been recovered, measuring 21 cm in length, 19 cm in width, and 6 cm in thickness. The tablet is in eroded condition and is broken into three pieces. It bears inscriptions on both sides, comprising a total of 30 lines. The text is in Sanskrit in the Nagari script, and has been identified as a *Dhāraṇī mantra* dated to the 11th-12th century CE.

S. No. 21, ACC No. 292

A wedge-shaped khondalite stone block has been recovered, measuring 26 cm in length and 17 cm in width. The stone is in good condition and contains five lines of inscription contained *Dhāraṇī mantra* dated to the 11th–12th century CE.



S. No. 22, ACC No. 293

A fragment of inscribed khondalite stone has been recovered, measuring 13 cm in length and 10 cm in height. The object is in good condition and contains five lines of inscription. The text is in Sanskrit, written in Nagari/Siddhamātrikā script, and has been deciphered as a *Dhāraṇī mantra* dated to the 8th -9th century CE.

S. No. 23, ACC No. 294

A votive stupa of khondalite stone has been unearthed and bears a two-line inscription on the *vedi* portion, is in an eroded condition. The inscription is in Sanskrit, likely written in the Nagari script contained Dhāraṇī *mantra*. The stupa is dated to the 11th –12th century CE.

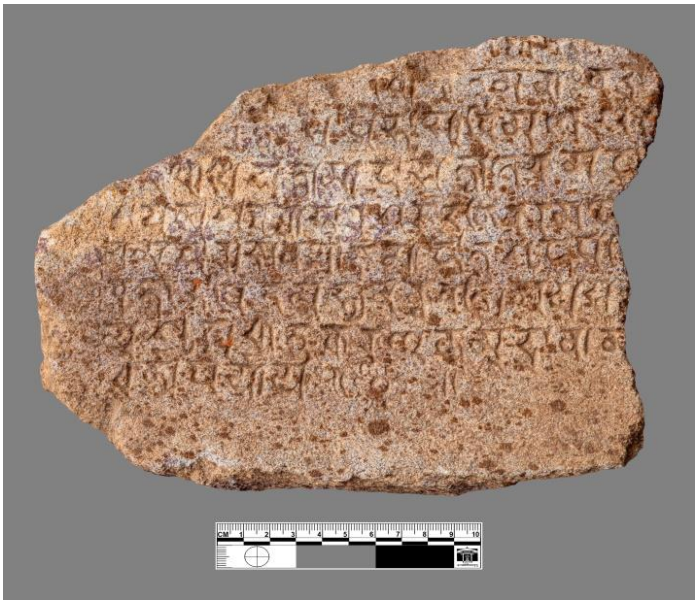
S. No. 24, ACC No. 295

The Dhāraṇī *mantra* inscribed stone fragment in an eroded condition and broken into two parts bears a three line inscription in Sanskrit in the Nagari script, dated to the 11th –12th century CE.

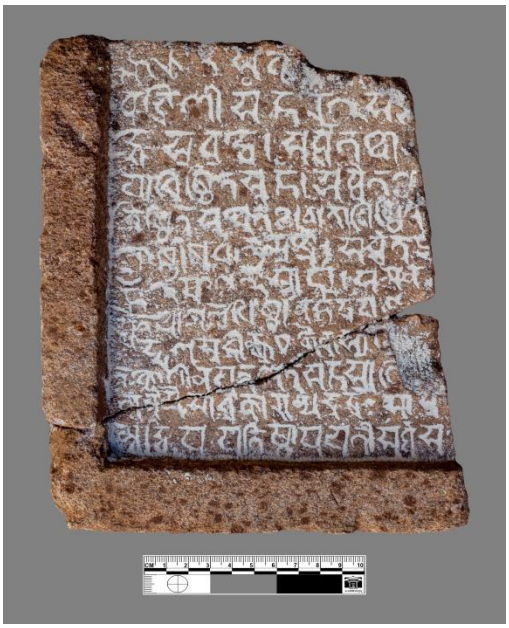
S. No. 25, ACC No. 296

An inscribed stone fragment made of khondalite stone has been discovered. The stone is in good condition and bears a five line inscription in Sanskrit language, written using Nagari script dated to the 11th–12th century CE.

S. No. 26, ACC No. 297



The stone fragment inscription contained Dhāraṇī mantra is made of khondalite stone, measuring 20 cm in length and 12 cm in width, has been found from excavation. The existed eight line inscription is in good condition in Sanskrit language, using Nagari script dated to the 11th –12th century CE.



S. No. 27, ACC No. 298

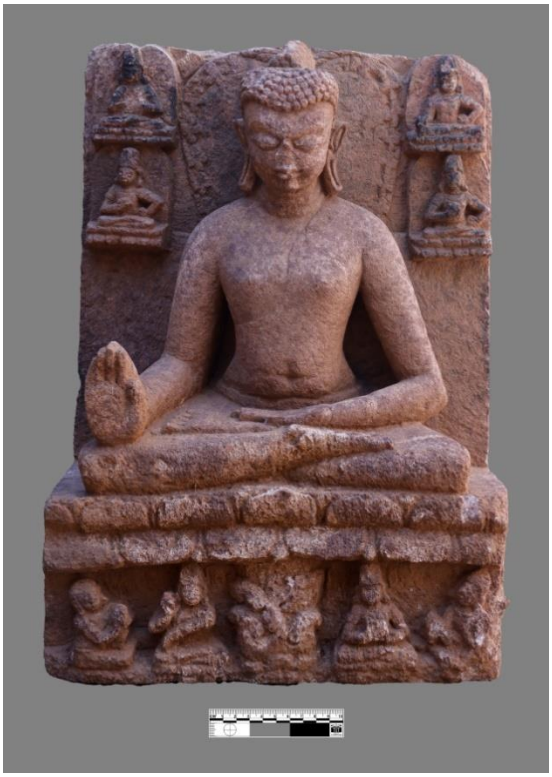
An inscribed stone fragment made of khondalite stone has been discovered in two pieces. The object is in good condition and bears an inscription in Sanskrit written in the Nagari script. The inscription is identified as a Dhāraṇī *mantra* dated to the 11th -12th century CE.

S. No. 28, ACC No. 299

A khondalite stone fragment bearing an inscription in Sanskrit, written in Nagari script, has been found. The object is in good condition and the inscription is identified as a Dhāraṇī *mantra* dated to the 11th – 12th century CE.

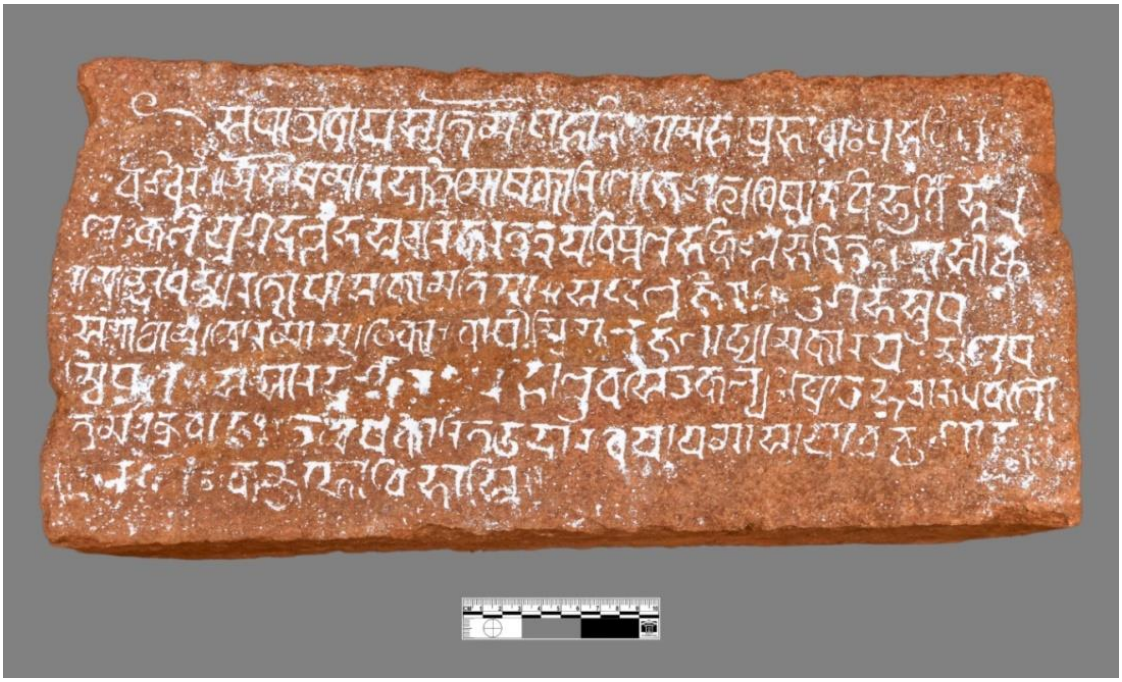
S. No. 29, ACC No. 250

A khondalite stone sculpture of Amoghasiddhi has been recovered. The object is in good condition and bears a two-line inscription on the back side, written in Sanskrit language using the Nagari script. The inscription is identified as a Dhāraṇī *mantra* dated to the 11th -12th century CE.



S. No. 301, ACC No. 300

The inscription measures 43 cm length X 22 cm width X 15 cm thickness eight line inscription is written in Sanskrit language and Nāgari characters of 9th/10th century CE states that Karachanda was a noble soul, who is in this age of Kali where the virtue is rare *(kali-yuga-durlabha-sucarikaḥ*), still performs deeds of dharma. Further it records that he renovated the stupa enshrined the sacred relics *(dhātu-garbha-stūpa)* which was dilapidated. He also erected a beautiful monastery, a stepped well and a pavilion near the site. All these achievements stand like a bridge across the vast ocean of worldly sufferings *(saṁsāra-durggata-mahārṇava-setu-kalpaḥ)*



TERRACOTTA PLAQUES

S. No. 01, ACC No. 64

An inscribed terracotta tablet has been recovered, measuring 3.07 cm in diameter and having thickness 0.09 cm, with a weight of 10.16 grams. The object dates to the 10th -11th century CE and is inscribed in Sanskrit language using the Nagari script. At the middle of the tablet a votive stupa has been embossed. The tablet has eroded and broken on one side, the tablet bears a nine line inscription containing a Dhāraṇī *mantra*/ Buddhist creed.

S. No. 02, ACC No. 72

A broken terracotta plaque having an inscription has been recovered, with measurement 1.3 cm in thickness and weighing 22.28 grams. The antiquity dates back to the 10th - 11th century CE. The inscription, composed in Sanskrit language and written in the Nagari script, is eroded and broken. The plaque contains a Dhāraṇī *mantra* in 12 lines.

S. No. 03, ACC No. 75

A broken piece of 9-line terracotta plaque has been recovered measuring 0.89 cm in thickness and weighing 9.18 grams. The fragment of plaque dates back to the 10th -11th century CE. The inscription is in Sanskrit language, written in the Nagari script. The fragment is eroded and the writings are illegible, it appears to be a Dhāraṇī *mantra*.

S. No. 04, ACC No. 159

This inscribed terracotta tablet measures 3.2 cm in diameter, 0.7 cm in thickness, and weighs 7.90 grams. Dating to the 10th–11th century CE, it is chipped on the upper sides. The tablet contains a 9 line Dhāraṇī *mantra*/Buddhist creed, written in Sanskrit language using the Nāgarī script. The human palm prints are visible on the back side of this tablet.

S. No. 05, ACC No. 163

This tablet made of terracotta, dated to the 10th -11th century CE, has a diameter of 4.8 cm, thickness of 1.4 cm, and weighs 37.69 grams. The tablet is in good condition, with a minor chipped out on one side. The tablet features a 7 line Dhāraṇī *mantra*/creed in Sanskrit language, inscribed in the Nāgarī script. A leaf impression is visible on the reverse.

S. No. 06, Acc. No. 170, 171, 172

Three inscribed terracotta plaques, Accession Nos. 171, 172, and 173, were unearthed from a same place and date to the 10th-11th century CE. Each plaque is oval shaped and bears inscriptions in Sanskrit language, written in the Nāgarī script, containing the Dhāraṇī *mantra*. The first plaque (Acc. No. 171) measures 5.8 cm in diameter, 1.1 cm in thickness, and weighs 33.02 grams. It is intact but exhibits cracks along the sides. The second plaque (Acc. No. 172) is slightly smaller, with a thickness of 0.9 cm and a weight of 21.90 grams; it is broken at the upper edge. The third plaque (Acc. No. 173) measures 5.2 cm in diameter, 0.9 cm in thickness, and weighs 22.70 grams. It has broken into two pieces and shows visible chipping on one corner.

S. No. 07, Acc. No. 173,174,175

Three inscribed terracotta plaques were recovered, assigned Accession Numbers 173, 174, and 175. All three plaques are oval in shape and dated back to the 10th–11th century CE. The first plaque (Acc. No. 173) measures 5.6 cm in diameter, 1.2 cm in thickness, and weighs 39.37 grams. It contains 11 lines of inscription, but they are not clearly visible. The second plaque (Acc. No. 174) is slightly larger in diameter at 5.7 cm, thinner at 0.9 cm, and weighs 27.43 grams, also bearing 11 lines of inscription. The third plaque (Acc. No. 175) is partially broken on both the upper and lower edges, with a thin, black-coloured layer visible on the back. It is 0.8 cm thick and weighs 18.39 grams and carries 11 lines of inscription as well. The text has been identified as a Dhāraṇī *mantra*. The inscriptions are written in Sanskrit using the Nagari script, which was commonly used in the early medieval period.

S. No. 08, ACC No. 158

An inscribed terracotta tablet made by using a stamping technique, measuring 2.2 cm in diameter and 0.8 cm in thickness has been discovered dated back to the 4th -5th century CE, is in good condition. The plaque contains a 3-line Dhāraṇī *mantra* written in Sanskrit language, using Gupta Brahmi script.

Text:

Ye dharmā hetu prabhavā hetu-

stesāṁ tathāgato hyavadata tesāṁ cha

yo nirōdha evamvādi mahāśramaṇaḥ

S. No. 09, Acc. No. 177

A half broken terracotta plaque has been recovered from the excavation. The object is circular in shape, measuring 8.3 cm in diameter, 1.2 cm in thickness, and weighing 80.08 grams. The inscription is arranged in a circular pattern around a central motif depicting four votive stupas. The script used is Nagri, and the language is Sanskrit. The inscribed text has been identified as a Dhāraṇī *mantra* a Buddhist creed. The artefact dated to the 10th -11th century CE.

S. No. 10, Acc. No. 180

An inscribed terracotta potsherd has been recovered from the excavation. The fragment measures 1.5 cm in length, 3.3 cm in width, and 0.8 cm in thickness, with a weight of 18.94 grams. The inscription contains four characters on one side written in Nagari script. The text has been deciphered as "BUDDHAMITRA." The language is Sanskrit language using Nagri script, and the object is dated to the 10th-11th century CE.

S. No. 11, Acc. No. 181

A fragment of a terracotta plaque has been recovered from the excavation. The object measures 6.8 cm in length, 1.6 cm in thickness, and weighs 38.35 grams. The plaque contains three lines inscription arranged in a circular pattern. The inscribed text has been identified as a Dhāraṇī *mantra*, a Buddhist creed. The plaque dates to the 10th-11th century CE.

S. No. 12, Acc. No. 185

An intact Dhāraṇī *mantra* terracotta plaque has been recovered during the excavation. The object is circular in shape that measures 7.8 cm in diameter, 1.4 cm in thickness, and weights 108 grams. The inscription is arranged in a circular pattern around a central motif depicting five votive stupas. The script of the inscription is Nagari, and the language is Sanskrit. This plaque dated to the 10^{th} -11^{th} century CE.

S. No. 13, Acc. No. 205

An inscribed potsherd has been found from the excavation measures 4.64 cm in length, 4.49 cm in width, 0.66 cm in thickness, and weighs 18.23 grams. The inscription, written in Nagari script, is in Sanskrit language and has been deciphered as "DINAKARANASYAH." The artefact is dated to the 10^{th} -11^{th} century CE.

S. No. 14, Acc. No. 199

An inscribed circular shape terracotta plaque has been recovered from the excavation measuring 89.36 cm in diameter, 17.10 cm in thickness, and weighing 137.14 grams. The inscription, written in Nagari script and in Sanskrit language, has been identified as a Dhāraṇī *mantra* or Buddhist creed. The object dated to the 10th-11th century CE.

S. No. 15, Acc. No. 238

This inscribed terracotta plaque circular in shape, measuring 8.07 cm in diameter, 1.9 cm in thickness, and weighing 121.3 grams. The inscription is in Sanskrit language written in Nagari script. The text has been identified as a Dhāraṇī *mantra* or Buddhist creed dated to the 10^{th} -11^{th} century CE.

The record of inscriptions in different scripts from the 4th-5th century CE to 11th-12th century CE are indicating towards the continuity of the Buddhist monastic establishment at Ratnagiri. These inscriptions are mostly composed of Buddhist sacred hymns known as *Dhārani*; a few describe the patronage of the noble class people and the ritual processes related to Vajrayana Buddhism.

Beads

The excavation yielded terracotta beads of different shapes and size like spherical, barrel, areca nut, and convex bicone. Out of which a smooth surfaced areca nut shape bead having deep groove on the neck and a string of iron inserted in the centre is significance, having D-2.1cm, H-1.9cm, and 9.11g in weight.

Beads

Beads

Beads

Discs

The mould-made and handmade well-fired circular designed discs have been retrieved from the excavation. It can be produced by mould or by using flaking and ground techniques to give it a circular shape. The terracotta disc has been retrieved while digging and can be interpreted as gaming or counting objects, weights and lids to cover *Surahi*.

Terracotta Disc

The discs produced from potsherds commonly known as pottery disc. The discs are also being collected in large quantity during excavation. The artefact is connected to well-known indigenous game where the player in turn needs to have a disc.

Terracotta Disc

Stone Disc

Dabber

The findings of a dabber from the excavation within the stupa complex which generally used to shape pots in leather hard condition, is tapered with slightly chipped working surface of convex base. It is intriguing to find such artefacts in secondary context. It might have brought to the site through outsiders or may be some specific ritual pots were prepared by the monks and nuns.

Spool

Besides such industrial artefact many crafts or textile tools also noticed from the site which corroborated the fact that the monastic establishment produce day to day objects required for their basic needs where they resided. The findings of spools suggest that the monks produce textile for themselves or ritual purpose.

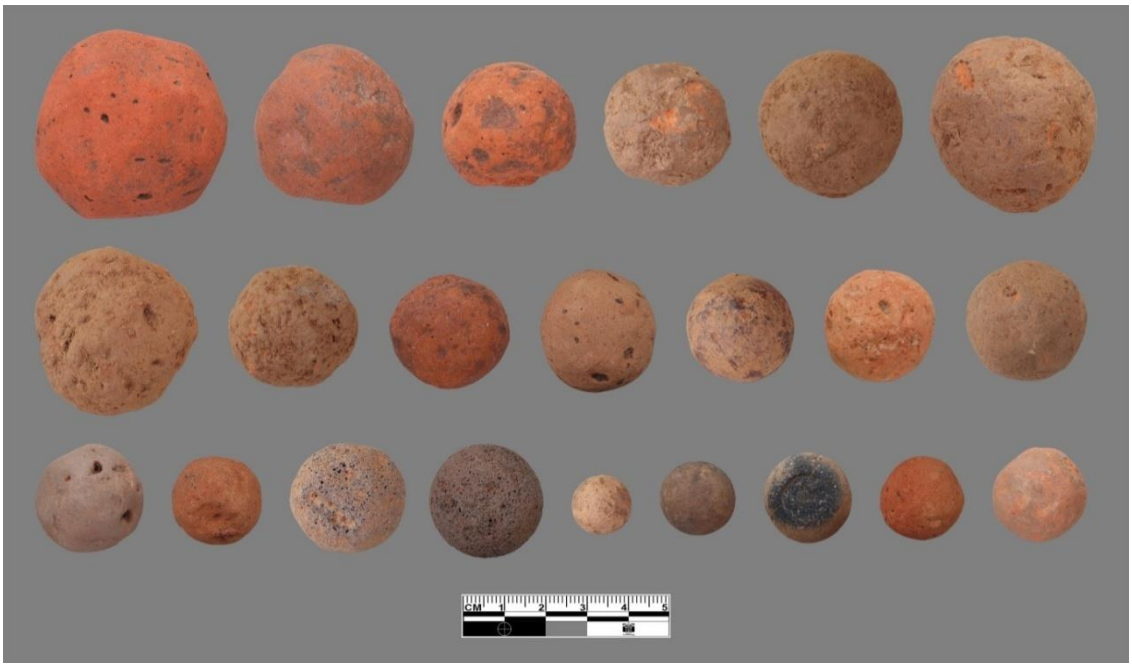
Pestle, Polisher and Saddle Quern

The grinding stones located at different areas suggest the presence of pestles and saddle quern. The findings of stone pestles and terracotta saddle quern suggests that residents of the monastery even prepared food for themselves. It might be used for medicinal purpose. Similarly, polishers with used marks are noteworthy that shedds some more light on their daily lives.

Sling Ball

The spherical objects known as sling ball have been found throughout the cultural deposit. These are made of terracotta as well as khondalite stone with some plain and decorated surfaces. These are of various size and weight. One sling ball is observed with a tiny motif probably of a bird.

Stone Sling Ball



Terracotta Sling Ball

Bangles

Broken pieces of bangles of both glass and lac along with an intact bangle have been found from the recent excavation. The different profiles/shapes of bangles

show not only about the age group but also body shapes. It has been observed from the sculptures that the divinities attired with bangles.

Terracotta Wheel

In the present spadework, 2 miniature terracotta wheels have been retrieved. These could be toy cart wheels suggesting the presence of children who either lived in the complex or had visited the monastic site.

Terracotta Animal Figurines

The terracotta images made out of clay have been continued throughout all the cultural periods. The site Ratnagiri has yielded terracotta figurines mostly of animals, were probably being used as gaming or ritualistic objects. The presence of figurines in these sites and their stylistic representation denotes the cultural and ideological affiliation.

Terracotta Plaques

Oval shaped terracotta plaques in both intact and fragmented condition have been found from the excavation. All of them are bearing the sacred Buddhist chants-dharani in Nagari script and Sanskrit language. One significant variety is depicted beautifully carved five stupas in a pattern of showing four cardinal directions. These are ornamented stupas surmounted by a *harmika* and chhatravali with dharani at the encircling edges. Some plaques are containing only dharani and without any decorative motifs. A small sized circular plaque/sealing has

been retrieved from the excavation in layer 3, bearing a legend in Brahmi script having Sanskrit language dated to c. 4th-5th century CE.

Stone Caskets

Two straight bowl shaped khondalite caskets have been recovered from the present excavation; one of them is with a lid and containing soil. The reliquary with lid resembles with a miniature stupa. The provenances of these caskets are found in near the brick stupa having cross plan.

Inscribed Potsherds

A bulk amount of red and grey wares pottery has been retrieved from the excavation. Among them, two inscribed potsherds have been collected, one is in grey and the other one is in red ware. The inscription on the grey potsherd reads as 'Buddhamitra' and another one is in red ware reads as '*Dinakarasyaha'*, both in Nagari script.

Metal Object

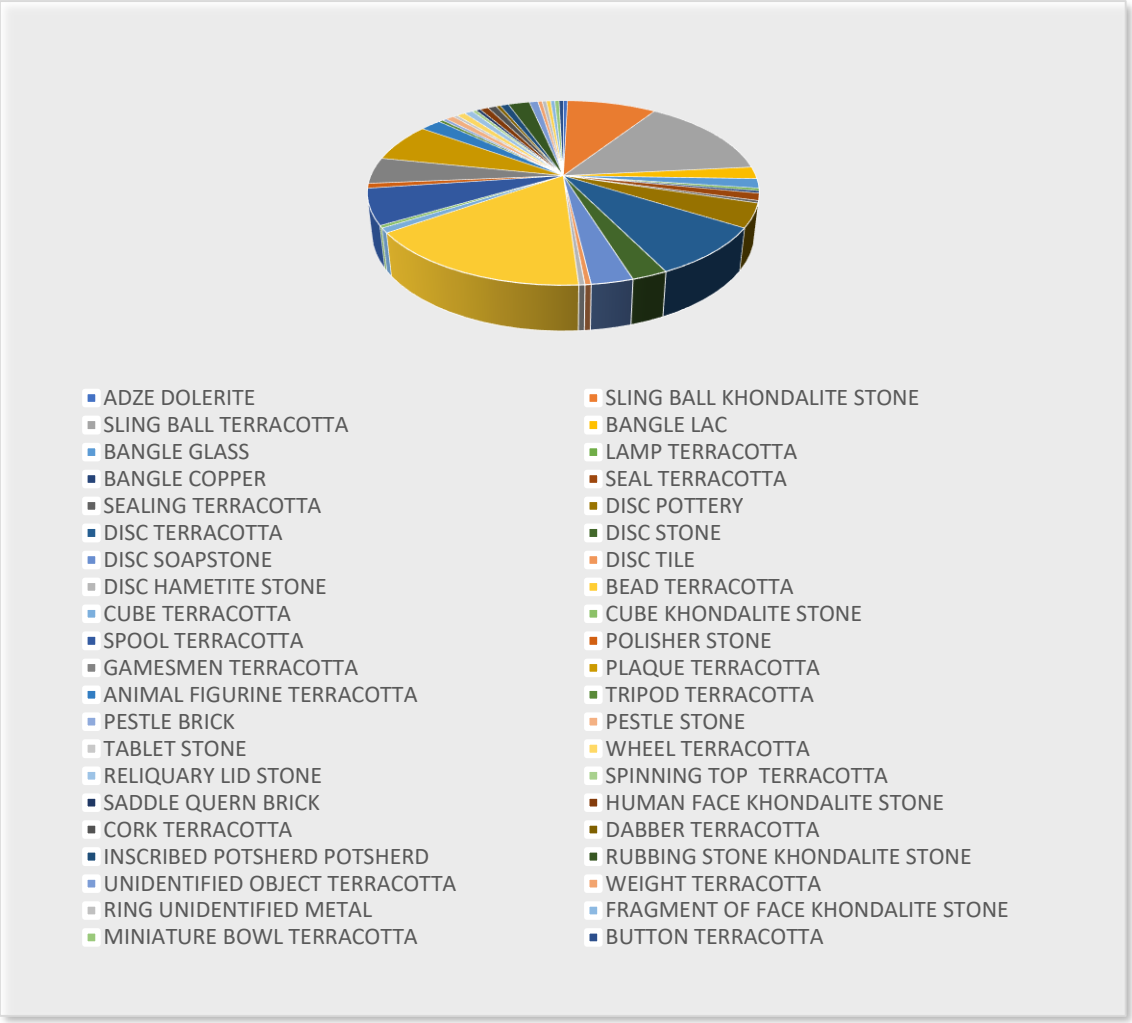
The spadework has revealed different kinds of metal objects from all cultural level. Iron objects are dominating among the other metal objects. Knowledge about the use of iron was common in early Odisha. In the Kondedda copper plate inscription of Sailodbhaba king Dharmaraja there is the earliest evidence of use of Iron for the manufacture of different type of weapons for the use in war. R.L Mitra stated in his work that short swords and dagger made of iron are used as favourite weapons in Odisha. There are many sculptures and divinities holding war objects like sword, dagger, bow and arrow etc shows the use of iron in that period. In the current excavation, some iron clamps found that are used to join the stone blocks, these iron clamps are used in structural stupas, stone flooring and they also used the iron made clamps to join the body part of the huge Buddha sculptures, The main purpose of these clamps are to join the stone blocks with each other and held them tightly. From the current excavation we have unearthed many sculptures like Vajra Gandha Tara, Arapchana, Achala and Marichi holding swords and other weapons which probably made of iron. They used iron as needle to make the perforation in the beads and

preparation of Aksh-mala. All these evidence shows that peoples and artisans in early Odisha had a deep knowledge of iron technology and production of iron implements of war and peace.

TABLE OF MINOR OBJECTS

S No.	Object	Material	Total No.
1	Adze	Dolerite	1
2	Sling Ball	Khondalite Stone	21
3	Sling Ball	Terracotta	34
4	Bangle	Lac	5
5	Bangle	Glass	4
6	Lamp	Terracotta	1
7	Bangle	Copper	1
8	Seal	Terracotta	3
9	Sealing	Terracotta	1
10	Disc	Pottery	10
11	Disc	Terracotta	21
12	Disc	Stone	6
13	Disc	Soapstone	7
14	Disc	Tile	1
15	Disc	Hematite Stone	1
16	Bead	Terracotta	39
17	Cube	Terracotta	2
18	Cube	Khondalite Stone	1
19	Spool	Terracotta	15
20	Polisher	Stone	2
21	Gamesmen	Terracotta	11

S No.	Object	Material	Total No.
22	Plaque	Terracotta	16
23	Animal Figurine	Terracotta	5
24	Tripod	Terracotta	1
25	Pestle	Brick	1
26	Pestle	Stone	2
27	Tablet	Stone	1
28	Wheel	Terracotta	2
29	Reliquary Lid	Stone	2
30	Spinning Top	Terracotta	1
31	Saddle Quern	Brick	1
32	Human Face	Khondalite Stone	2
33	Cork	Terracotta	2
34	Dabber	Terracotta	1
35	Inscribed Potsherd	Potsherd	2
36	Rubbing Stone	Khondalite Stone	5
37	Unidentified Object	Terracotta	2
38	Weight	Terracotta	1
39	Ring	Unidentified Metal	1
40	Ring	Stone	1
41	Fragment Of Face	Khondalite Stone	1
42	Miniature Bowl	Terracotta	1
43	Button	Terracotta	1
		Total No of Minor Objects	239
		Total Number of Major objects (Inscriptions and Sculptures)	62
		Grand Total	301



Stupa

The excavation revealed remains of structural as well as votive stupas associated with Buddhist pantheon. According to Buddhist text Mahaparinibbana Sutta, Gautama Buddha himself said that stupas should be erected over his and his disciples' remains.

The structure of Stupa consists of a circular platform, a hemispherical dome over it and a square shaped enclosure called *harmika* with superimposed umbrellas on it. These umbrella is known as *yashti* with *chhatravali*. It is surrounded by a railing and at places, these railings are seen with a gateway.

These Stupas are broadly categorised into four types, *Sharirika, Paribhogika, Uddeshika* and *Manasi* Stupas. The Sharirika Stupas were those which were erected over the corporeal relics of Buddha and his chief disciples. The second one, *Paribhogika* Stupas were constructed over belongings of both Buddha and his

disciples or other monks. Stupas which were built for the commemoration of particular events from Tathagata's (Gautama Buddha) journey of life. The fourth one, Manasi or Votive Stupas were donated by the followers at sacred places for attaining merit and fulfilment of vows and wishes.

Votive Stupa

These stupas are typically small, miniature versions of larger stupas, built as offerings by monks and pilgrims. There are two major types of votive stupas: monolithic votive stupas which are single, unified pieces of stone carved into the shape of a stupa and structural votive stupas, which are built using multiple components, such as stones or bricks, to create the structure. Both types of votive stupas hold significant cultural and religious importance in Buddhism, serving as symbols of devotion and faith.

The history and evolution of Votive Stupas play a vital role in Buddhism and are present in numerous Buddhist sites both within and beyond the country. Odisha may be one of the epicentres for the construction of votive stupa as evident from Langudi. Palaeographic and stylistic studies suggest that the beginning of such creativity dated back to 2nd century BCE and gradually reached its zenith in 11th -12th century CE. The Votive Stupas at Udayagiri exhibit a more developed form compared to those at Lalitagiri and Langudi, though they remain relatively simple in design. The most refined and abundantly found examples come from the present excavated site Ratnagiri, where the Votive Stupas are intricately adorned with decorative patterns, motifs, and numerous sculptures carved into niches.

Votive Stupas at Ratanagiri

In Odisha, Ratnagiri Buddhist site possess out numbers of votive stupas in comparison to rest Buddhist site. The Stupa is considered a symbol of Mahaparinirvana. The history and evolution of Votive-stupa play a significant role in Buddhism and the presence of outnumbered in Buddhist sites of Odisha not only suggests its immense potentiality but also shows its evolution, creativity, attachments and expression of the devotees towards Lord Buddha and Buddhism.

During the excavation a large number of Votive Stupas have been retrieved, documented and categorised as structural and monolithic. The structures are made out

of multiple layers bricks or stone. During the documentation it has been observed that 90% of the votive stupa are monolithic while about 10% of the votive stupa are structural made out of stone blocks. One of the features noticed is that most of the votive stupa housed Buddhist divinities within a niche portion of the *medhi*. The prominent divinities are of Avalokiteshvara, Padmapani, Lokeshwara, Amoghapasha Vajrasattva, Arpaccana, and goddess Tara, Bhrikuti, Tarodbhava Kurkula, Marichi, Chunda. The representation of Avalokiteshvara and Tara are mostly seen. Few stupa housed deities such Amoghhasiddhī in *Abhaya mudrā* in the northern niche, Ratnasambhava in *varada mudrā* in the southern direction, Akśobhya in *bhūmisparśa mudra* in eastern and Amitābha in *dhyān mudrā* housed in the western direction niche.

Avalokiteshvara

The statue of Avalokiteshvara Bodhisattva is most visible in the niches of the Votive Stupa. The Bodhisattva is seated in Vajraparyakasana and holds a lotus flower in his right hand.

Vajrsattva

Vajrasattava is represented seated on a lotus in vajraparyankasana, in his right hand he holds a vajra close to his chest and in his left hand, he holds a ghanta resting on his thigh.

Arpaccana

Arapchana, also known as Sadyonubhava-Arapchana or Sadyonubhava-Manjusri, radiates magnificence like the full moon. He has a serene, smiling face and is adorned with princely ornaments. Seated in the vajraparyankasana posture on a visva-padma, he holds a raised sword in his right hand, symbolizing wisdom and clarity. In his left hand, positioned near his chest, he carries a manuscript, signifying knowledge and learning.

Bhrukuti

The three-eyed goddess emanates from Amitabha. She holds the *tridandi* and the *kamandalu* with her left hands, the upper right hand holds a rosary while the other right is in the varad mudra

Tara

Tara is also recognized to be the saviour goddess protecting her worshippers from all the evils of existence. Tara depicted seated in Vajraparyakasana with a flower in her right hand at the Votive Stupa.

Parnasabari

Parnsabari is the goddess who prevents epidemic outbreaks and assures safety to the terror-stricken. She is depicted in a yellow complexion with three faces, three eyes, and six arms; her face at the center is blue, the right white, and the left red, and has a smiling face. She is adorned with jewels, and holds a vajra, a parasu, and an arrow in her right hand and in her left Tarjani with the noose, cluster of leaves and the bow. Her jatamukata bears the image of Aksobhaya. In some instances Parnasabri is shown standing in pratyalidha posture on the moon over the white lotus, trampling Vighnas (Ganesha) under her feet.

Marichi

Marichi is the Buddhist counterpart of Surya. The goddess is depicted either human headed or three headed (left head of the goddess is sow-like) and six armed forms are shown standing in the pratyalida posture either on a chariot the charioteer being rahu or on a plain pedestal. The right hand carries a needle, an arrow and thunderbolt. Two of the left hands hold the branch of an ashoka tree and a bow, the three being in the tarjani mudra with or without a thread ending in a noose. She drives a chariot drawn by three, five or seven pigs / horses; she has a female charioteer who has no legs or Rahu who has no body.

Tarodbhava Kurukulla

The goddess is shown with two, four, six or eight arms when she is four arms she is known as Tarodbhava Kurukulla. She is four-armed seated in vajraparyankasana, left two arms displays abhaya mudra and the arrow and in the right to arms she carries the bow and a red lotus.

Vajra Gandha Tara

The deity attire with of Jatamukuta, long circular earrings, necklace (galahara) around her neck, armlets (bajubandha), bracelet (keyur) can be named as Vajra-Gandha-Tara.

Most of the divinities found from the sites are housed within the votive stupa which makes unique and significant features of Ratnagiri and exhibits the site's association with Vajrayana. The votive stupas are architecturally significant as observed few stupas are designed as Kalingan temple architecture and shows religious syncretism of that time at the site. These khondalite stone built stupas are plain as well as highly curved and some are housed with different Buddhist divinities. The excavation yielded two types of Votive Stupa i.e Monolithic and Structural. In Monolithic Votive Stupa, the components are vedhi, medhi, niches with frames, anda, harmika and chhatravali.

Chhatra/ chhatravali

During the excavation, several types of *chatravalis* (umbrella and spiral like structures usually placed on top of stupas) were found, showing different designs and styles. Some were shaped like a semicircular disc with a band at the base, and were topped with a lotus petal design at the center. Another type had a short shaft supporting five thin discs, each getting smaller towards the top. One *chatravali* was made of three thick circular bands with sloping edges and was topped with a capping stone. Another example had nine circular rings, arranged one above the other, getting smaller in size as they went up. A similar design had seven rings, also arranged in a decreasing order.

Sculptural Art

The tradition of sculptural art is as old as the material culture of human society. The medium of the production of sculptures are of clay, wood, stone etc. The best way to express the human ideology in the form of idol has come to notice from the paleolithic period. The earliest evidence a sculpture produce from clay in Odisha has been noticed in Hikudi a Neolithic Site in Subarnapur district, in the middle Mahanadi valley, the animal and anthromorphic figurine from Golbaisasan in Khurda dist., Bharatihuda in Cuttack district and other Chalcolithic sites of the state exhibit the beginning of sculptural art in terracotta. Stone as a medium of expression of sculptural

art comes to notice from the 2nd-3rd cent BCE in Odisha. The earliest stone sculptures of Yaksha, rock cut elephant of Dhauli dated to c. 3rd cent BCE in Bhubaneswar followed by the sculptural panel of Ranigumpha in Udayagiri hill, Bhubaneswar and the 5th-6th century Asanpat Nataraj image found from Keonjhar are noteworthy findings to study the beginning and spread of sculptural art in the state. In Buddhist context Langudi is the earliest example where symbolic representation of Buddha and Buddhism gradully developed towards rock cut sculptural art followed by individual masterpiece of sculptural art dated to 1st-2nd cent BCE. The Buddhist site of Lalitagiri, Udayagiri and other sites have rich depository of sculptures dated from the early part of the commnera to 12th century CE. The excacvated site Ratnagiri exhibits the best specimen of sculptural art from 7th Century CE to 12th century CE.

Ratnagiri is particularly famous for its highly refined Mahayana and Vajrayana sculptures of Buddhism, the sculptural art discovered here represents a mature phase of Buddhist iconography, enriched with profound philosophical symbolism and a high degree of aesthetic excellence. The sculptures of Buddha and Bodhisattvas to Avalokiteshvara and esoteric deities like Tara, Marichi, Chunda provide invaluable insights into the religious, artistic and cultural milieu of early historic period. All the sculptures retrieved in this excavation are further defined as.

Colossal head of Buddha

(Acc: 243)

A colossal head of Buddha (L- 1.25 m × W- 1.07 m) lies on the ground, where the downward facial portion is noticeable. There is a decorative flower of four petals on top of its *ushnisha*, which has four rows of curls, and the rest of the head has thirteen rows of curls. Its elongated earlobes have a length of 64 cm. Also, a large neck fragment with three auspicious lines known as rekhil is present in situ. Additionally, two monumental feet have been retrieved from the trench.

(Acc: 244)

An east-facing colossal head of Buddha, measuring L- 0.92 m × W- 0.76 m × H- 1.20 m, is exposed. The facial expression is exuding a contemplative aura, with petal-shaped, half-closed eyes and elegantly bow-shaped eyebrows. The craftsmanship is remarkable, with a prominent chin and full, rounded cheeks and a sharp nose that add to the serene countenance of Buddha. The elongated earlobes, measuring 66 cm, further enhanced its spiritual symbolism. The head and the ushnisha are meticulously decorated with twenty rows of curls, fourteen of which are on the head itself, while the ushnisha features an additional six rows. One auspicious line around the neck and a broken urna is observed.

(Acc: 245)

Another colossal Buddha head facing west, measures L- 1.02 m × W- 1.05 × H- 1.25 m. The compassionate and contemplative expression is conveyed through its half-closed eyes, full lips and sharp nose. The elongated earlobes, measuring 0.66 m, feature long perforations, enhancing its spiritual symbolism. A circular curly urna adorns the forehead, while the head and ushnisha are intricately decorated with nineteen rows of curls, thirteen on the head itself and an additional six on the ushnisha. Two auspicious lines encircle the neck, adding to the face's sacred aura.

Manjuśrī (Acc: 246)

Apart from this, a sculpture of Manjuśī measuring H-45 cm × W-36 cm has been unearthed. The sculpture represents Manjuśrī with one head and two arms, seated in lalitasana pose, crown with a shikhandaka kakapakasha headgear, although the face has been eroded over time. A halo is visible behind the head, and the figure adorns ear studs in both ears. A beaded upavita is draped from left to right across the chest, and three auspicious lines are visible around the neck, accompanied by a pendant. Bajubandha is worn in both hands, with the partially broken right hand. The left hand rests comfortably on a pedestal behind the left leg, and a Nilotpala flower emerges from it. A book is placed on top of the Nilotpala, although it is broken and obliterated, it exhibits gracefulness.

Manjuśrī (Acc: 247)

The sculpture features a single-faced, two-handed figure in standing dwibhangi pose measures L: 84cm x W:39cm Th: 20cm. The figure wears a crown with a sikhandaka kakapakasha headgear. The round face has protruding eyes with bow-shaped eyebrows, a broken nose, and a fuller upper lip. The elongated earlobes and three auspicious lines around the neck are notable features. A tiger nail pendant adorns the neck, and a *bajubanda* is worn on both hands. A thread-like upavita is draped from left to right. The figure's lower garment is diaphanous, and the left leg is positioned forward. Unfortunately, both hands are broken. A book is placed on a Nilotpala flower near the

figure. Beside the right leg, a female attendant is seated with folded legs in anjali mudra.

Image of Vajrasattva as Bana Madhava

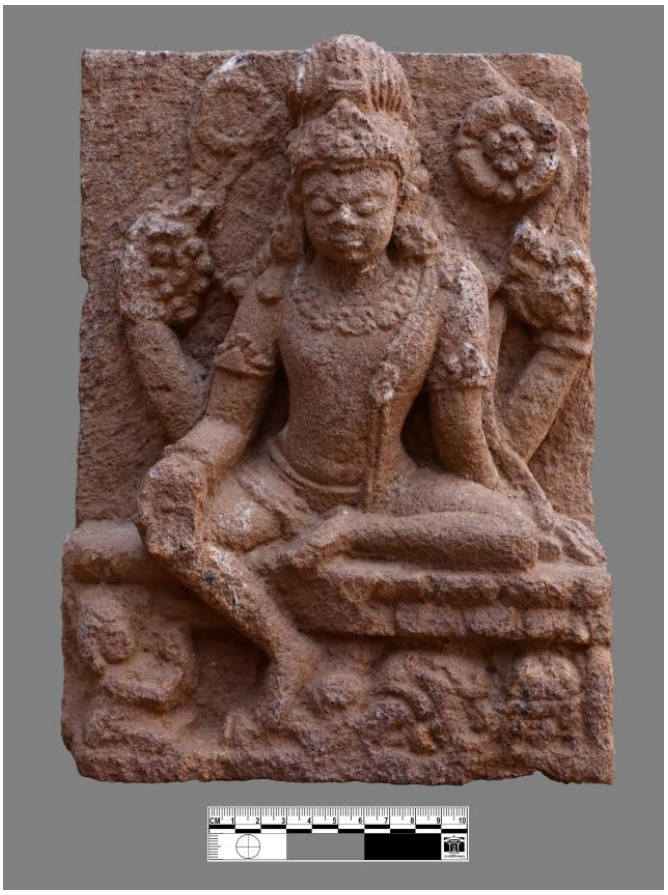
Bejewelled two-armed Vajrasattva seated in the paryankasana attitude on double petal lotus. Vajra sattva carries a thunderbolt in his right palm, held upwards against the chest, and a bell in the left hand resting on the thigh. Around the head is an elongated-oval halo, conspicuously narrow towards the bottom. his facial expression is serene, exuding a contemplative and meditative aura. His eyes are half-closed, with bow-shaped eyebrows, a damaged nose and mouth, and elongated earlobes. Three-string beaded kanthahara, thread-like upavita. The above pedestrian shows two lions facing forward, a female attendant is seated with folded legs in anjali mudra (Village deity of Parvatapanasua).

Gana (Acc: 248)

The sculpture appears to be carved from a single block of khondalite stone measures H: 60 cm x W: 52 cm x Th: 26 cm. The figure is squatting in a ritualistic posture with knees bent outward and feet placed flat, often referred to as utkutikasana. The body appears muscular, with a pot belly; the eyebrows are in bow-shaped and their eyes protrude. His nose is broken; Large, disc-like earrings are visible on both ears. Hands were rested on the knees, possibly once displaying a mudra. The head shows a

stylized coiffure with long, matted hair. The chest is adorned with a tube-shaped bead necklace. A represented genital part indicates the sculpture's association with fertility or tantric practices.

Amoghpasa Lokeshwar (Acc: 249)



Three-eyed and four-handed Amoghapasa Lokeshwar is sitting in Lalitasana on a double-petal lotus with his right leg resting on a lotus flower (H: 28cm W: 22cm Th: 18 cm). Right lower arm in varad mudra, right upper hand holding akshmala and pasa, left upper hand holding sprinkler, and lower left hand resting on the seat behind his thigh holding the stalk of the padma. The hair is tied in a bun with locks falling over the shoulders. It has eroded an effigy of Dhyani-Buddha Amitabha on it. There is a circular halo behind the head. Below the lotus pedestal, to the right side, is shown a devotee kneeling and looking upwards with her hands in anjali-mudra. The left side displays some objects of worship. Amoghapasa, known as the Bodhisattva with the unfailing noose, is a form of Lokesvara who uses his noose—symbolizing compassion—to catch and save all livingbeings. The text Amoghapasahrdaya Dharani, translated by R. O. Meisezahl, says that those who respect Amoghapasa and recite the text receive twenty benefits during their life, and eight more important ones at the time of death.

Buddha in Abhay mudra or Amoghsiddhi (Acc: 250)

An image of Buddha in *Abhay mudra* or Amoghsiddhi (H: 50 cm W: 36cm Th: 26 cm). This image of the Buddha seated in vajra-paryankasana over a double petal lotus is draped in an uttarasanga which covers the left side of the body and an antarvasa with a part of it spread on the seat. The left palm is placed on his lap while the right one is represented by *abhaya-mudra*. Behind the head is an oval halo, above which is the depiction of the flowers. On either side of the main image are depicted four Bodhisattvas as attending figures. Below the lotus pedestal, two deities are shown, on the right and left side, devotees kneeling and looking towards them with their hands in *anjali-mudra*. On the back side of the image, two lines of inscription of nagari characters, it is possibly a dharni *mantra* belonging to the 10th-11th CE, were eroded.

Tara (Acc: 251)

A one-headed, two-armed tara is seated in lalitasan on a double-petal lotus, with her left leg folded and right leg resting on a lotus flower. The figure wears a tight-fitting kanchuki and is adorned with necklaces, earrings, baju band, keyur, upvita, and a headdress. The face is calm, symmetrical, and smiling, projecting a sense of inner peace and spiritual grace. Behind the head, one circular halo, right hand in varad mudra and left hand resting on the seat behind her thigh, holds the stalk of the Utpala. Below the lotus pedestal, one devotee is kneeling and looking towards with her hands in Anjali mudra and some ritualistic things are shown

(H:34, W:19, Th:10)

Unidentified Goddess (Acc: 252)

One-headed two-armed goddess seated in Lalitasan on a double-petal lotus, her left leg is folded, and her right leg is resting on a lotus flower (H:27, W:18, Th:9). She has a broken face, bun-shaped headgear, wearing necklace, a bajuband, upvita, and mekhala. The right hand is in Varada mudra and the left hand is broken, holding an unidentified object. Behind the head of the deity, a circular halo and the stylistically blowing clothes are shown. Below the lotus pedestal, one devotee is kneeling and looking towards with her hands in Anjali mudra and some ritualistic things are shown.

Buddha in Bhumisparsh Mudra or Akshobhya (Acc: 253)

Buddha in bhumisparsh mudra seated in paryakasan, face is broken, ushnisha eroded, elongated earlobes, right hand is in bhumisparsh mudra and left palm is placed on his lap (H: 41cm W:23cm Th: 15cm)

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Standing Buddha (Acc: 254)

Sculpture of Buddha where the left part of the face is broken and both hands from the elbow are broken, and is legs from the knee portion are broken. The lower garment shown is a sculpture. He has a smiling, chubby face, thick upper and lower lips, and has flattened nose with closed eyes (H: 53cm W: 34cm Th: 15cm).

Nag Purush (Acc: 255)

A five-serpent hooded deity holding a kalash, his hair is styled in a coiffure of a bun and shown with a headband with the pendant at the middle of the forehead (H: 30cm W: 57cm Th 40cm). Earlobes wearing long, rounded earrings, big beaded kanthhara, decorated bajuband, keyur in both hand he is shown wearing uttariya decorated mekhala. Face expression serene, reflective, with a smile. This sculpture was rested on lion pedestal of the rectangular chaitya hall.

Buddha in bhumisparsh mudra (Acc: 256)

Buddha is seated under a pipal tree (Ficus religiosa), seated in vajraparyakasan on a double-petal lotus. the face is chipped out from lips, ushnisha, elongated earlobes, right hand in bhumisparsh mudra and left palm is placed on his lap (H:52 cm W: 34cm Th: 23cm). Behind the head, it has an elliptical halo. On both sides, flying gandharvas holding garlands in their hands are shown. Below the lotus pedestal, one attending devotee is kneeling and looking towards with her hands in Anjali mudra and some ritualistic things are shown.

Bust of Buddha (Acc: 257)

The Buddha sculpture has an elliptical halo. He showed ushnisha. Oval-shaped face having pointed bow-shaped eyebrows, nose part bit chipped, have protruding lips. Draped in a diaphanous antarvasa and an uttarasanga which is shown on his left chest, shoulder and arm. On both sides, flying gandharvas holding garlands in their hands are shown (H: 41cm W: 55cm Th: 17cm).

Buddha in Dhyan Mudra or Amitabha (Acc: 258)

Buddha is seated in Vajraparyakasan on a double-petal lotus and has a circular, decorated halo and Torana behind him (H: 20cm, W:12cm, Th:7cm). The right portion of the head is broken, and facial expression and features are not prominently visible. Both hands are in the Dhyana mudra. Below the lotus pedestal, floral motifs are there, and below it devotees are shown doing some kind of ritualistic things. The corner of the sculpture has a flower motif.

Head of Buddha (Acc: 259)

A broken head part of a Buddha, with ushnisha and nose part, is broken. He has very sharp eyebrows, and his eyes are closed, an urna is visible on his forehead, his thin lips, and a smiling face, and he has elongated earlobes. A broken part of a

Broken Buddha in Dhyan Mudra (Acc: 260)

A circular halo is visible behind his head, his right portion of face is broken, his left eye is closed, his nose and lips are eroded, elongated ear lobed. The right hand is placed on his lap in dhyan mudra, and the left portion of the body is broken (H: 21 W: 10cm Th: 5cm).

Lower Part of the Deity (Acc: 261)

The bust part of the sculpture is broken. The right hand is in varada mudra, and the left is placed on the lap and holding a bowl. The upper and lower garments are visible. Seating in Vajraparyakasan on a double-petal lotus, one devotee is kneeling and looking upwards with his hands in Anjali mudra, and lotus stems are visible near the devotee (H:31cm W: 31cm Th: 16cm).

The bust part of the sculpture is broken. The deity is standing on a double-petal lotus, wearing a translucent lower garment and mekhla. The stem of a flower is visible (H: 13cm W: 13 Th: 17cm)

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Bust of Buddha Acc. No: 263

A half-elliptical halo is visible, the Buddha is seen with *ushnisha* and closed eyes, broken nose, lips, and elongated earlobes. Draped in a diaphanous *antarvasa* and an *uttarasanga*, which is shown on his left chest and shoulder (H: 35cm W: 29cm Th: 26cm). Flying *gandharava* can be seen on the left side of the sculpture holding a garland in his hand.

Lower Part of the Deity Acc. No: 264

Standing female deity, wearing a lower garment and *mekhala*. Only the right hand wearing the keyur is visible, and the palm is in varad mudra. On both sides, some kinds of floral motifs are seen (H:35cm W: 27cm Th: 22cm).

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Lower part of Deity Acc, no: 265

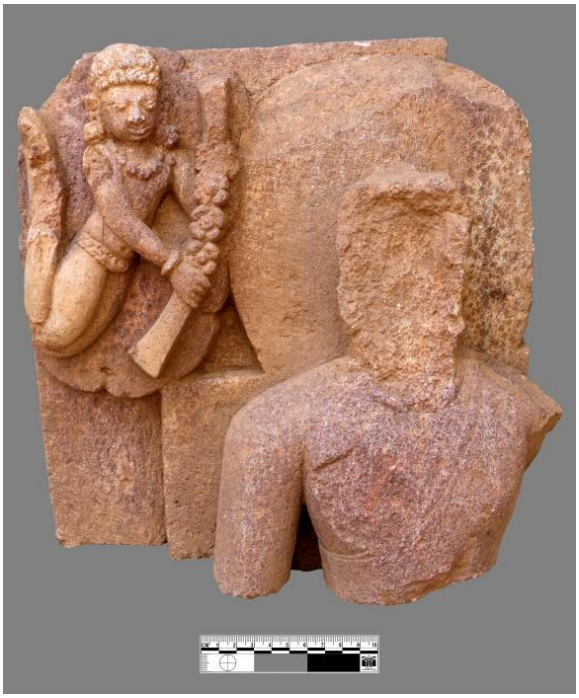
A deity seated in Lalitasan and seen wearing a lower garment, upvita, and mekhala. Left hand is behind the thigh and holding the stem of the flower, and the right hand is broken. An object used for ritualistic purposes is seen on the right side, below the pedestal H:22cm W:

Seated Buddha Acc No.: 266

Buddha is sitting in a decorated broken niche, on a double-petal lotus in varad mudra. He is shown in ushnish, eyes are closed, has a long, pointed nose, thin lips, and elongated earlobes H: 15cm W: 17cm Th: 22.

Seated Deity Acc No.: 267

The head part of the sculpture is broken. A deity seated in Lalitasan on double-petal lotus and wearing a lower garment, bajuband, keyur, and mekhala. The left hand is behind the thigh and holding the stem of the flower, and the right hand is in varad mudra (H: 25cm W: 26cm Th: 23cm).



Bust of Buddha Acc. No: 268

Elliptical halo behind head. Broken face, diaphanous antarvasa is seen and a uttarasanga which is shown on his left chest and shoulder. On the right side of the sculpture seen a flying gandharava is seen holding flower of garland (H: 32cm W: 29cm Th: 18cm).

Marichi Acc No: 269

A broken sculpture of a marichi in pratyayalidha posture. A charioteer driving seven sow/pigs is seen. Both hands are broken. Wearing a lower garment, mekhala and nupur on the right foot H:18cm, W:16.5cm, Th: 7cm.

The bust part of the sculpture shown with one head, three eyes, and four hands, wearing Jatamukuta, long circular earrings, gala haar, decorated Bajuband, Keyur, and Upvita. In his upper right hand, he is holding Akshmala, and his lower right and left hands are broken, and he is holding a sprinkler in his upper left hand. In the right upper side corner, a flying Gandharva is there holding a garland in his hand

Mahavairochana Acc. 271

A deity is sitting in a Vajraparayakasana on a double-petal lotus in Bodhayangi mudra (Right hand holding the index finger of the left hand). He is shown wearing Kiritimukut, circular beaded earrings, long beaded gala haar, decorated Bujaband, keyur in both hands, and decorated upvit. Four attendants, two male and two female, are there on either side of the sculpture. And four attendants are sitting below the lotus pedestal (H:49cm W: 33cm Th: 25cm).

Vajra-Gandha-Tara

The deity attire with of Jatamukuta, long circular earrings, necklace (galahara) around her neck, armlets (bajubandha), bracelet (keyur) can be named as Vajra-Gandha-Tara found in the northern niche of the stupa having four armed, seated in the vajraparyakasana, on a double-petalled lotus flanked by a devotee and nidhi over a tripod pedestal. The right lower hand holds a sword, right upper hand holds a vajra while the lower left hand holding a conch and the upper left hand holding a pasa. Such image has not been documented at any sites of Odisha and make its significant in the present excavation at the site.

Significant Findings from the Excavation

Shrine complex

The significant findings from the complex are of three colossal heads of Buddha measures 1.27m 1.25m, 1.28m and four neck portion and fragments of body parts found in situ inside the complex. In the southern part of the complex, there is a damaged stone built structural stupa has been documented having diameter of 1.15m in front of the Lion pedestal. Around 50 numbers of monolithic votive stupas having height from 0.30 m to 1.20m with niches housed Buddhist divinities such as Tara, Chunda, Manjusri, Dhyani Buddha etc. Less number of ceramic assemblages revealed from the excavation within the shrine complex. The pottery dominated by grey ware followed by red ware of different shape and size, most of them are semi oxidized to oxidized, thin to medium in section, fine to medium grain structures.

Colossal head of Buddha

View of Lion Pedestal within the Chaitya complex

In the north-west and south-west corner of the chaitya complex two small shrine chamber made out of bricks masonry have been unearthed. The north-west side shrine

facing toward east measures 2.50m L x 2.50m W (square on plan) having an entrance measuring 80cm, the thickness of the wall is 60cm, maximum 12 courses of brick has been exposed.

Stupa with Caskets

A brick stupa having square plan having cross wall within it (cross in square), have been exposed and the wall measuring at present 4.85m x 3.65 m. and **contain stone caskets**, such type of stupa plan also noticed in Lauriya Nandangarh in Bihar, Chaukhandi in Uttar Pradesh, Shyam Sunder Ashram Tilla in Tripura, those can be assigned to a time bracket of 3rd century BCE to 10th century CE.

Plan of the Stupa

Stone caskets

Significant Findings

The tangible evidence of fusion of art between Brahmanism and Buddhism are two interesting monolithic stupa, one designed as a temple of Kalingan order consists of a *Rekha vimana* with a pyramidal Jagmohan housed goddesses Tara within, and another stupa is designed as a *panchayatan* temple, adorned with miniature stupa in its four corner and housed 2 male and 2 female divinities in four directions.

In situ image of votive stupa

Side view of votive stupa

Panchayatan Votive Stupa

Buddhist deity Chunda

Two Armed Chunda

Four Armed Chunda

The excavation retrieved different form of Chunda (two armed, four armed, six armed). Chunda the buddhist goddess. Description of Chunda is found in Nispanayogavali and Sadhanamala mentioned, seated in the vajraparyakasan attitude with one headed and two, four, and six handed goddess. 'Two handed Chunda arrayed in the dhyana mudra with bowl (Patra) on the palm'; 'Four handed Chunda, the principal hands of the goddess are in the dhyana mudra with bowl on right palm, and right upper hand hold roassary and left upper hand weapon which described in sadhanas (Buddhist mantra)

Six Armed Chunda

mention the deity as Vishv-varja or Vishwa-padm, 'the Six handed goddess, principal hand are in the dhyana mudra with bowl on the right palm, right middle arm holding parsu and upper right hand in tarjni mudra, left upper hand hold water pot and middle left holding roassary. **For the first time two armed Chunda has been documented from the present excavation.**

Vajra Veena Saraswati

Vajravinasaraswati is a Buddhist Goddess who is a form of Saraswati, has two arms, her distinctive feature is that she is seen holding a Veena (a type of stringed musical instrument) in both her hands and playing it. The hair styled depicted with side bun, there is a (Kanthamala) necklace around the neck, armlets (bajubandh) and keyur (bracelet) in the hands and legs are kept in a crisscross posture.

Arapchana

Arapchana, seated in the vajraparyankasana posture on a visva-padma, the right hand holding a sword in raised pose, the left hand holding a manuscript close to chest. The stylistically blowing cloths behind exhibit him as a hero in motion. His hair style known as *shrikhandak-kakaphasha*, the deity wears beaded Kanthmala, karnakundal, Bajubandh, and a lower garment.

Vajra- Gandha Tara

The deity attire with of Jatamukuta, long circular earrings, necklace (galahara) around her neck, armlets (bajubandha), bracelet (keyur) can be named as Vajra-Gandha-Tara found in the northern niche of the stupa having four armed, seated in the vajraparyakasana, on a double-petalled lotus flanked by a devotee and nidhi over a tripod pedestal. The right lower hand holds a sword, right upper hand holds a vajra while the lower left hand holding a conch and the upper left hand holding a pasa. Such image has not been documented at any sites of Odisha and make its significant in the present excavation at the site.

Mārīcī

The three-headed and six-handed Mārīcī stands in *pratyālīḍha* posture on a chariot drawn by three horses, with a charioteer (damaged). Her left face is sow-like, while the middle and right faces are human. In her left hands, she holds a bow, Ashoka leaves, and in Tarjani Mudra. Her right hands hold a Vajra, Khadga, and a needle resting on her thigh. Her hairstyle arrayed in Kiritamukuta style wear a Kanthmala, Bajubandh, Kamarbandh, and a lower garment.

Terracotta pedestal

General view of the excavated trenches (before removal of baulk line)

General view of the excavated trenches (After removal of baulk line)

Ratnagiri site witnesses the development of art and architecture on Buddhism from Mahayana to Vajrayana. The findings of hundred numbers of monolithic to ornamented votive stupas made out of brick and stone of different size and dimension also housing divinities within the niches are significant. The syncretism of Brahamanical and Buddhist art and architecture clearly noticed in the site. The attributes described in the literature for Brahmanical divinities found within the divinities of Buddhist sect. The facial features of the sculptures noticed in the sites resembles with the local inhabitants indicate production of these art by local artisans made out from khandolite stone retrieved from Assia hill range. The contribution of Bhaumakaras' for the growth of art and culture and the role of women queens for the sake of Buddhism in this site is reflected from each piece of art. The site is a perfect place to study the fusion of Buddhist and Brahmanical art and architectural and to know the beginning and growth of Vajrayan Buddhism in this part of the Bharatbarsha.

Chapter IV

Excavated Structures

The amalgamation of different materials and structures of varying nature adorns the hilly landscape archaeological site of Ratnagiri affiliated to Buddhism, as evidenced by the associated materials. The site witnessed the building activities in the form of Monasteries, Stupas, Chaitya complex, minor shrines, and other structures such as platforms, walls, floors, etc., dated from the early historic period to the 13th century CE.

The building materials mostly used in the structures are bricks and Khondalite stones. In the present excavation, it has been noticed that bricks are preferably used for the construction of walls. The stone members, as observed, have been used in the entrance and in the floors of many structures. The spadework revealed different structural phases within the operational area. One of the notable observations from the building material is that most of the structures presently exposed were rebuilt by modifying the earlier structure, as evident from the broken and reused bricks. The bricks used in structures are of various shapes and sizes, having plain to moulded designs. The mortar used in the brick structures is prepared from well-levigated clay and a very thin layer has been applied while joining the bricks with each other, for fine joints, mud slurry is used as the joining material. The architects might have mixed organic glue, like Belgiri/or Alvera juice, to increase the stickiness of the clay. Wherever the stone blocks are used, the fusion and clamping method in iron has been applied to join the stone blocks. In a few places, brick jelly has been used for flooring and as decorative moulding in the base part of structures. In some instances, a rock bed has been hewed out for the construction of structures mixed with stone blocks and bricks to form a composite wall. The evidence of deliberate use of well-dressed bedrock as a supporting base upon which a brick wall has been constructed at the site demonstrates resource management by the builders.

Structure -1

The structure measures 5.5 meters in length and follows a square plan having a cross wall within it (cross within square), with each wall exhibiting a substantial thickness of 1.3 meters. The surviving height of the structure is 1 meter. At the core of the stupa, the foundation lies in a cross design constructed with bricks arranged diagonally within the square layout. Each arm of the cross measures 74 cm in

thickness, and the diagonal span of the entire structure is 7.10 meters. The cross itself is composed of 5 courses of brickwork, indicating a stable construction technique intended to support the superstructure above. An open space measuring 1 meter in width with a narrow passage may be used as a *pradakshina patha*/circumambulatory passage commonly found around stupas, used for a ritual walk.

The bricks used in the structures are well-fired and fine in texture, of different shapes and sizes, including rectangular, square, and wedge-shaped, decorative, cutting, and slanting-edge bricks. The maximum and minimum sizes of a brick evident from the trench are L- 42 cm × W- 27 cm × T- 7 cm and L- 24 cm × W- 15.5 cm × T- 7 cm, respectively.

Structure -2

The Structure located adjacent to the south-facing section in Trench YC1, measures 2.5 meters in length and 1.6 meters in width. However, the structure has not been fully exposed during the current phase of excavation. Further excavation is expected to provide a better understanding of its full dimensions and layout. The orientation of the structure indicates that it extends northward beyond the limits of the presently excavated area. The structure may be a small shrine.

Structure – 3

A rectangular brick platform/structure has been unearthed in Trench YE, situated along the south-facing section of the trench. The exposed portion of the structure measures 12 meters in length, 3 meters in width, and its existing height is 1 meter. The height comprises 15 well-laid courses of bricks, arranged in a stretcher bond technique and bonded using mud mortar (mud slurry). The structure appears to be massive and heavy, oriented east to west.

Structure – 4

A rectangular platform was uncovered in trench XE1, measuring 8.30 meters in length, 6.30 meters in breadth, and 1.40 meters in height. The structure comprises 20 courses of bricks and is filled with bricks and broken bricks in its core. The platform has a horizontal moulding at the centre.

Structure- 5

After excavation in trenches XD1 and YD1, all brick masonry has been unearthed. The exposed portion of the wall measures 20 meters in length, approximately 90 centimetres in width, and 1.3 meters in height with 17 courses of bricks. The structure appears to extend beyond the northern and southern limits of the excavation area. Its full extent has not yet been revealed. The bricks in this wall is primarily laid in a stretcher bond pattern. However, the exposed wall has a joint, which shows two phases of work or may be of a later addition to

expand the wall toward north and south. This joint is constructed in a stack bond pattern, differing from the surrounding brick pattern. The wall is tilted/bulging at the centre, but is in a partially good state of preservation and straight alignment, while the east-facing side is irregular and not parallel as the west side. The wall extends beyond the current excavation limits, needed further excavation to have a better understanding of the structure.

Structure- 6

A small structure unearthed in trenches XE1 and YE1, appearing to be a base part of stupa constructed from stone blocks. The structure is square in plan, measuring 1.68 meters in length and width. Its base moulding is designed as '*khura*' out-turned curved. The western part of the structure is broken and missing.

Structure -7

In trench XA1, structural remains were unearthed, indicating that multiphase construction activities had occurred. On the western side of the trench, a stone platform was uncovered, measuring 6.64 meters in length and 3.5 meters in width (STR-B). This platform is constructed using large dressed stone blocks, measuring 69 cm in length, 24 cm in width, and 39 cm in thickness. Beneath this stone platform lies a squarish brick structure, identified as the base of a stupa (STR-A). This structure measures 3.90 meters on each side with a height of 42 cm. On top of the stone platform, there is another brick platform, measuring 5.34 meters in length and 1.9 meters in width. The bricks used in this structure measure 25 cm in length, 20 cm in width, and 5 cm in thickness.

In the southwestern part, another brick platform was uncovered, with dimensions of 0.88 meters in length,

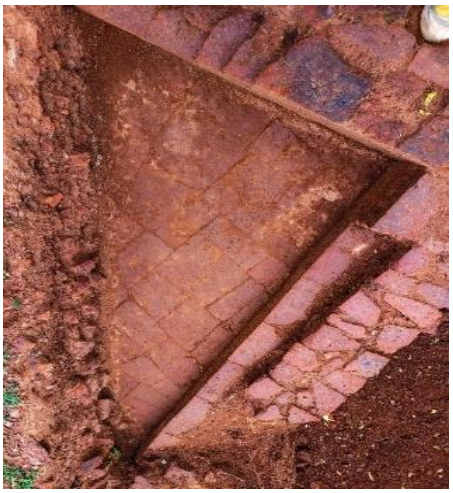
1meter in width, and a height of 28cm. This platform is constructed in a *pancharatha* plan on two stone blocks, each measuring 69cm in length, 41cm in width, and 25cm in thickness. Towards the east side, an additional brick platform was exposed, measuring 1.73 meters in length, 1.71meters in width, and 36cm in height. Adjacent to this, a brick floor running in an east-west orientation was uncovered, with dimensions of 3.84meters in length and 3.78meters in width. In the northeastern corner of the trench, a small stone platform was uncovered, measuring 1.64 meters in length and 1.5 meters in width.

Structure – 8

In the eastern part of trench XB1, a stone masonry platform has been unearthed. The structure measures 7.30 meters in length, 2.80 meters in width, and 0.57 meters in height. It is made of 3 courses of finely dressed stone blocks, carefully laid one after another using a forging technique. This method involves the use of iron pins and clamps to securely bind the stones together and to enhance the stability of the structure. The size of the stone blocks varies, with the largest block measuring 72 cm in length, 46 cm in width, and 27 cm in thickness, while the smallest measures 40 cm in length, 37 cm in width, and 23 cm in thickness.

Structure – 9

In trench XA2, partially a brick-built structure has been exposed, characterised by bricks laid in a mosaic pattern. The exposed portion of the flooring measures approximately 2.25 meters in length and 1.40 meters in width. Further excavation can clear the picture of the structure.



Structure -10

A rectangular plan Chaitya complex made of brick and stone has been exposed. The structure measures 19.60 meters in length and 6.65 meters in width, having 1mt thickness brick wall, with entrances identified on both the northern and eastern sides. The masonry of the complex consists of both stone and brick, with stone primarily used for the flooring and pedestal, while brick was used for constructing the walls. The walls preserve a maximum of 20 courses and a minimum of 5 courses of brick. The bricks used in the wall are of different sizes and shapes. The width of the wall measures approximately 1 meter. The wall was constructed using fine-faced bricks along its outer surfaces, while the core of the wall was filled with broken bricks. The bonding material used in the brick courses is mud mortar/mud slurry, composed of fine clay, which may be mixed with organic glue to improve binding properties and durability.

At the chaitya complex, a circular stone structure is unearthed. Beneath this structure, there is an adjoining circular platform of wedge-shaped brick masonry. The diameter of this whole structure is 1.15 m.

Structural stupa in the prayer hall

Lion Pedestal in the chaitya Griha

The highly decorated lion pedestal has been uncovered during the excavation within the chaitya complex. It is decorated with anthropomorphic figurines, intricate floral designs, a *Nāga Puru*sha (serpent deity) holding a *kalaś*a (a ceremonial pot), and a mythical lion with a calm expression. This pedestal/platform was used to place 4 to 5 colossal Buddha figurines in the seating position. This existing panel measures 2.70 m in length, 45 cm in width, and 60 cm in height. It might have been around 20mt long to house 4 to 5 seated Buddha. The present excavation retrieved 4 neck portions and 3 colossal heads and broken parts of the sculptures, which justify the presence of such huge sculptures of Lord Buddha in a seated posture.

Structure- 11

In trench XE1, two square-shaped stone platforms were exposed on the southwestern side of the trench, both constructed using finely dressed khondalite stone blocks. These platforms display iron dowel marks, indicating the use of metal fittings to secure the stone blocks during construction.

The first structural base from the north, consists of three courses of khondalite stone and measures 3 meters by 3 meters. Adjacent to this, towards the southeastern direction, a second platform has been exposed. This platform comprises two courses of khondalite stone and measures 4 meters in length and 4 meters in width.

Structure - 12

In trench YD1, at the north-west corner of the Chaitya complex, a small shrine chamber constructed in brick was unearthed. This square plan structure measures 2.50 meters in both length and width. The shrine features an entrance on the eastern side, measuring 80 cm in width. The walls of the chamber are 60 cm thick, and a maximum of 12 courses of brick have been preserved.

Structure – 13

In trench YC1, on the northern side of the Chaitya complex, a small shrine constructed of brick and stone masonry was unearthed. This east-facing structure measures 2.70 meters in length, 2.66 meters in width, and 0.91 meters in height. The entrance to the shrine, measuring 1.25 meters in width, and the size of the walls varies in thickness from 75 cm to 65 cm. Up to 13 courses of brick have been preserved. Inside the shrine, a decorated stone pedestal and a brick floor were discovered in situ. The pedestal measures 1.10 meters in length, 0.44 meters in width, and 0.33 meters in height. The brick floor measures 110 cm by 95 cm. The entrance stone member still has the impression of door sockets and latch depression.

(PEDESTAL)

Structure-14

A structure was unearthed by excavations in trench YA1 and YB1. This is a large shrine chamber, measuring approximately 13 meters in length and 5.85 meters in breadth.

The east, west, and south walls of the structure have been exposed. The southern wall contains a rock-cut feature, which appears to have been deliberately used of the bedrock and incorporated into the wall's construction. This use of the natural rock bed suggests an intentional adaptation of the landscape for architectural purposes. The wall preserved a maximum of 17 layers to 6 layers. The structure witnessed multiple phases of renovation as observed from the reuse of bricks and stone architectural members. The eastern wall appears to be part of a secondary phase of construction. It includes a stone structure embedded within the brickwork, and 175 cm thick brick wall is built directly over an earlier stone structure, indicating multiple phases of activity.

B. Building Materials

Bricks

Bricks were used in constructing various structures from the ancient period. In the excavated site, most of the structures are made out of burnt bricks. The bricks used in the structures are well-fired and fine in texture, of different shapes and sizes, including rectangular, square, and wedge-shaped, decorative, cutting, and slanting-edge bricks. The maximum and minimum sizes of a brick evident from the trench are L- 42 cm × W- 27 cm × T- 7 cm and L- 24 cm × W- 15.5 cm × T- 7 cm, respectively. These specially shaped bricks were commonly used in the architecture, such as the *khura* portion of brick-built stupas, monastery walls, circular structures and shrines. In most structures, bricks are used in the stretcher bond technique.

Stones

The site showcases a combination of stone and brick construction, which reflects available materials and architectural techniques. The primary type of stone used for construction is khondalite, a metamorphic rock that is available in the Asia Hill Range. It has been extensively used in the construction of walls, floors, and stupas. In addition to khondalite, chlorite stone has also been identified at the site, particularly in the making of sculptures and selected architectural members. The dressed khondalite blocks were meticulously shaped and used to build various parts of the structures and sculptures. To ensure the stability of these stone constructions, iron dowels were employed to join the blocks securely. These dowels helped to lock the stones in place,

preventing displacement and enhancing the structural integrity of the monument over time.

Stone Chips

Stone chips were used in the preparation of the foundation and flooring. Since the area is located on uneven, hilly terrain, creating a flat and stable surface for building, the architect used stone chips to level the ground for the formation of a solid base for structures.

Bricks Jelly

Apart from the stone and brick, brick jelly was also used as a construction material at the site, particularly in flooring and architectural moulding. Brick jelly, made by mixing small fragments of broken bricks with adhesive, glue, or mud slurry to create a durable and compact surface. The material was prominently used in the flooring of the chaitya and shrine structure, where the layer of brick jelly can be observed, indicating its significance in both structural and functional aspects. It provided a smooth and firm walking surface that could withstand regular use. Moreover, brick jelly was used as an apron and moulding in the structures. A notable example can be seen at the north-west corner of the chaitya-griha, where a small shrine is surrounded by an apron made of brick jelly.

Mortar

Several structural remains made of bricks were uncovered from the excavation. The brick structures featured both rectangular and square-shaped bricks. To bind these bricks together, a form of mortar was used. This mortar appears to be a mixture comprising mud, powder of brick and mud slurry. To understand the exact composition and construction technique, a sample of the mortar has been sent to a laboratory for analysis. The test results will reveal the ingredients used in the mortar and provide insights into the traditional methods employed in its preparation. Preliminary observations may suggest the presence of organic materials, indicating a sophisticated understanding of natural binding material in construction.

Tiles

The present excavation at Ratnagiri has unearthed a huge number of roof and floor tiles. The tiles were manufactured in a reduced firing technique, which gave them

a grey colour. A major feature of these tiles is the presence of husk impressions, suggesting that paddy was mixed with the clay to give more strength to the tiles. All the tiles maintain uniformity in size and have a curved moulding design on one end to hold a wooden batten. Some tiles have traces of lime on their upper surface, which may be applied to prevent water leakage from the roof or may be used as a binding material.

Constructional Techniques

The construction techniques observed at the site demonstrate a high level of architectural planning and craftsmanship. One of the prominent architectural forms is the stupa, which typically follows a distinctive design and sequence. The stupa begins with a square or rectangular base plan, providing a strong and stable foundation. As the structure rises, it transitions into a circular elevation, and after a certain height, it adopts a hemispherical form, characteristic of the traditional stupa design. This combination of geometric shapes, such as square, circular, and hemispherical, not only ensured structural stability but also held symbolic religious meaning, representing the cosmic and spiritual order in Buddhist architecture.

Apart from stupas, several shrines and sculptural elements at the site reflect intricate stoneworking techniques. Many shrines are constructed on well-carved pedestals, often richly decorated with artistic motifs. For instance, in the Chaitya complex, one can observe pedestals featuring lions, snake hooded deities, incense burners, and lotus motifs, all carved in high relief. The use of dressed khondalite stone blocks, precisely cut and fitted, often joined with iron dowels, allowed for the creation of large, stable structures without the need for mortar. In areas where uneven terrain posed construction challenges, materials like stone chips and brick jelly were used to level the ground and construct floors, especially in the chaitya and monastic areas. The site reflects a mature construction tradition that combined engineering innovation, material efficiency, and artistic excellence.

Structural Phases

At this site, four distinct structural phases are associated with different historical periods, with their own cultural significance. These phases are layered one over the other, with later constructions often built by dismantling or breaking earlier ones. The first three phases are closely associated with Buddhist architecture and activities. However, the fourth and most recent phase is attributed to the Mahakala tradition,

which remains active at the site even today. This final phase marks a significant transitional period. It reflects the decline of Buddhism in the region and the gradual emergence and establishment of Brahminical practices. The architectural and cultural shift observed in this phase highlights the religious changes and transformations that took place at the site over time.

Phase I

The earliest occupational phase at this site is associated with the Mahayana phase of Buddhism, dating prior to the 3rd-4th to 7th century CE. This phase is primarily reflected in the architectural remains and associated material culture. Although direct evidence of buildings from this period is limited, significant clues come from the bricks used in the earliest structures. Many of these bricks are secondary in use, and some bear finger impressions, a typical feature of Kushan period bricks, suggesting reuse of materials from the 1st–2nd century CE. Despite the lack of substantial architectural remains from the 2nd–3rd centuries CE, ceramic evidence strongly supports occupation to the above-mentioned period.

Phase II

The second structural phase at the site is associated with the transitional phase of Mahayana and Vajrayana traditions of Buddhism, dating to approximately the 7th -8th century CE. This phase is marked by significant architectural modifications, where earlier structures from the Hinayana phase were deliberately destroyed to make way for new constructions. The site witnesses the transition in trench XA1. The remains of the earlier Hinayana structure were reused in the construction of a new building, showing a clear pattern of adaptation and continuity. Bricks from the previous phase were reused, and the newer Mahayanic structure was built directly on top of the older remains. The key architectural feature of this phase is observed in the trench XA1 and YB2, which belongs to the Mahayana period. The partially exposed apsidal structure at YB2 has been seen as superimposed by a rectangular shrine complex, possibly during the last part of the 7th century CE.

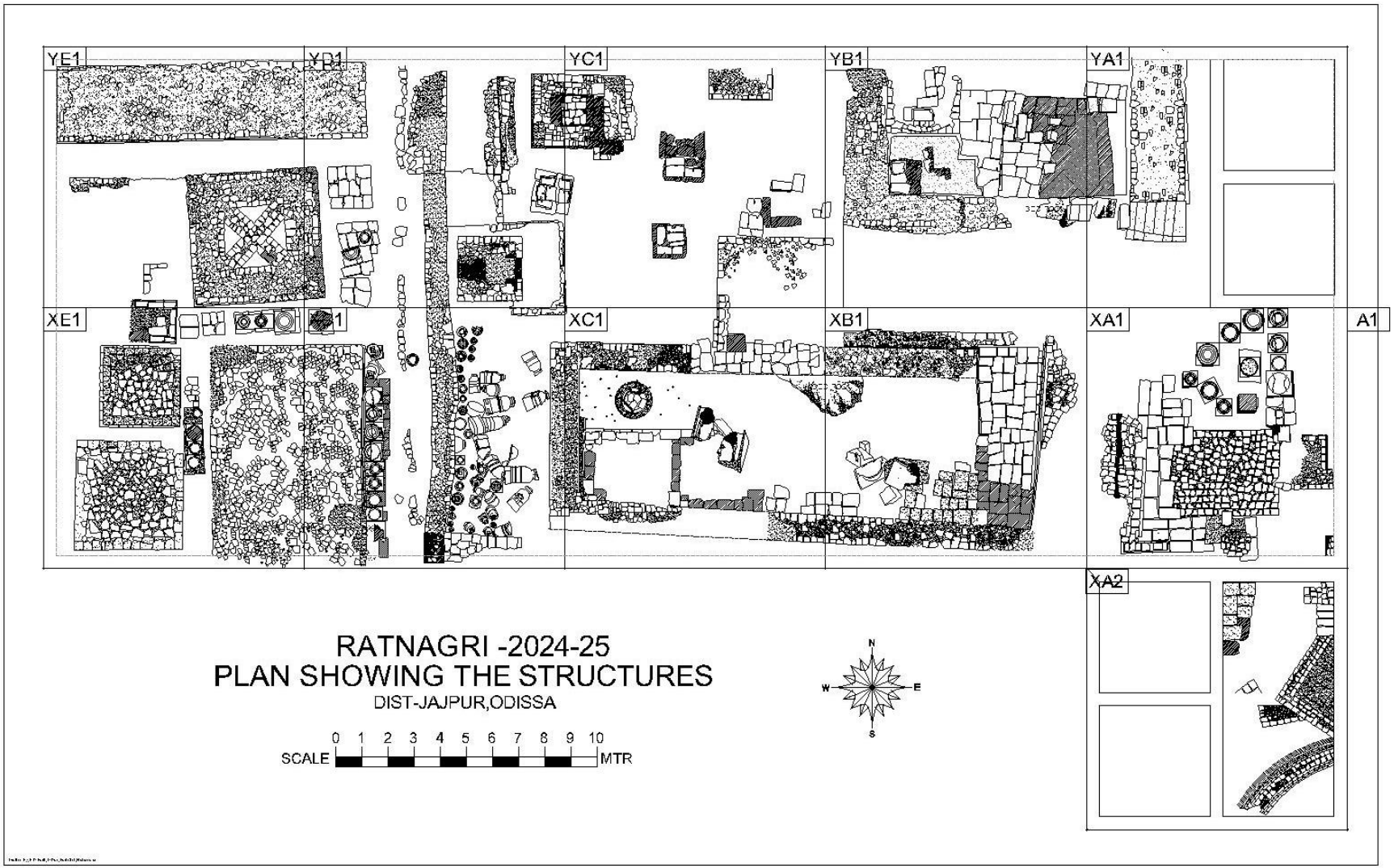
Phase III

The third phase of structural activities at the site is associated with the Vajrayana phase of Buddhism, starting from the 7th –8th century CE to the 10th century CE. This period marks a significant transformation in both religious practice and

architectural expression. With the rise of Vajrayana Buddhism, the earlier Mahayana structures were largely destroyed to make way for more expansive and elaborate constructions. One of the most prominent developments during this phase was the construction of a large rectangular *Chaityagriha*, indicating the growing importance of ritual spaces in Vajrayana tradition. Architectural modifications continued through this period, as noticed from the pradakshina patha **(**circumambulatory path) around the Chatyagriha made of brick jelly flooring.

Phase IV

The fourth and final structural phase at the site is associated with the rise of Brahmanical influence and the establishment of the Mahakala temple which marks a significant ideological and religious transformation at the site. The temple was constructed over the ruins of earlier Buddhist structures by incorporating earlier phase architectural remains around the later part of the 12th century CE or at the beginning of the 13th century CE. The architectural remains from this period are secondary in nature, composed largely of reused stone and materials.



RATNAGRI -2024-25
PLAN SHOWING THE STRUCTURES
DIST-JAJPUR,ODISSA

Chapter-V

Pottery Assemblage

The spadework has yielded potteries belong to the early historical period to medieval period. On the basis of the colour, fabric, texture, shape and type etc., the pottery assemblage of Ratnagiri is basically divided into four phases. The pottery assemblage dominated by Grey ware followed by Red ware, Red slipped ware; Black slipped ware, Blackish grey ware. Most of them are plain and few shreds are with geometric linear incised designs on the rim portion. The notable pottery shapes include pitcher, dish and bowl and lamps. It has been observed that the ceramic assemblages are medium to fine in fabric and texture, oxidized to semi oxidised in nature. The domination of grey ware increased with the passing of time due to adaptation of reduced firing technique which resulted in production of mass scale of grey ware in different cultural periods.

Phase-I (4th century- 7th century CE)

The period I marked with the presence of well oxidised red ware ceramics associated with inscribed terracotta ceiling dated from 4th century CE to 7th century CE. The presence of Red ware is more than the succeeding phases, however the appearance of the Grey ware has been also noticed but comparatively less in quantity than the Phase-II. Various types of lid cum basin, handi of medium to thick fabric, spouted pots, vase, pitcher, storage jars, basins, dish with sagar base, dish with internal corrugation, shallow and deep bowl with footed base, carrinated handi, carrinated basin with sagar base, decorated lamps etc, are noticed in the ceramic assemblage of the period. Storage jars with globular body are found in maximum quantity. Well levitated clay used for making of these potteries, these are of medium to fine in texture and well oxidised in nature shows the production of these pottery by giving more than 600-700^{o}C temperature. The decorative pattern arrayed on the pottery are of incised linear pattern, appliqué with incised, vertical slanting strokes, curvilinear concentric incised grooves, triangular pinching with vertical strokes, slanting impressed pattern, appliqué with incision, strokes by applying impress technique, diamond shape pinching, appliqué rope pattern. Such decorative patterns have also observed within the pottery assemblage of the later period.

Fig.1- Comprise carrinated Handi, storage jars of Gray ware, single shred of basin of red ware & vase. Storage jars are represented with splayed out externally thickened flattened top & internally depressed grooved rim short neck and oblique shoulder, globular body. Outer surface is decorated with series of horizontal bands followed by geometric pattern incised motive. Medium fabric made out of well levigated clay and well fired. Redware basin is of wide mouthed, externally thickened featureless rim with triangle incised decoration on top, globular body treated with red slip. Vase of Red slipped ware decorated with vertical incised lines and horizontal bands on neck and shoulder.

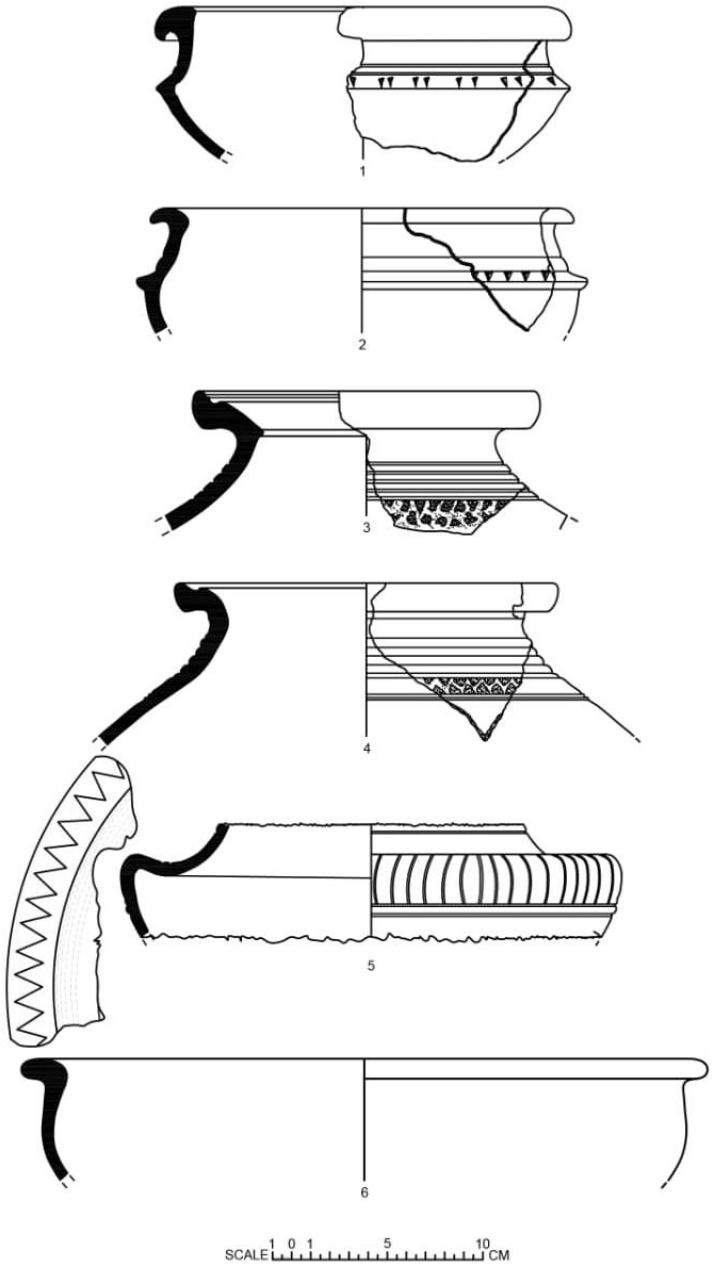


Fig**.2-** Comprise bowls and leads of red ware with featureless rim, externally and internally grooved, tapering sides and flat base of medium fabric devoid of any surface treatment and well fired.

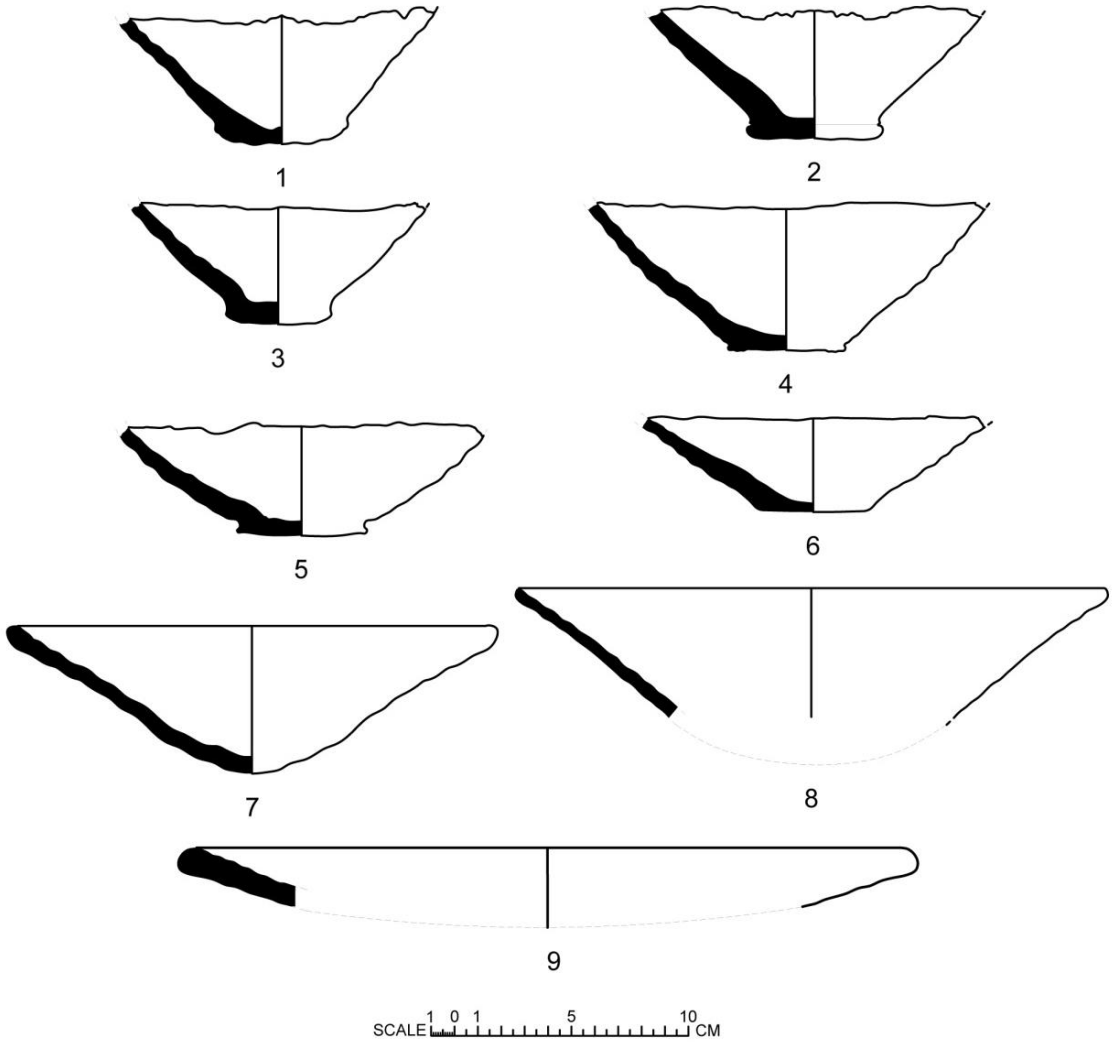


Fig.3- comprise carrinated basins of gray ware and red ware, with splayed out, slightly drooping and horizontal lines with internally notch rims; short neck with horizontal bands; prominent to normal carination followed by horizontal lines on shoulder; generally devoid of any surface treatment however evidence of black slip also noticed in one instance, medium fabric, medium to well fired.

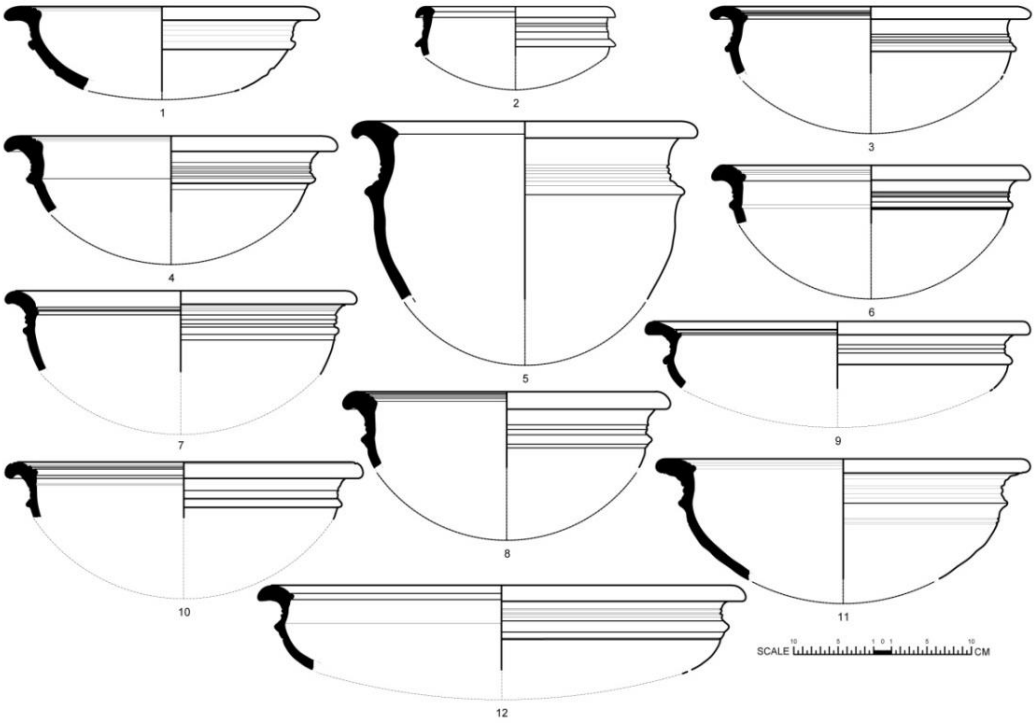


Fig.4- comprise carrinated basins of gray ware and red ware, with splayed out, slightly drooping and horizontal lines with internally notch rims; short neck with horizontal bands; prominent carination followed by horizontal lines on shoulder; generally devoid of any surface treatment however evidence of rustication is also noticed in one instance, medium fabric , medium to well fired.

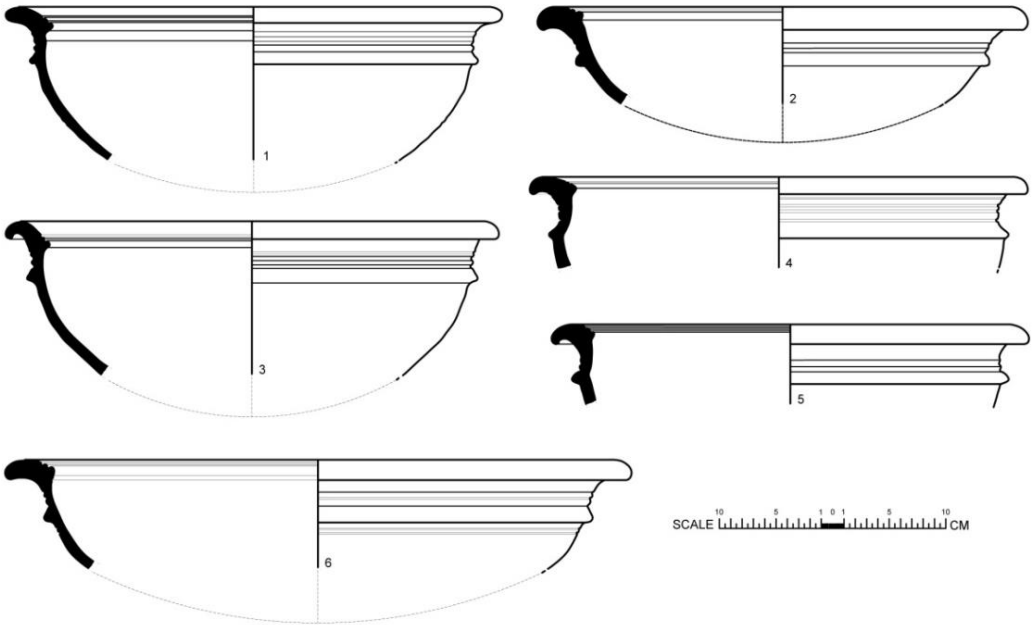


Fig. 5- Comprise Red ware storage jars with variety of rims of thickened out turned, out turned with undercut, splayed out internally depressed grooved; squat, concave neck; oblique shoulder and globular body. Horizontal grooved lines on shoulder and devoid of any surface treatment and well fired.

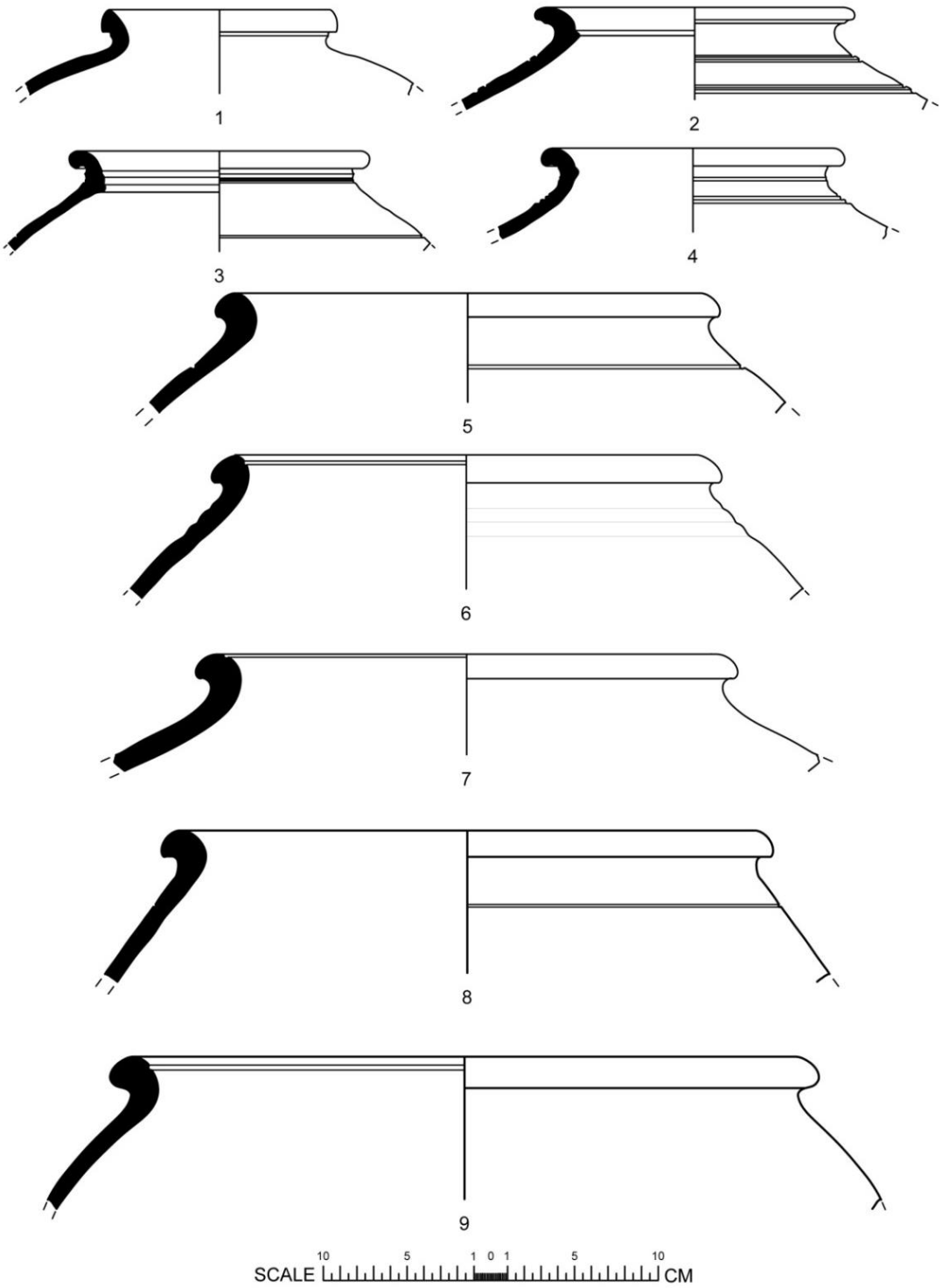


Fig. 6- Comprise storage jars of gray ware, with out-curved slightly drooping rims, thick-end, flared out rims, internally grooved, short to squat necks, oblique shoulder with horizontal bands, globular body, medium fabric well fired.

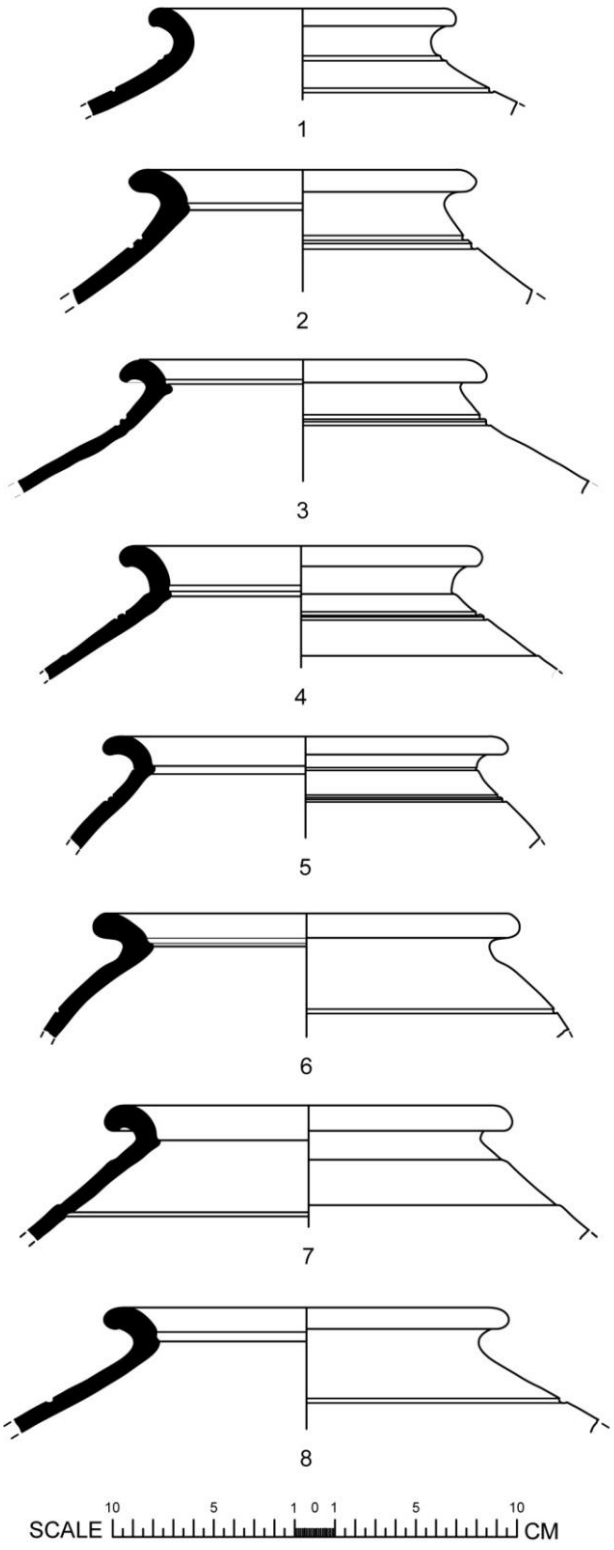


Fig. 7- Plate 7 comprise Gray ware wide mouthed storage Jars of medium fabrics, made out of levigated clay, medium fired, short flaring rim with slightly internal notch with a distinguished horizontal mild groove; neck distinguished with a cut and oblique shoulder and externally undercut for proper handling and horizontal grooved decorations, globular body globular body, devoid of any surface treatment, medium fabric and well fired.

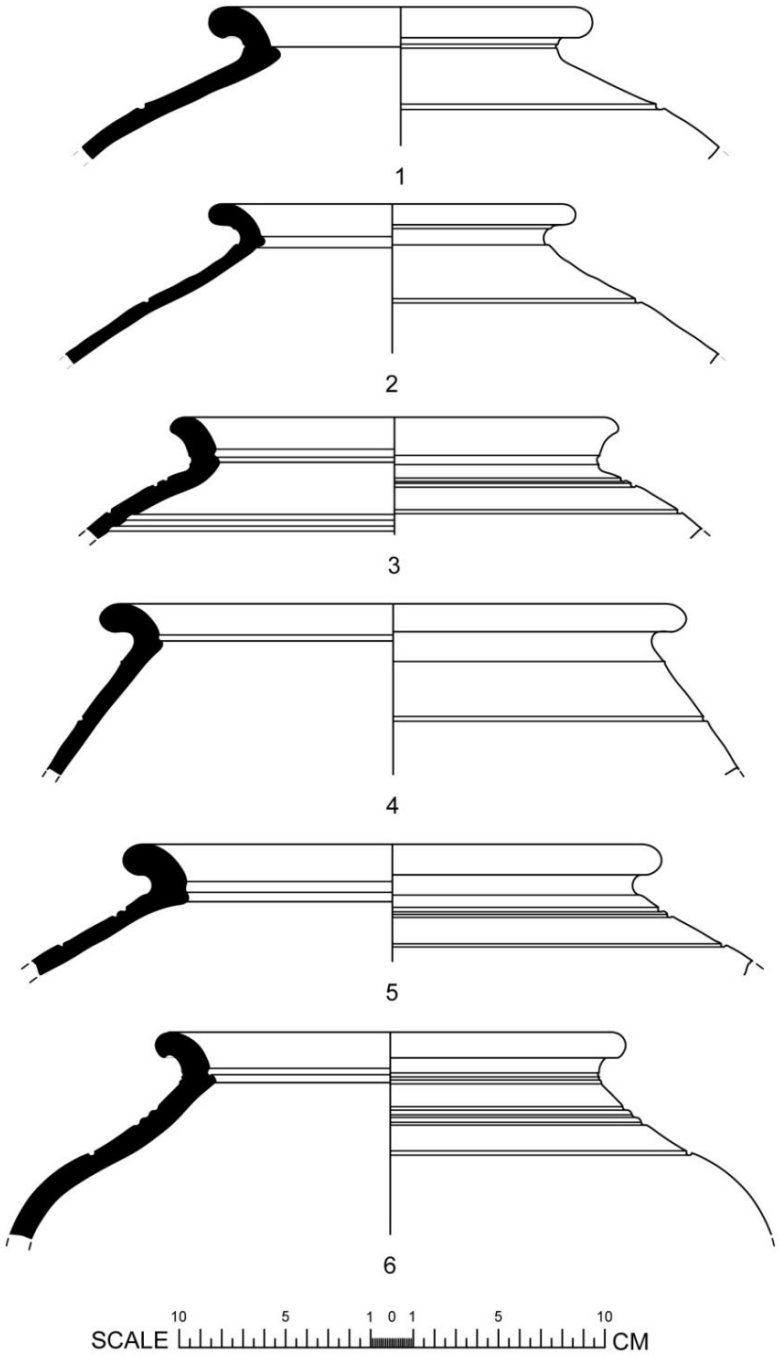
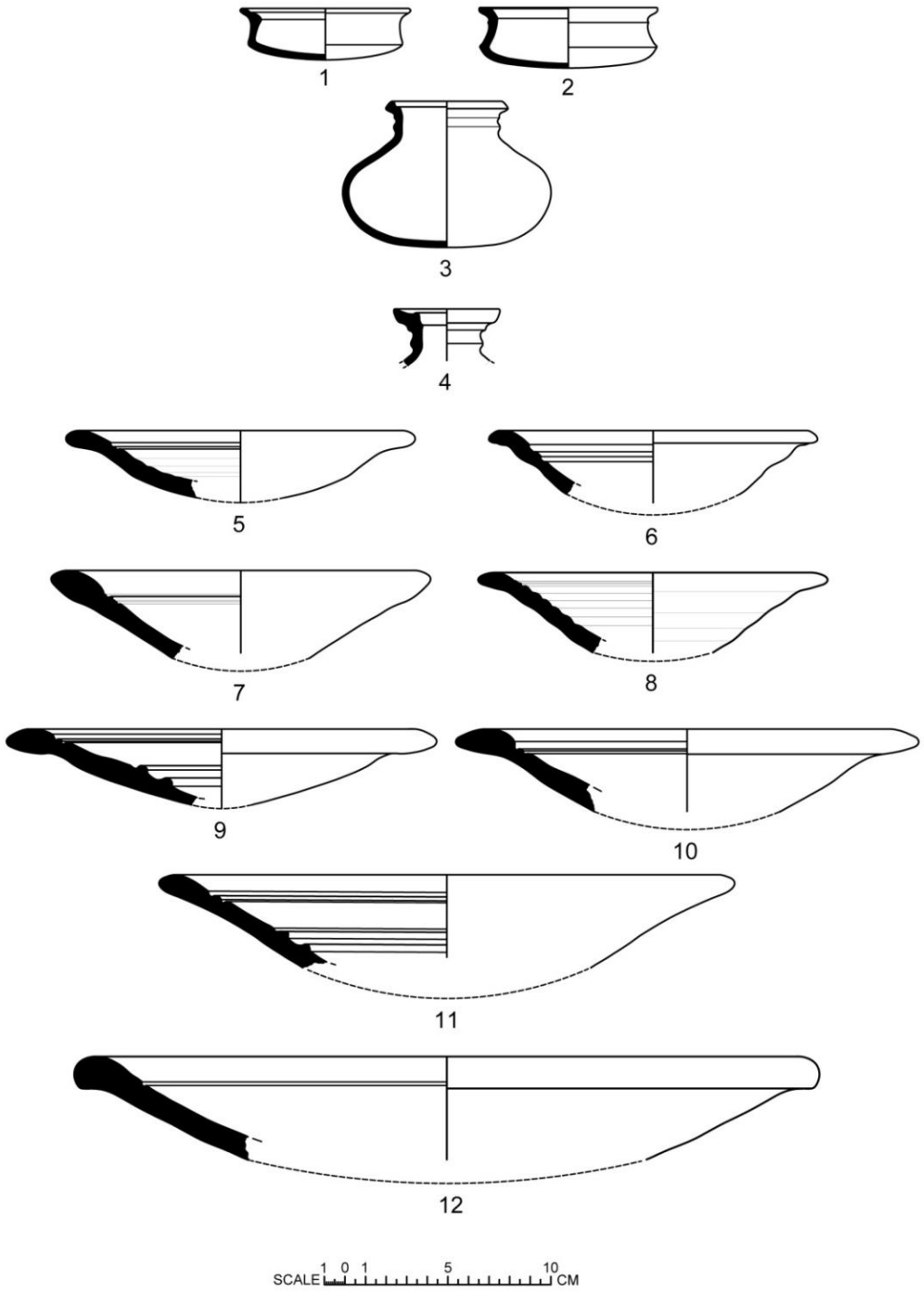


Fig. 8- comprise leads, bowls and miniature vase of gray ware, Leads are of thick-end, featureless splayed-out rims, horizontal bands or grooves with plane exterior, treated with black slip in few cases. Medium fabric well fired. Bowls are of sharpened flared rim, vertical body with slightly flattened base, well fired thin fabric. Miniature vase is of incurved thickend, internally depressed rim, short neck and globular body, thin fabric and well fired.



Phase-II (7th century to 10th century CE)

The phase –II ceramic assemblage comes within the time bracket of 7th century CE to 10th century CE. A good quantity of sprinklers was procured from this phase. The pottery assemblage of this phase exhibits better quality of burnished, polished and finished surface. The incised decorations found on the potteries are similar as the earlier phase. The red ware and grey ware are mostly equal in quantity and also in shape. The red ware potteries have more variation in shape and size in comparison to grey ware. The measure shapes in both the ware are of pitcher, *surahi*, storage jars with wide & narrow mouth, slightly elongated body, basins of different sizes, spouted pots, dish with sagar base, shallow and deep bowl with footed base, carrinated handi and basin, vase, lamps and lids etc. noticed in this phase. These potteries were made out of levitated clay and are of medium to fine grain in texture. The red ware were fired at a temperature more than 700^{o}C, which shows they are oxidized in nature while the grey ware may be produced in reduced firing technique. Various decorative patterns seen in the period are of linear pattern, applique with incised, vertical slanting lines, curvilinear concentric incised grooves, triangular pinching with vertical strokes, slanting impressed pattern, appliqué with incision, strokes by applying impress technique, diamond shape pinching, appliqué with rope pattern as mostly in phase I. Few inscribed potsherds with legends in Nagari script have been found in this period.

Fig. 9- comprise bowls of red ware, with featureless slightly incurved rims, tapering sides internally plain or corrugated surface and flattened ridged base. Medium fabric, devoid of any surface treatment and well fired.

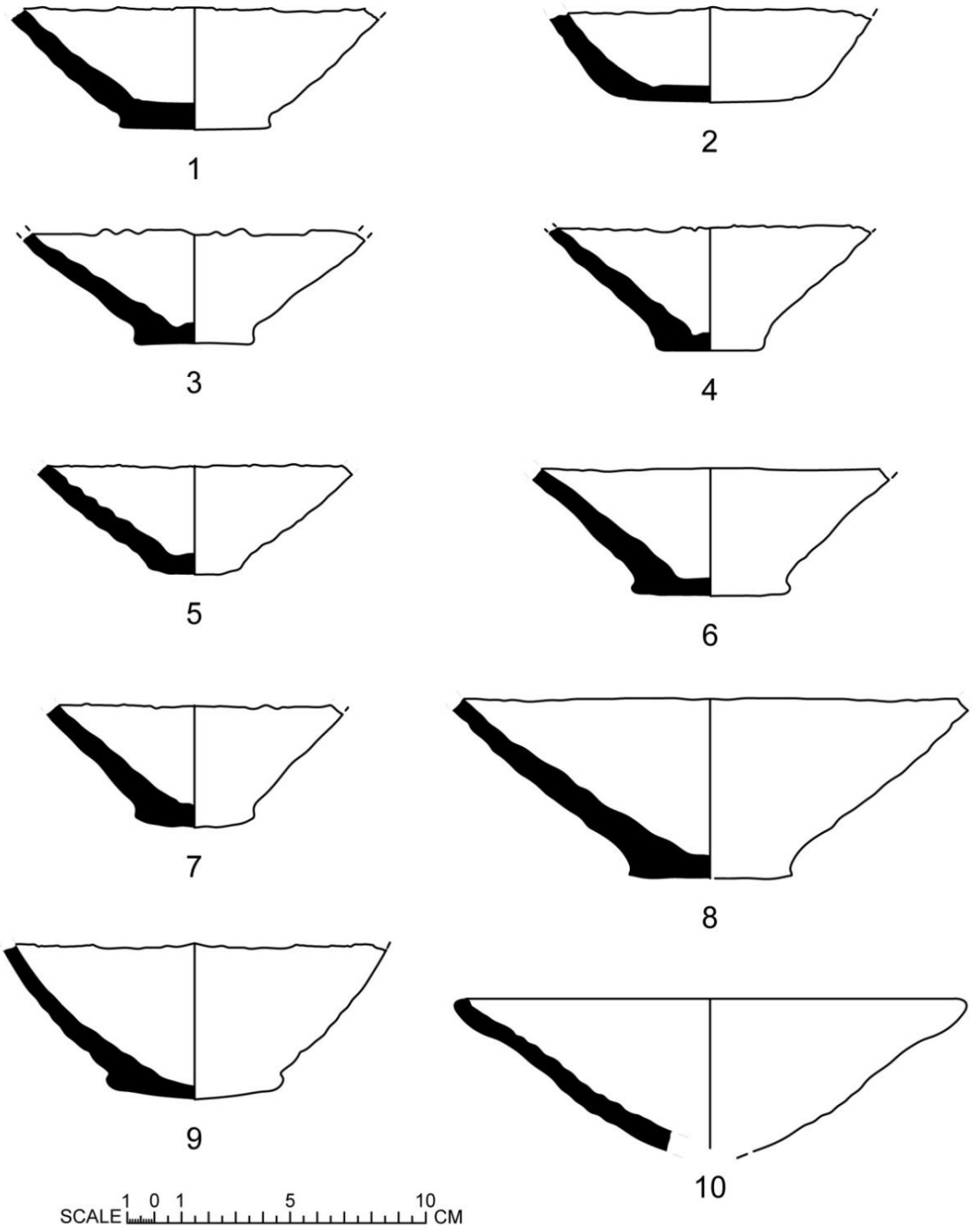


Fig. 10- comprise carrinated basins of gray ware and red ware, with splayed out, slightly drooping, thumb nailed rim, internally notch and horizontal lines at neck; variant of short neck with horizontal bands; prominent to normal carination followed by horizontal lines on shoulder; generally devoid of any surface treatment, medium fabric, medium to well fired.

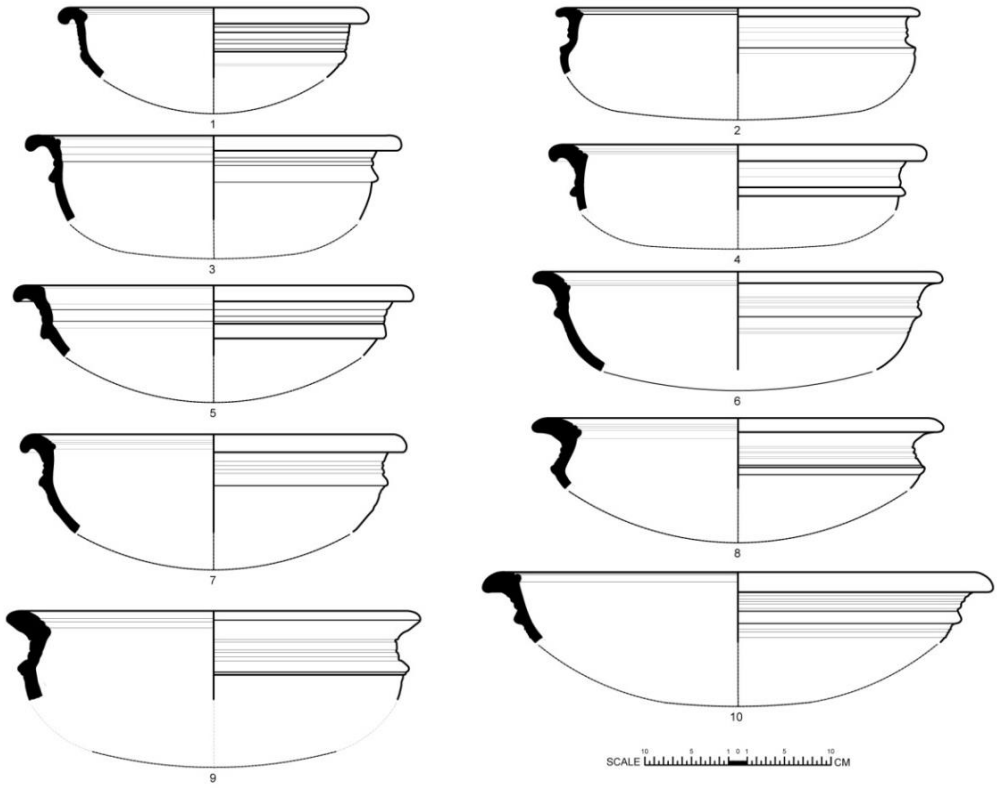


Fig. 11- comprise wide mouthed storage jars of red ware with thickend splayed out rims and thickend flaring mild drooped rims, squat neck with oblique shoulder and globular body. Horizontal groove is on shoulder, medium fabric and well fired.

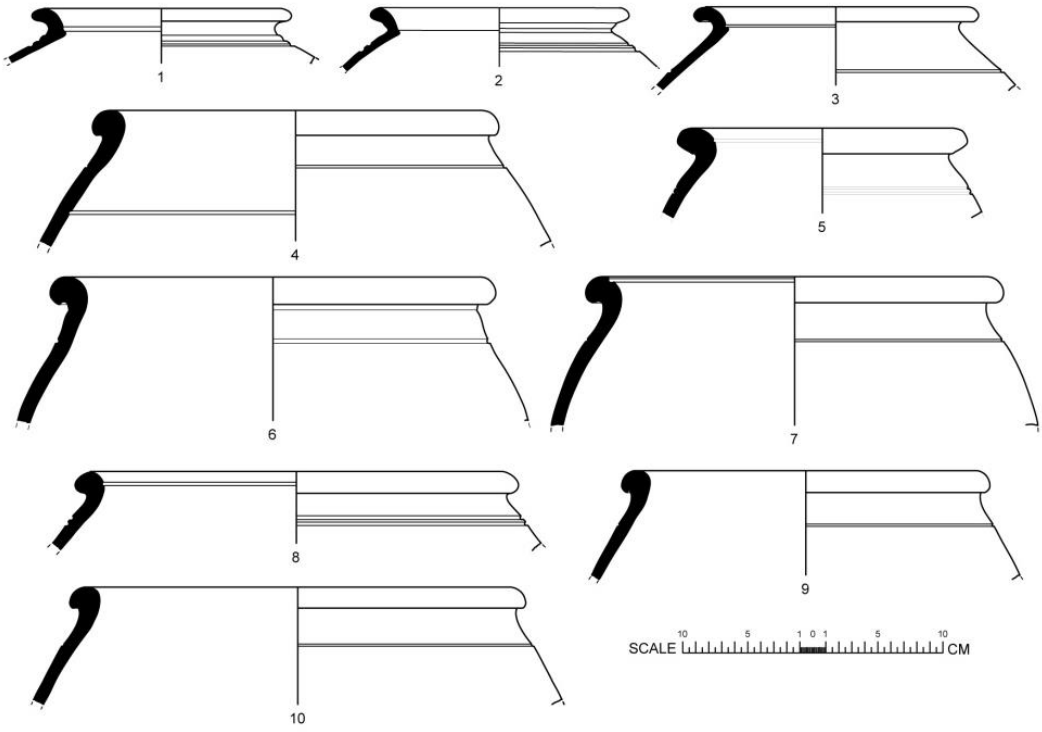


Fig. 12- comprises leads and bowls of gray ware. Leads are of featureless splayed out rim with internal and external grooved and tapering sides, medium fabric and well fired. Bowls are of features less externally incurved with internal depressed horizontal groove and round base, medium fabric and well fired.

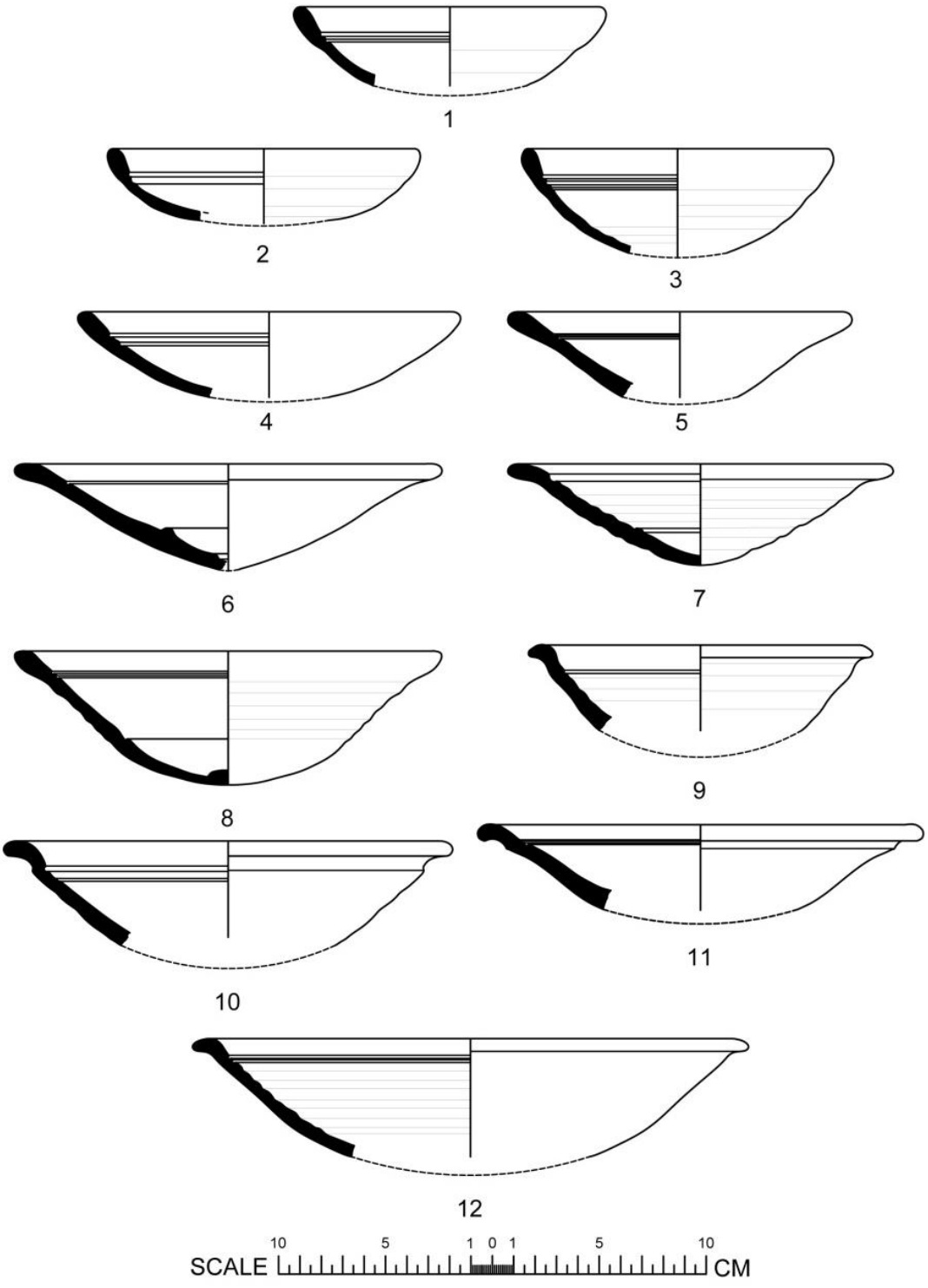


Fig. 13- comprises ritualistic miniature potteries like vase, lamp, bowl on stand incense burner and lead of red ware and grey ware. The variant rims of miniature pots like incense burner, lamps, vase are of featureless, clubbed, thickend, sharpened, splayed out externally and internally depressed rim with flat or round base. All are of thin and coarse fabric and devoid of any surface treatment and ill to well fired.

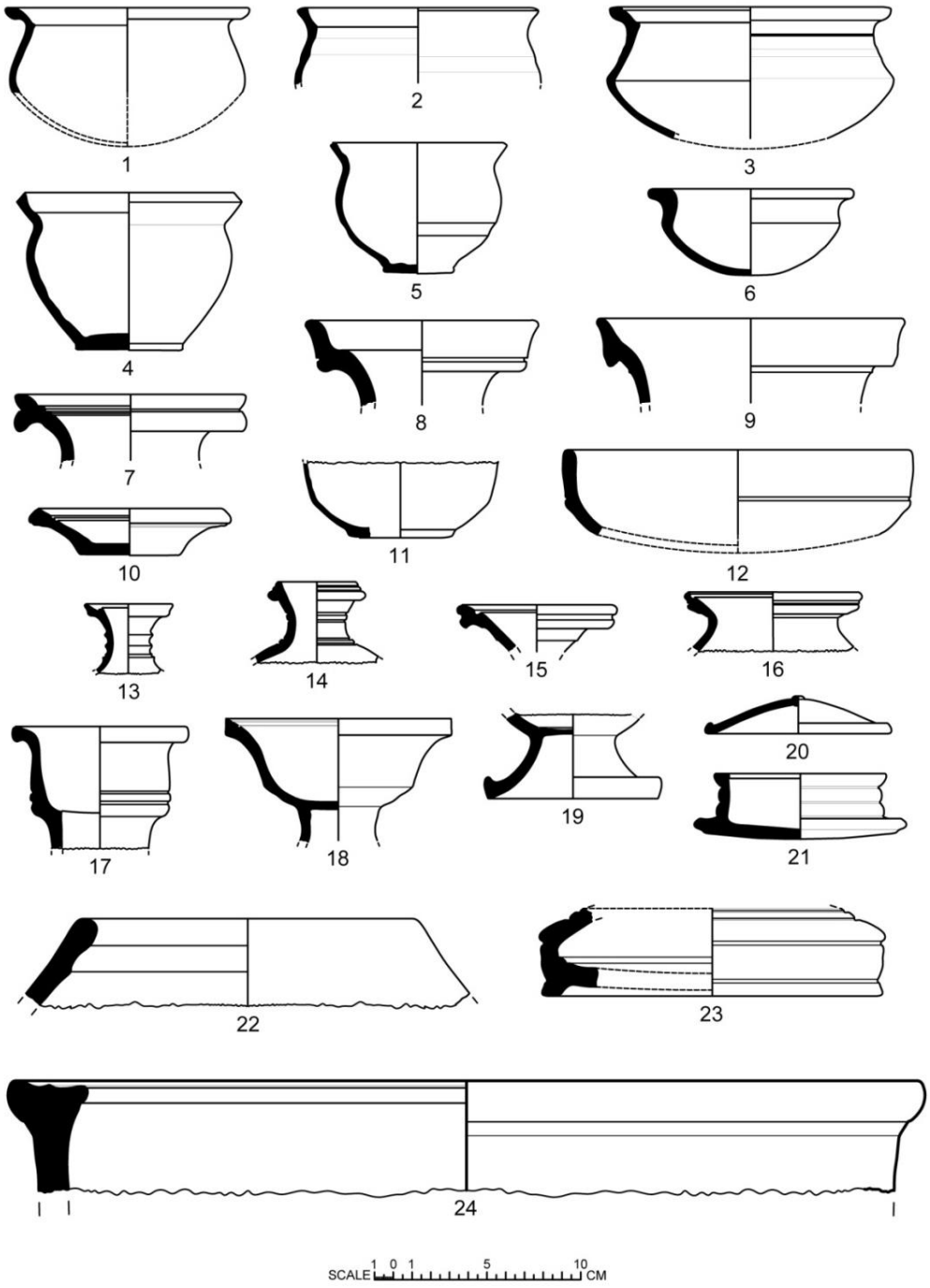


Fig. 14- storage jars of red ware with flaring, feature less rim with horizontal grooved rim, squat neck, slightly oblique shoulder and globular body, horizontal thick line on shoulder, medium fabric and well fired. Storage jars of gray ware with splayed out thickened and sharpened rim with internally notched and externally horizontal lines, squat neck and globular body horizontal thick line on shoulder, medium fabric and well fired.

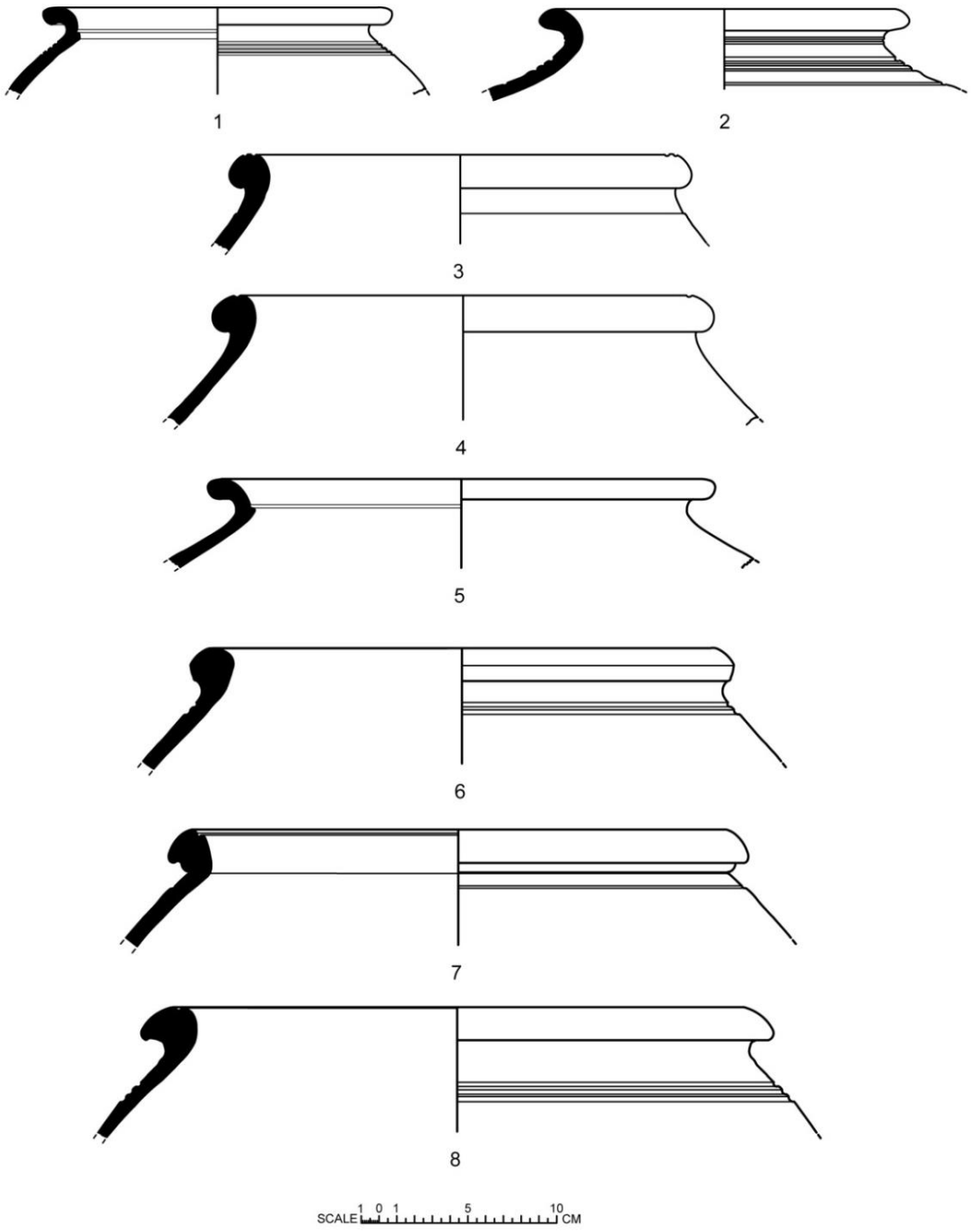


Fig. 15-Comprise basins, different type of bowls and leads of gray ware and red ware. The rims of basins are of splayed out, slightly drooping, internally notch and

horizontal lines at neck; prominent to normal carination followed by horizontal lines on shoulder; generally devoid of any surface treatment, medium fabric, well fired. The gray ware bowls are of featureless closing rim and rounded body, splayed out featureless and incurved sides, another bowl is something special and internally thickened rim and straight sides with horizontal lines on shoulder, all are of fine fabric and well fired. Leads are of gray ware having an ink pot like central cup with featureless vertical flattened top rim and horizontally flanged base with a slightly incurved sharpened edge top and decorated with incised slanting lines, of medium fabric, devoid of any surface treatment and well fired.

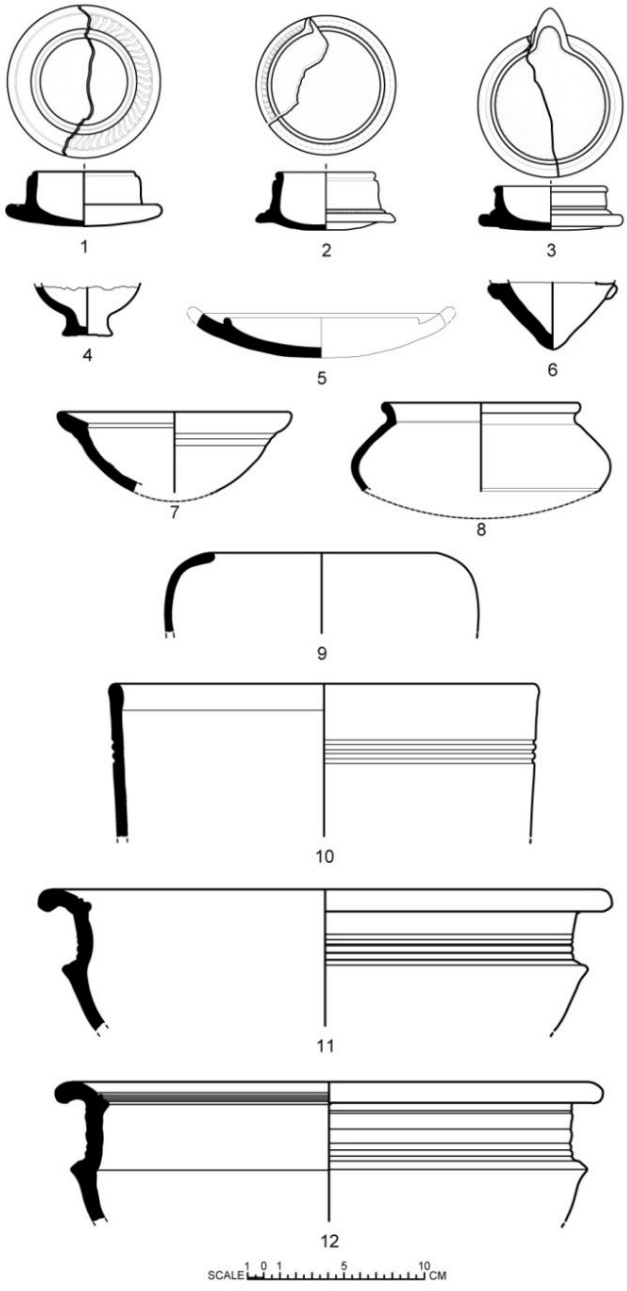
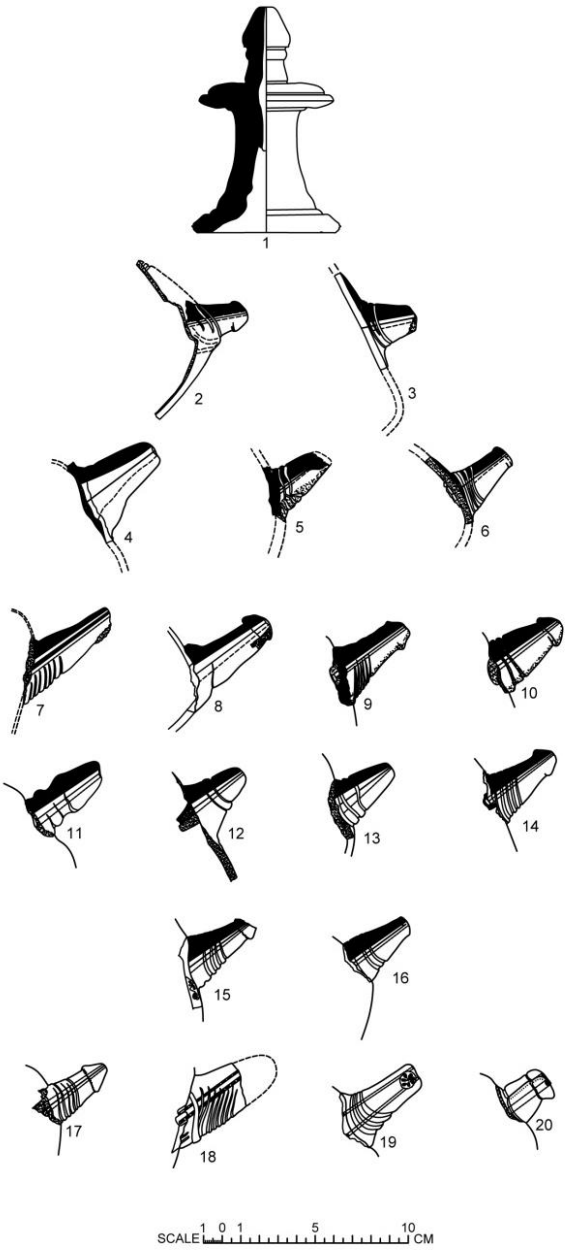


Fig. 16-comprise spout of red ware and gray ware. Spouts of red ware are of single and double perforations, double perforation spouts are decorated with dot like incised decorations and series of horizontal grooves at base and tapering, some are of knobbed tip design and others are plain. Spouts of gray ware are of single perforations, series of horizontal grooves at base and tapering, some are of knobbed tip design and others are plain. Apart from these Single specimen of red ware sprinkler from the site is represented with sharpened, flanged rim and conical knobbed openings, prepared by luting technique, all are of medium fabric, treated with red and black colour slip and well fired.



Phase-III (10th century-12th century CE)

The pottery assemblage of this period dominated by grey ware which shows more acquaintance towards reduced firing technique which is more economical in fuel consumption or the demand have been increased for producing large quantity of potteries both of ritual and secular in nature in short time or the potter adopted this reduced firing technique due to scarcity of firewood. The potteries of this phase are made from levitated clay and are of medium to fine in texture and semi oxidised to oxidised in nature. The major types represented of handi, carrinated handi, kadhai, basin, pitcher, disc, shallow bowl with footed base, vase, decorated lamps etc. Although the spouts are rarely evident in this phase, but different types of spouts with single and double perforation along with applique and incised designs are found. The number of decorated potsherds of this period outnumbered the potteries of earlier phase. The decorative patterns noticed are same as seen in the phase II, only decorative designs potteries are outnumbered and the percentage of producing terracotta lamps of plain and decorative are more in number in comparison to the earlier period.

Fig. 17- comprise lamp with channel of red ware and gray ware. lamps are having an ink pot like central cup with featureless vertical flattened top or externally and internally depressed rim and channel; horizontally flanged base with a slightly incurved sharpened, and decorated with incised slanting lines and applique designs are of medium fabric, devoid of any surface treatment and well fired.

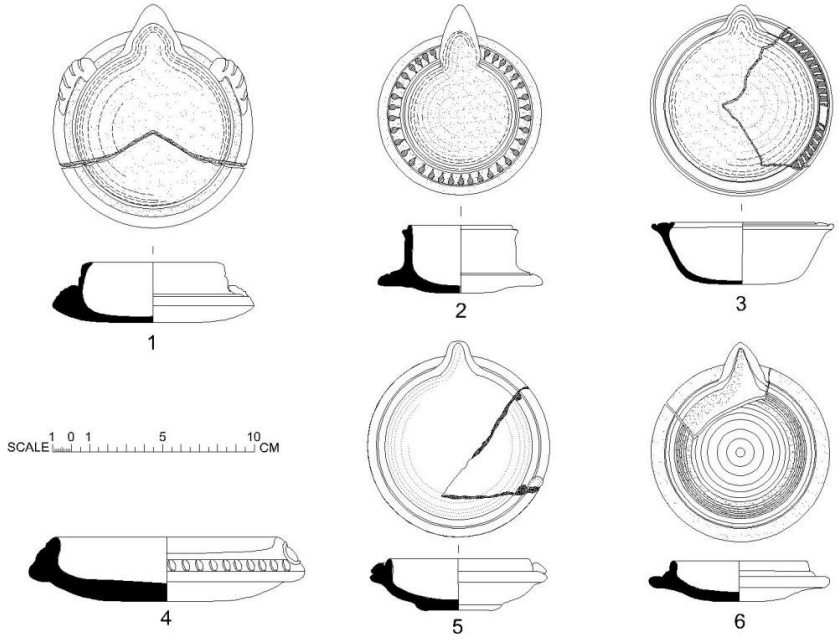


Fig. 18- Comprise Lamps of gray ware and red ware, gray ware lamps are having an ink pot like central cup with featureless vertical flattened top or externally and internally depressed rim and channel; horizontally flanged base with a slightly incurved sharpened to flatten edge top, and decorated with incised slanting lines, of medium fabric, devoid of any surface treatment and well fired.

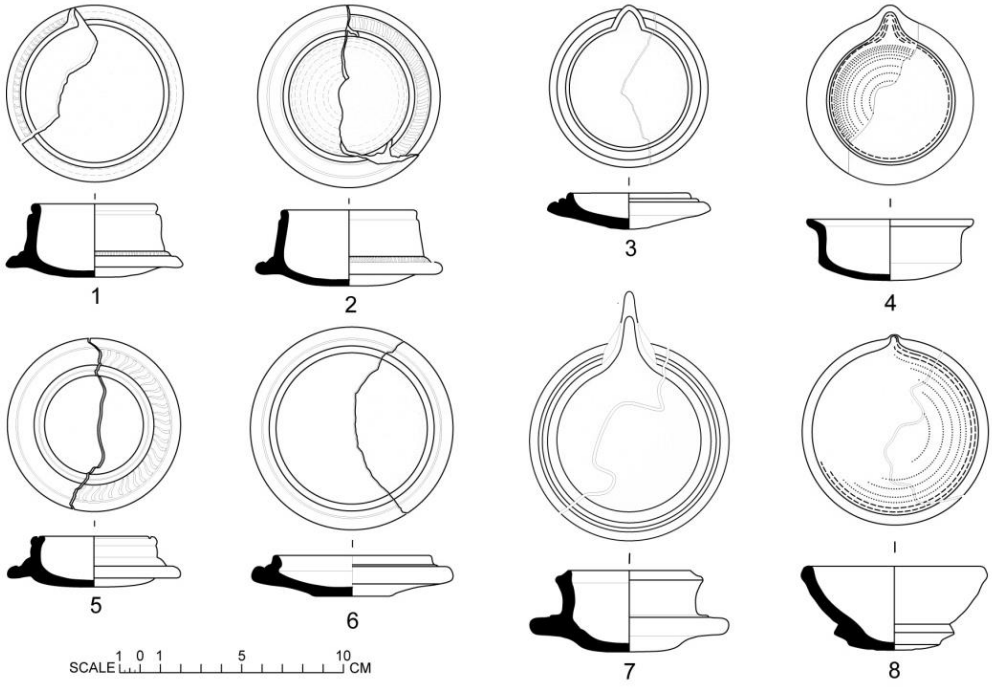


Fig. 19- Comprise storage jars and basins of different types gray wares with different incised decorations on rim and body. The shapes are repetition of as described earlier. The incised decorations are mainly geometric patterns like slanting lines, triangle patterns, group of horizontal lines etc. All are of medium fabric and well fired.

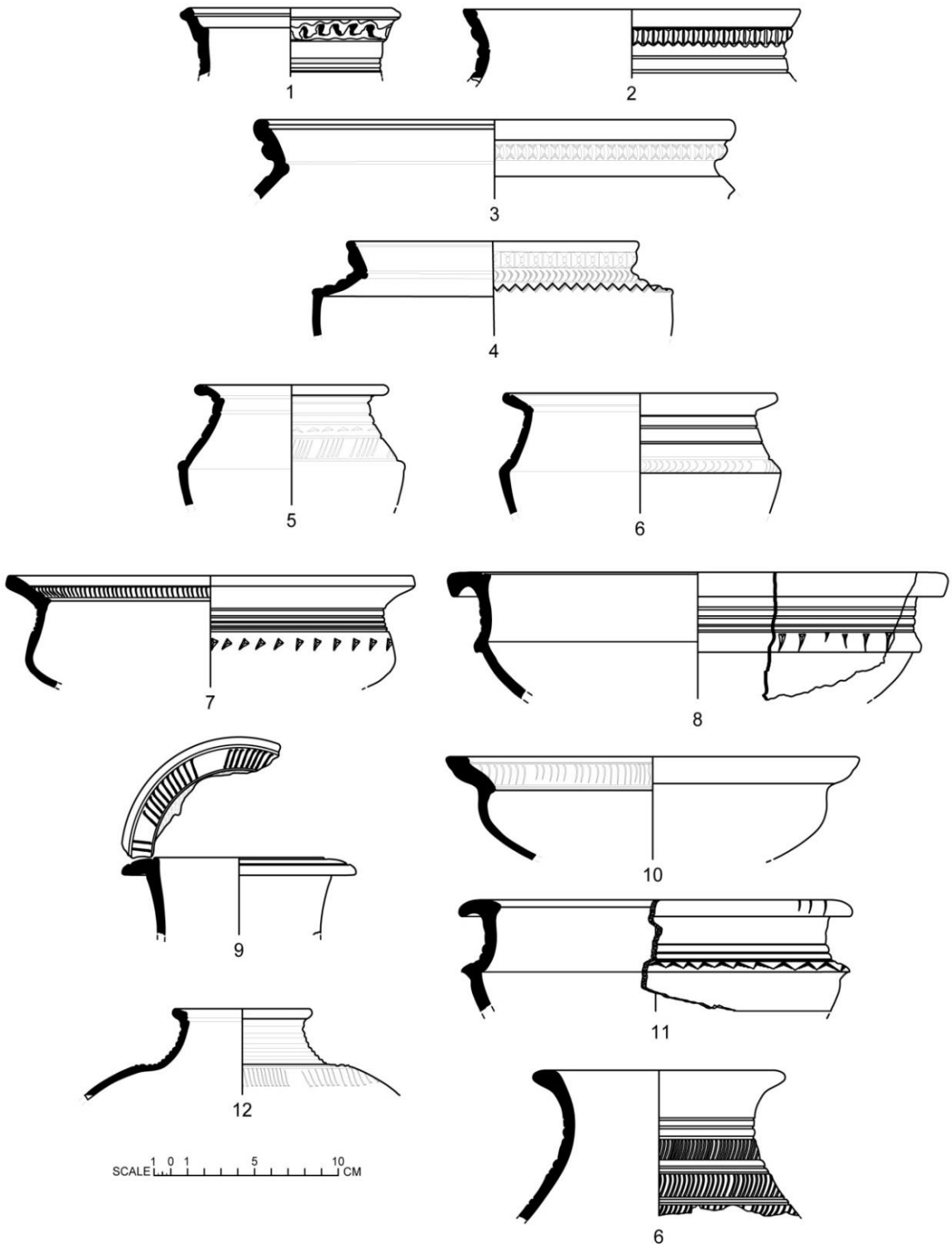

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Fig. 20- comprise basin of gray ware with splayed out sharpened or flattened rims with incised decoration of geometrical patterns. Medium fabric, devoid of any surface treatment and well fired.

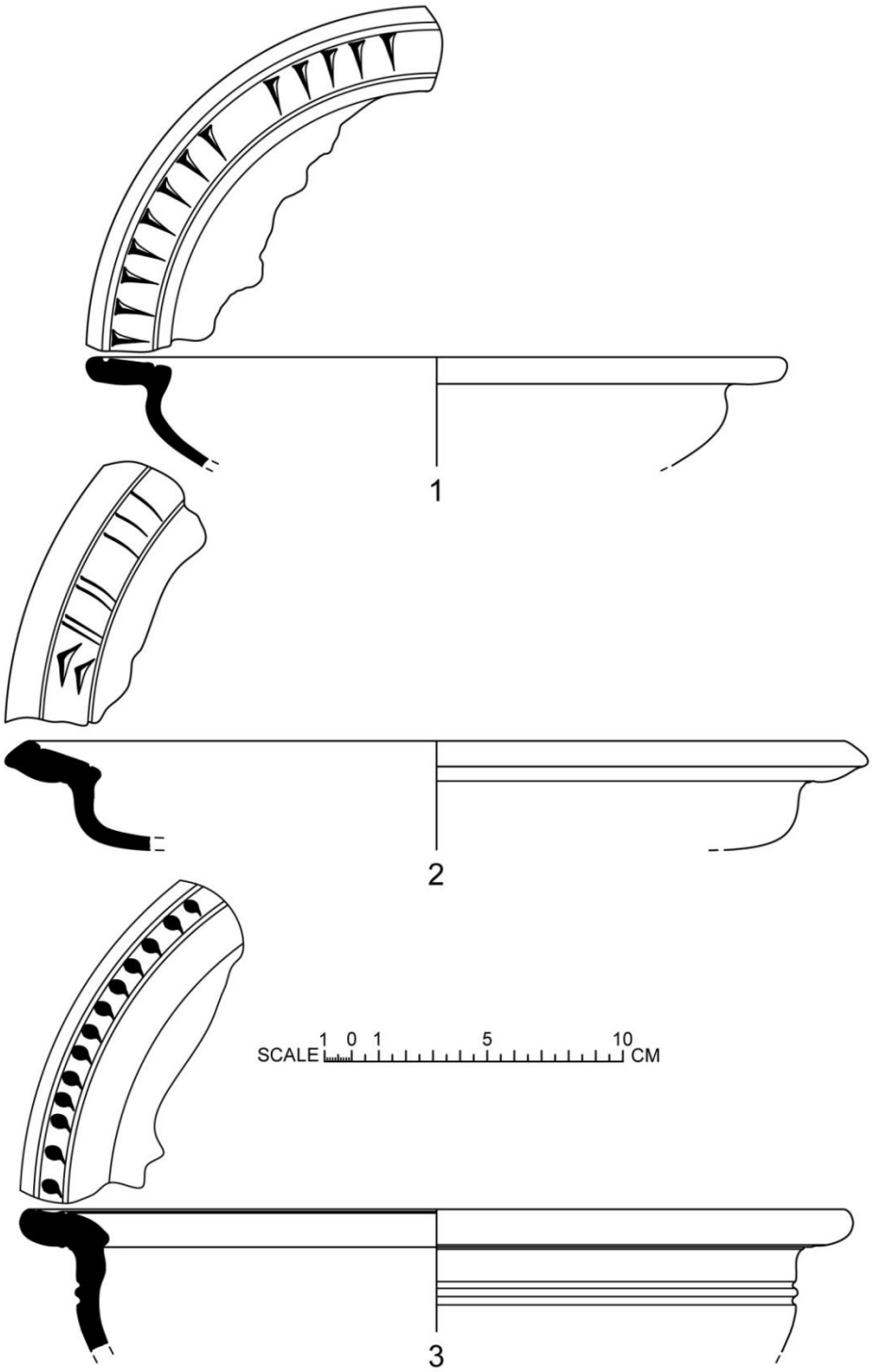


Fig. 21- comprise bowls of red ware, with featureless slightly incurved rims, tapering or slightly round sides, internally plain or corrugated surface and flattened ridged base. Medium fabric, devoid of any surface treatment and well fired.

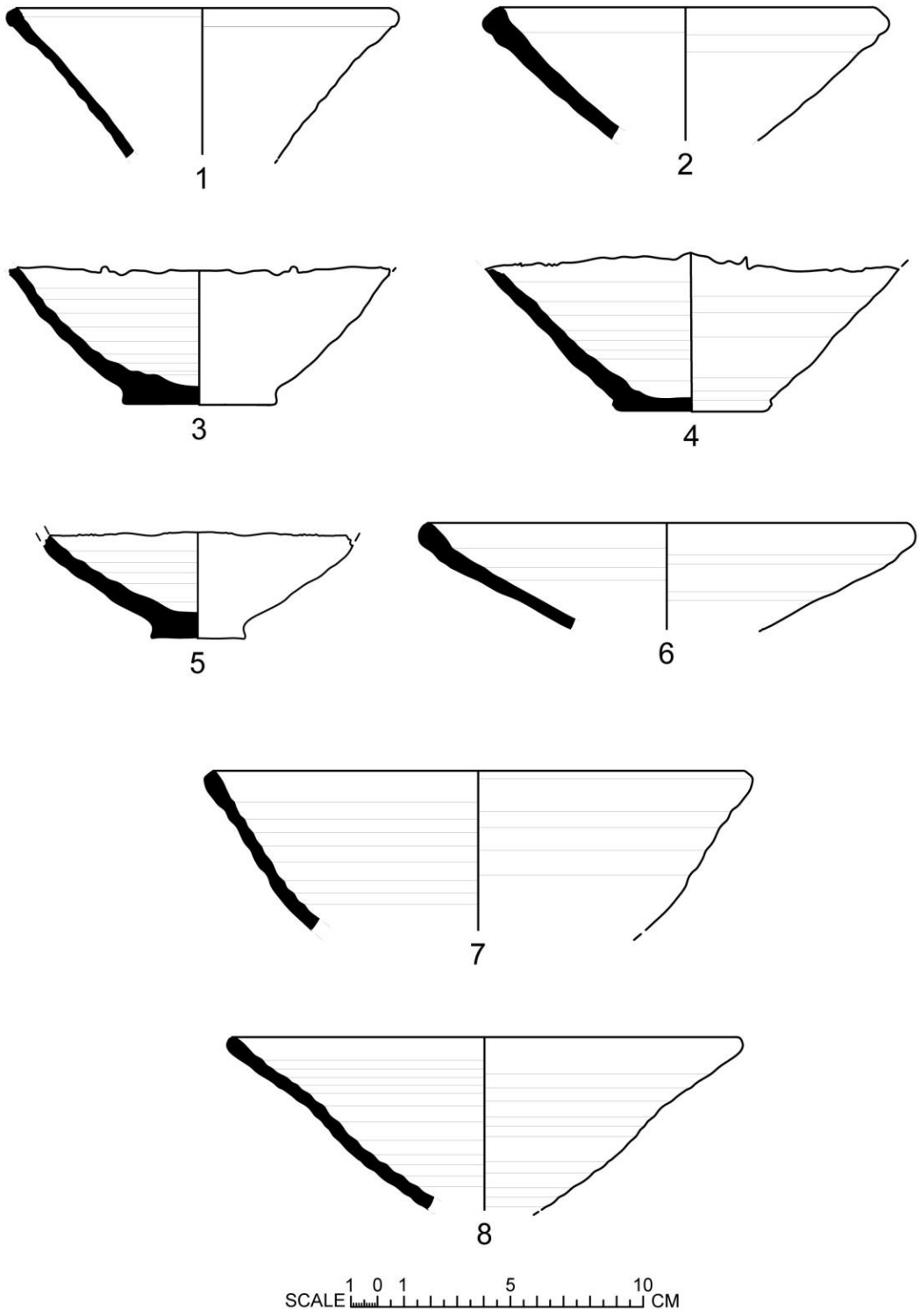


Fig. 22- comprise wide mouthed different types storage jars of grey ware with flared, flattened top, internally depressed externally mild corrugated rim, short neck and globular body of medium fabric, well fired. Splayed out internally and externally depressed rim of medium fabric and well fired. Externally thickened, horizontal single groove on top of the rim, short neck, corrugation on shoulder and globular body medium fabric and well fired. Thickened, internally depressed horizontal grooved rim with short neck and corrugated shoulder, globular body, medium fabric and well fired.

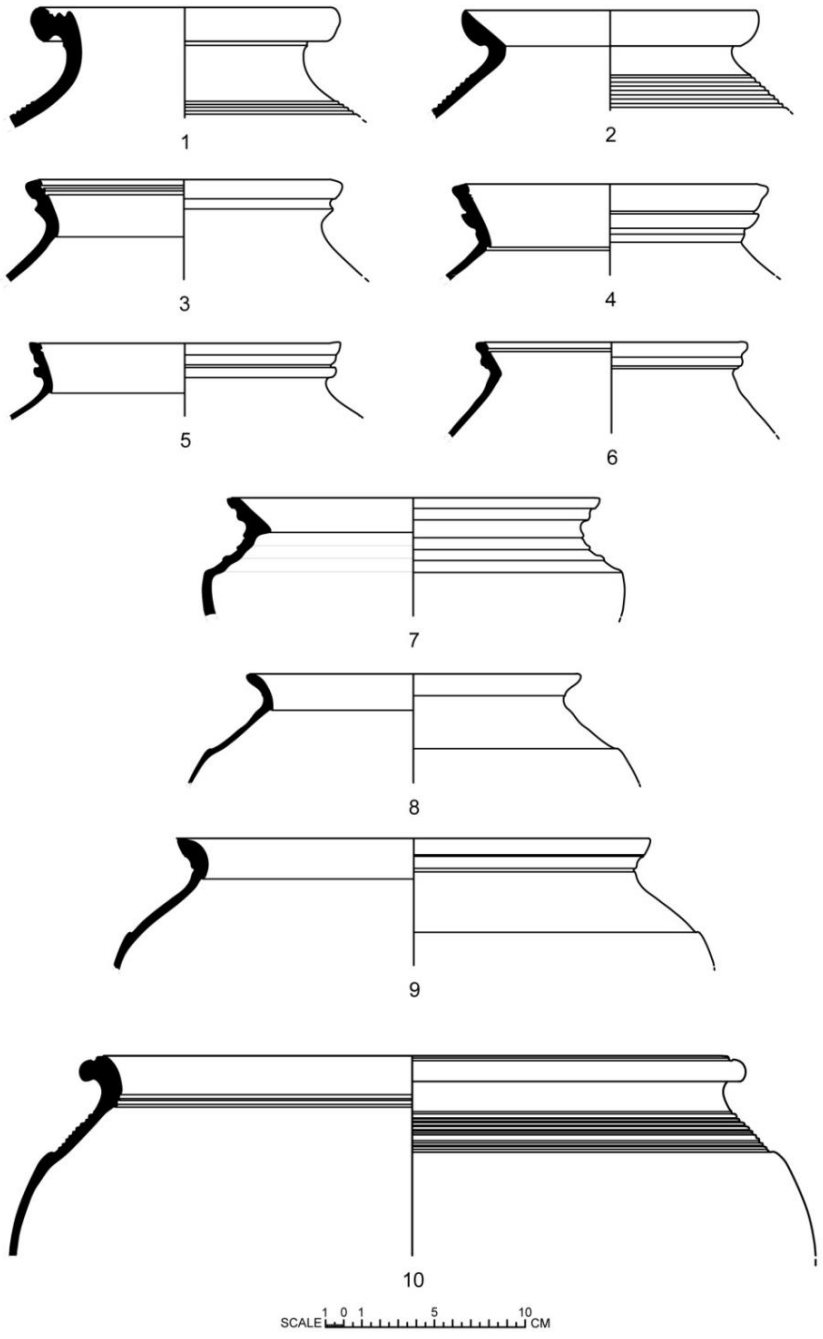


Fig. 23 - comprise wide mouthed medium size storage jars of grey ware with thickend splayed out rims ; thickend flaring horizontal grooved and internally thickened rims, necks are of squat, short and high with oblique shoulder and globular body. Horizontal groove is on shoulder, medium fabric and well fired.

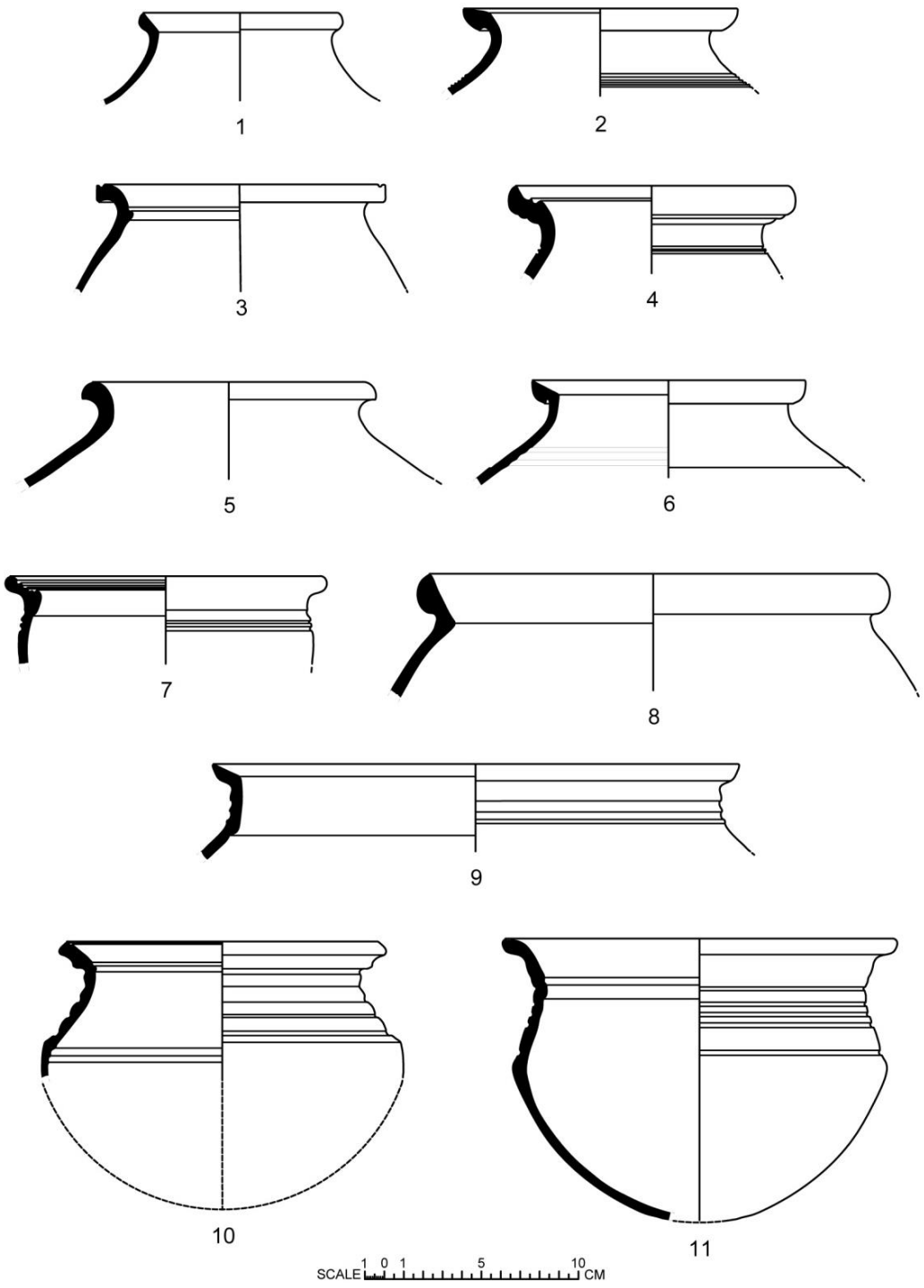


Fig. 24- Comprise basin of red ware and gray ware having splayed out rim or splayed out internally curved with horizontal lines on flat top rim, horizontal lines on shoulder and mild carination. Medium fabric, devoid of any surface treatment and well fired.

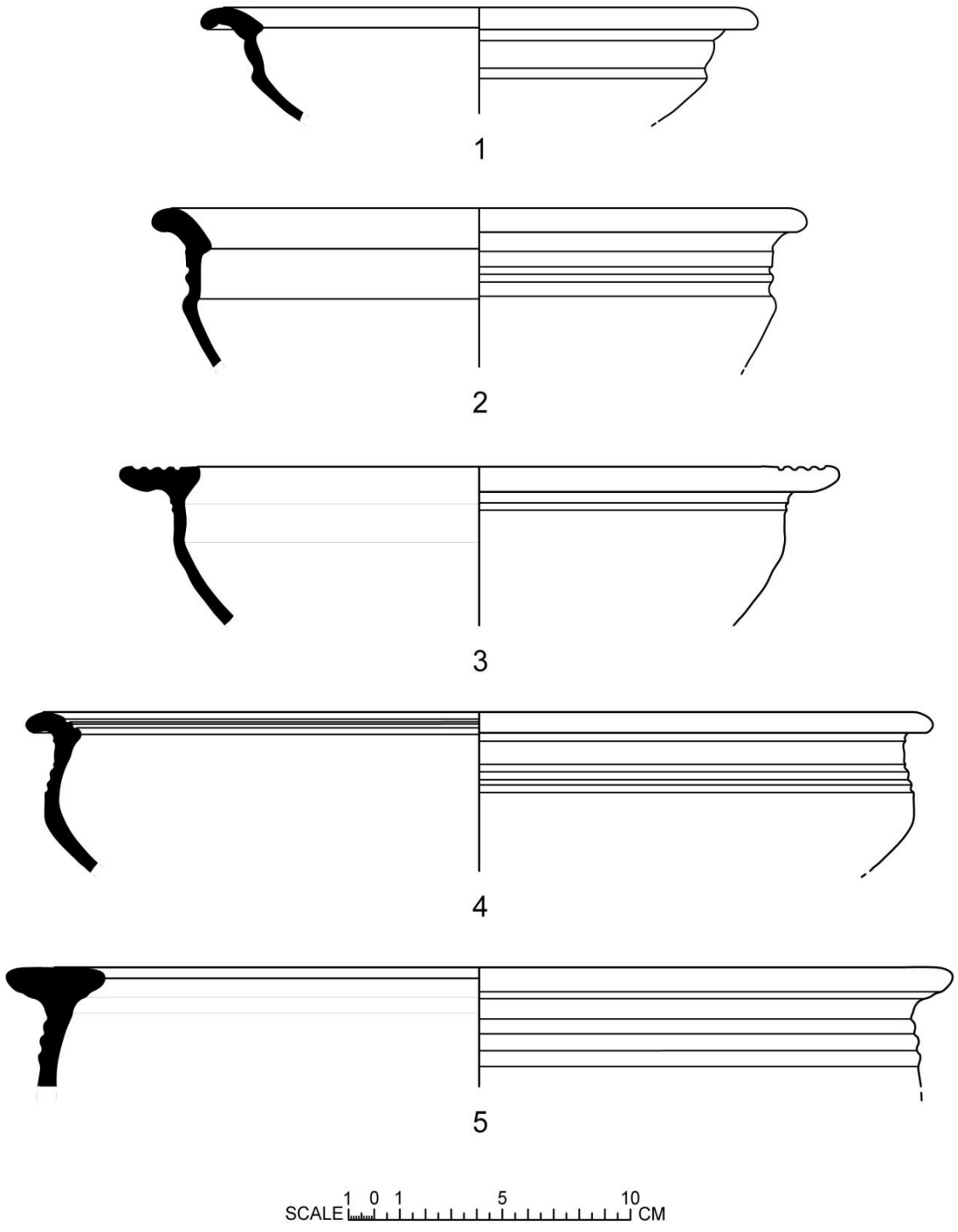


Fig. 25- (variant of Fig. 28), comprise high neck vase of Gray ware and red ware. The variant of rims are of splayed out plain and externally and internally depressed rims with horizontal lines on top of the rim and set of horizontal lines on the neck, incised decorations comprise vertical lines on the outer, lower edge of rim. All are of medium fabric and well fired.

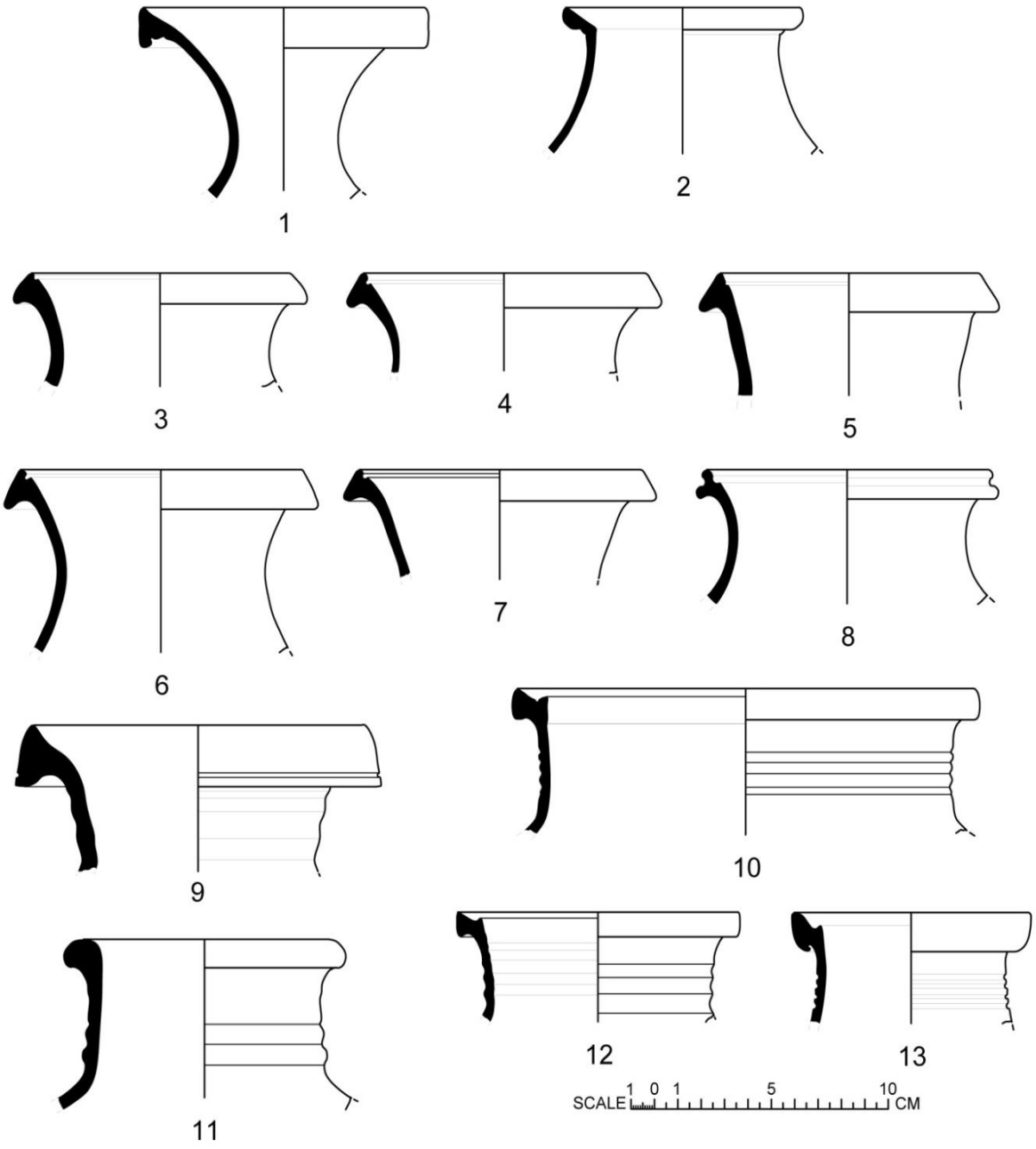


Fig. 26-Comprise bowls, leads of gray ware. Bowls are of featureless internally curved with a horizontal base lined rim and tapering sides, flat base. Both internal and external body is plain, well fired and medium fabric. Leads are of thick-end, featureless splayed-out rims, horizontal bands or grooves with plane exterior, treated with black slip in few cases.

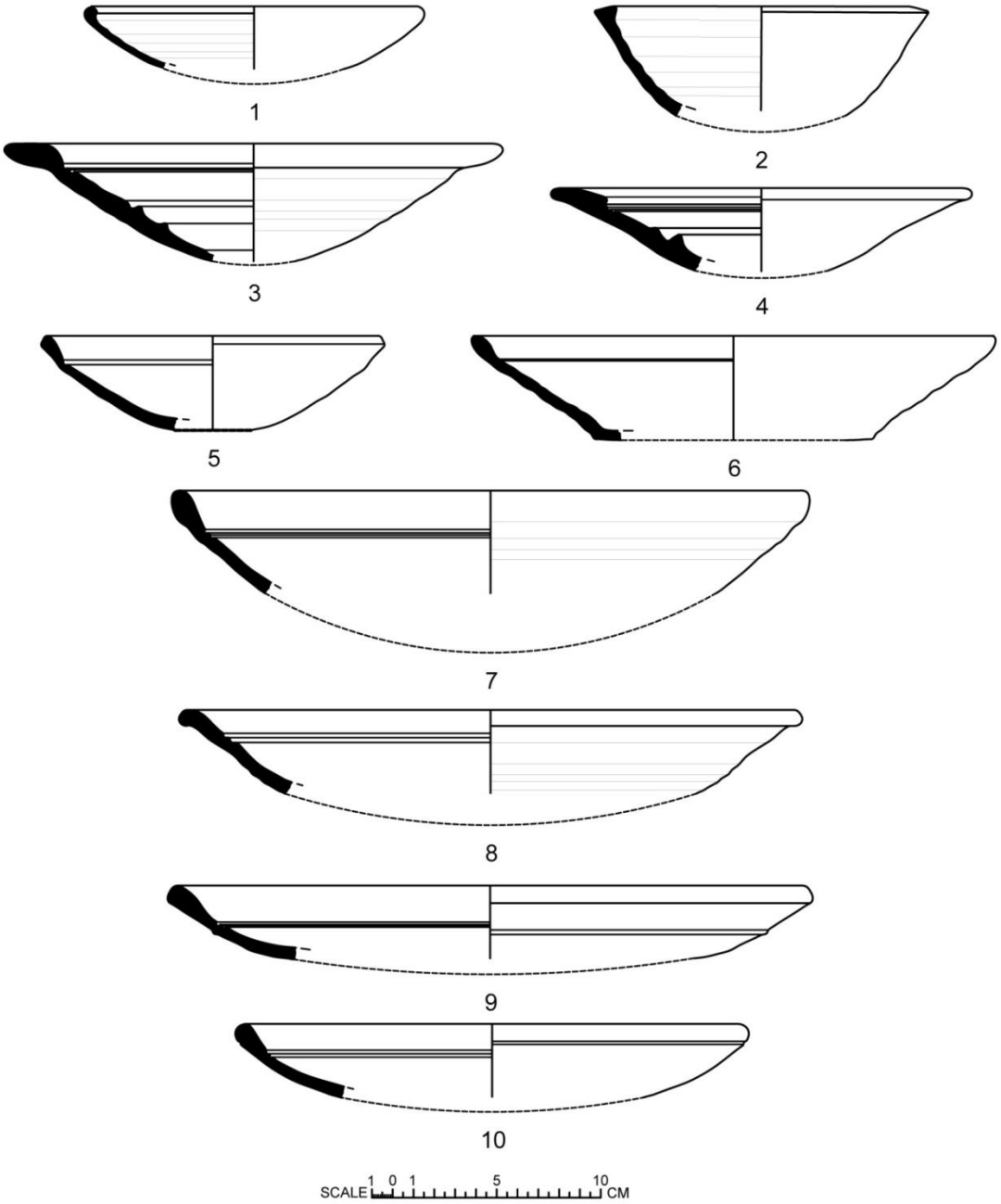


Fig. 27- Comprise vase/storage jars and bowls of red ware. The variants of storage jars are represented with thickened splayed out rim, squat neck, oblique shoulder; out curved , thickened, flattened top, internally ridged rim, rounded body; featureless closing, internally ridged rim, oblique body; shallow bowls are having featureless flat top, incurved closing rim, tapering body and flat base. All are of medium fabric and well fired.

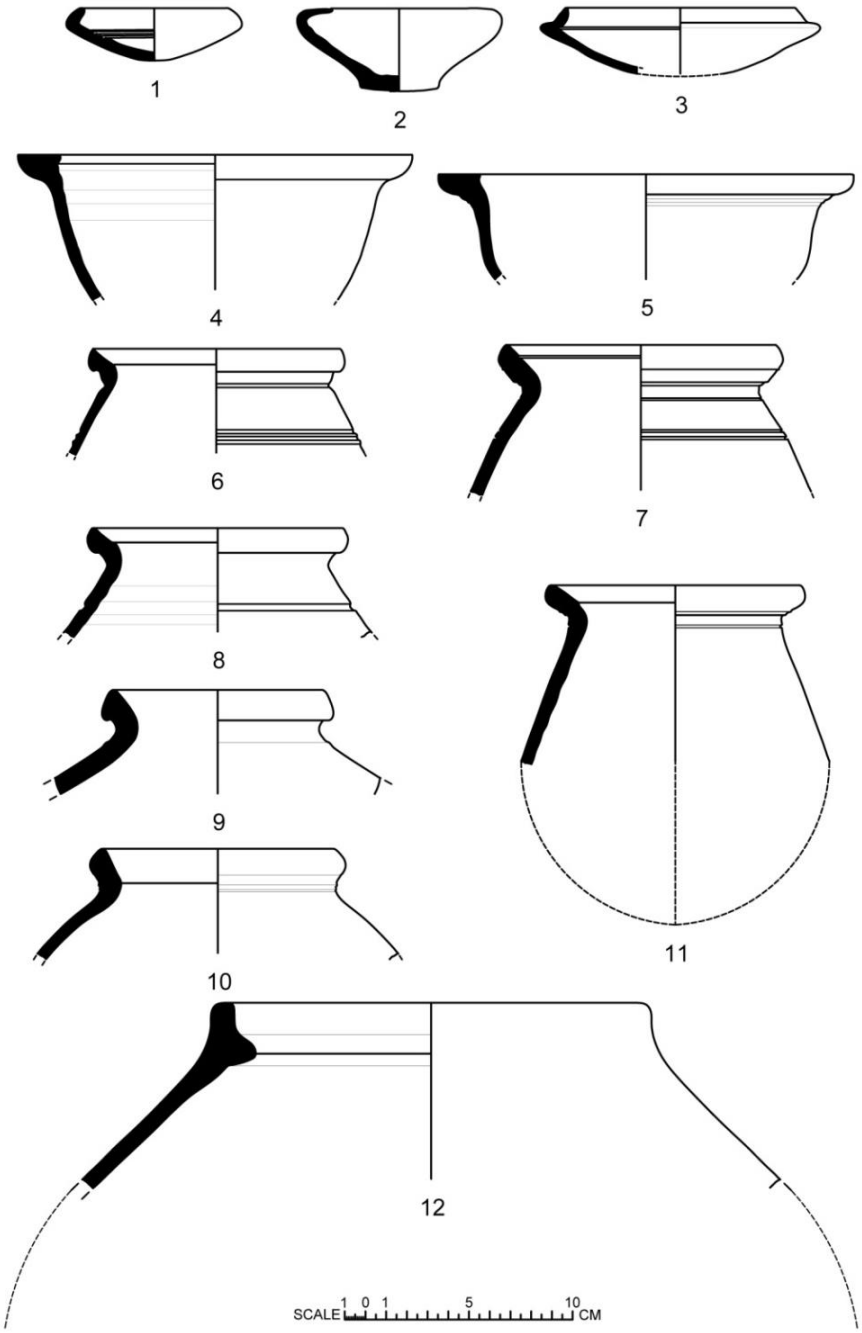


Fig. 28- comprise high neck vase of Gray ware and red ware. The variant of rims are of splayed out plain and clubbed rims with horizontal lines on top of the rim and set of horizontal lines on the neck, incised decorations comprise vertical lines on the outer, lower edge of rim . All are of medium fabric and well fired.

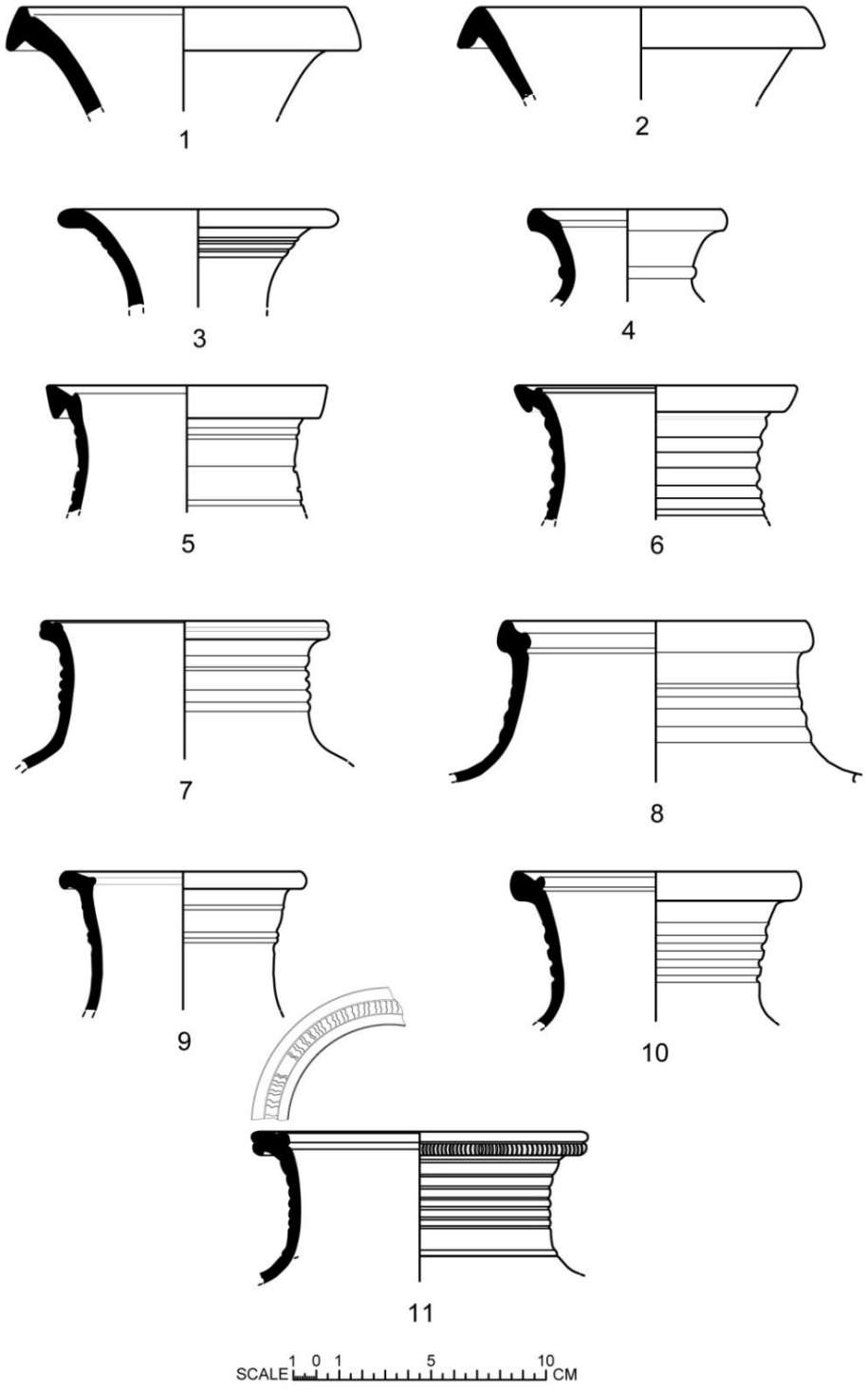
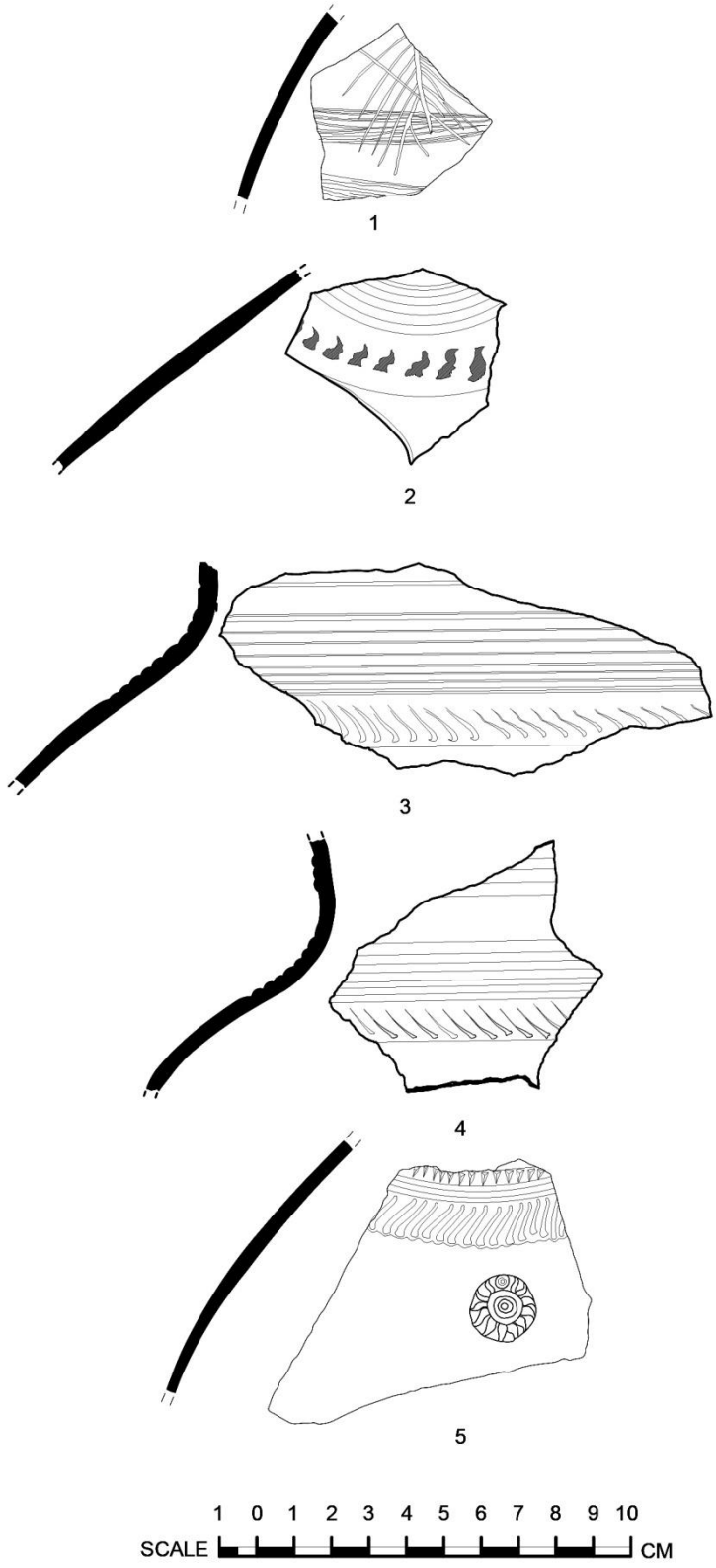


Fig. 29- comprise vase of fine fabric gray ware, decorated with geometrical patterns on neck and shoulder. In one instance black painting of slanting line pattern like grass is noticed. Fine fabric and well fired.



Phase-IV (12th century to 13th century CE)

The pottery assemblage dominated by dark grey ware medium to coarse fabric, semi oxidised in nature. Such types of potteries found confined to the present excavated stupa area, where previously Mahakal Temple was existed. Except this area, other part of the mound is bereft of such pottery assemblage. As per the archaeological remains, the cultural activity was assigned to 12th -13th century CE onward. The diagnostic shapes are of deep and shallow bowl, storage jar, vase, handi, dish, basin, carrinated vase, high neck and miniature pots and large number of terracotta lamps. These potteries are of medium texture, which are made out of locally available clay. Reduced firing technique was used for the preparation of these potteries. The basic decorative patterns are of vertical pinching strokes which resembles with a decorated necklace on the neck portion of the pot, series of parallel horizontal lines on neck, shoulder and rim portion have been documented.

Fig. 30- comprise lamp with channel of red ware and gray ware. Lamps are having an ink pot like central cup with featureless vertical flattened top or externally and internally depressed rim and channel; horizontally flanged base with a slightly incurved sharpened, and decorated with incised slanting lines, one specimen of red ware lamp is of featureless flat incurved rim and flat base, medium fabric, devoid of any surface treatment and well fired.

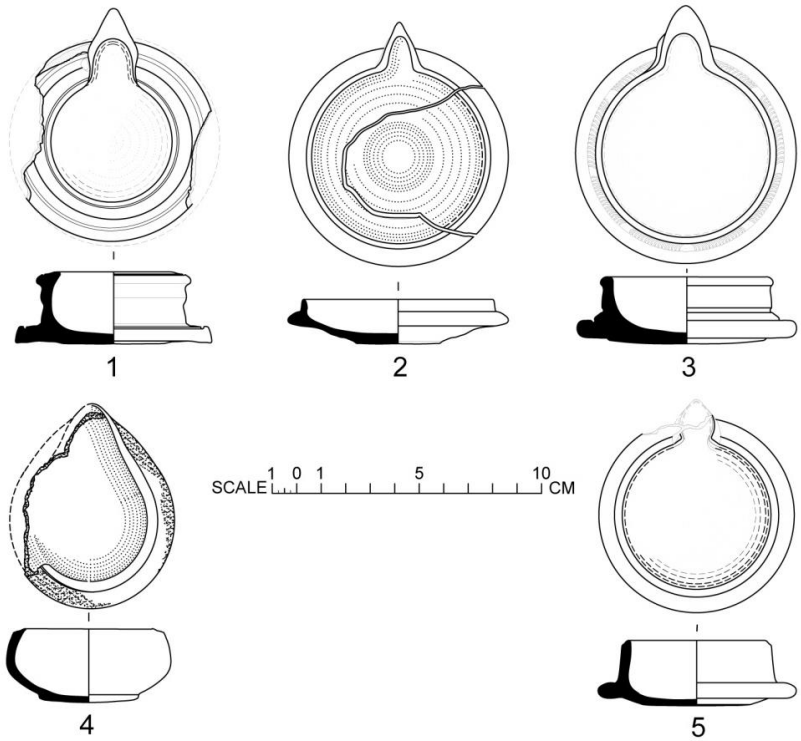


Fig. 31-comprise storage jars represented with splayed out depressed, ridged rim, splayed out thickend rim, squat neck, round body, medium fabric and ill fired.

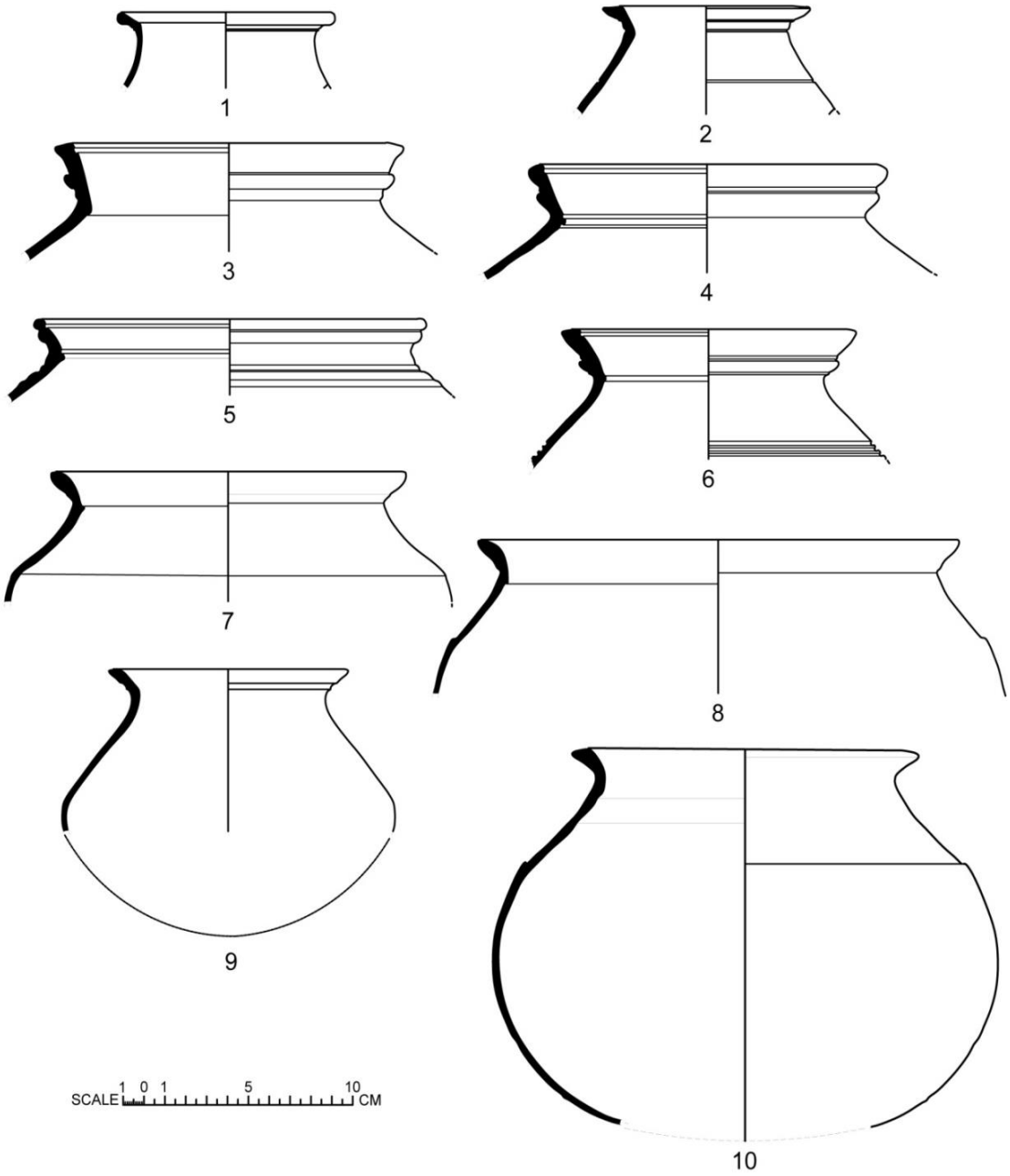


Fig. 32- comprise storage jars of grey ware, the variant rims are of thickened flared or featureless flared flat topped rim with internal grove at upper and lower part of rim with a ridge at lower part of rim, devoid of any surface treatment and well fired.

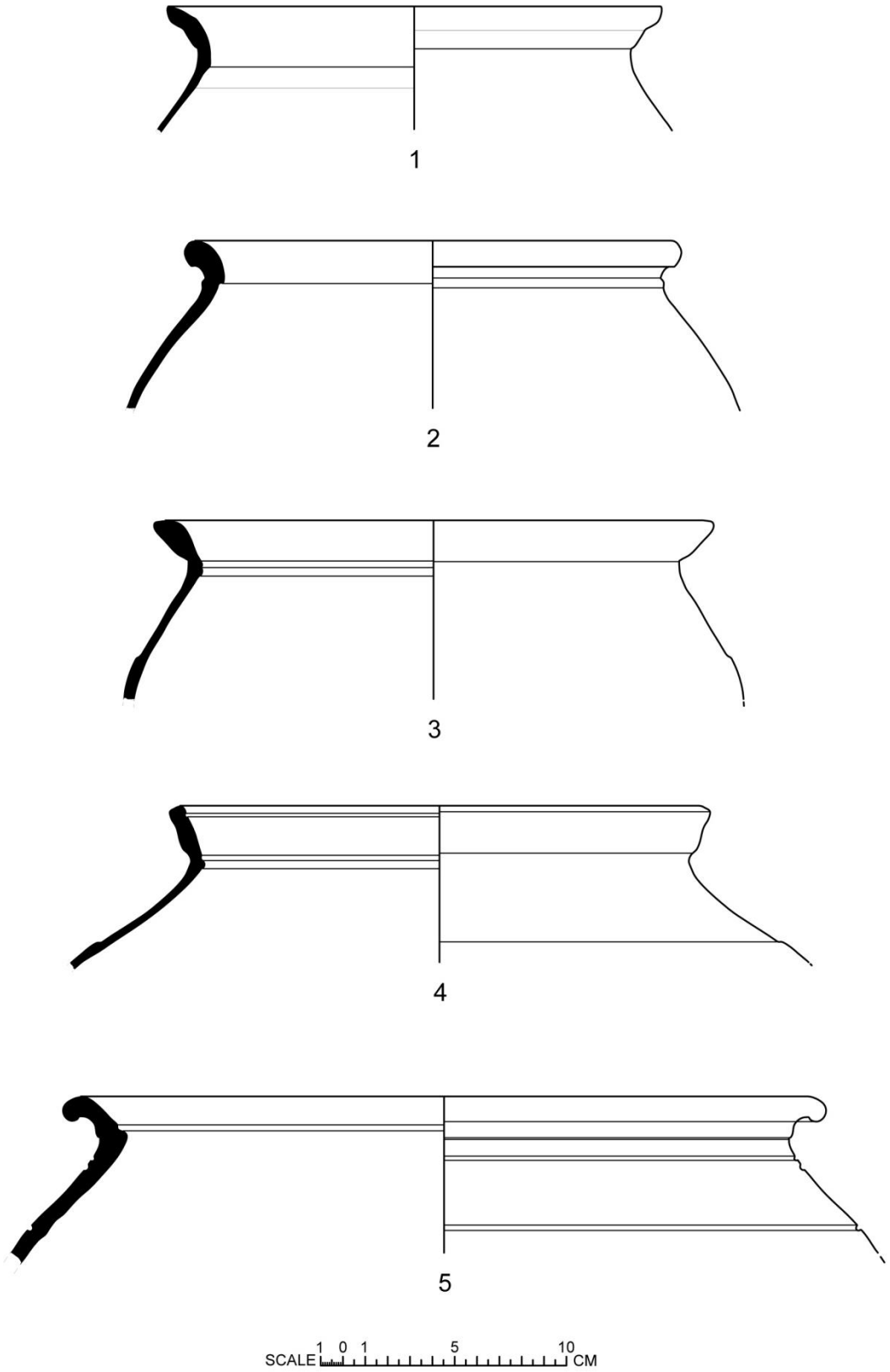


Fig. 33- comprise basins of gray ware represented with splayed out and internally bevelled rim and incurved body; nail headed splayed out, internally beveled rim; having an incurved tapered body, coarse medium fabric, devoid of any surface treatment and well fired.

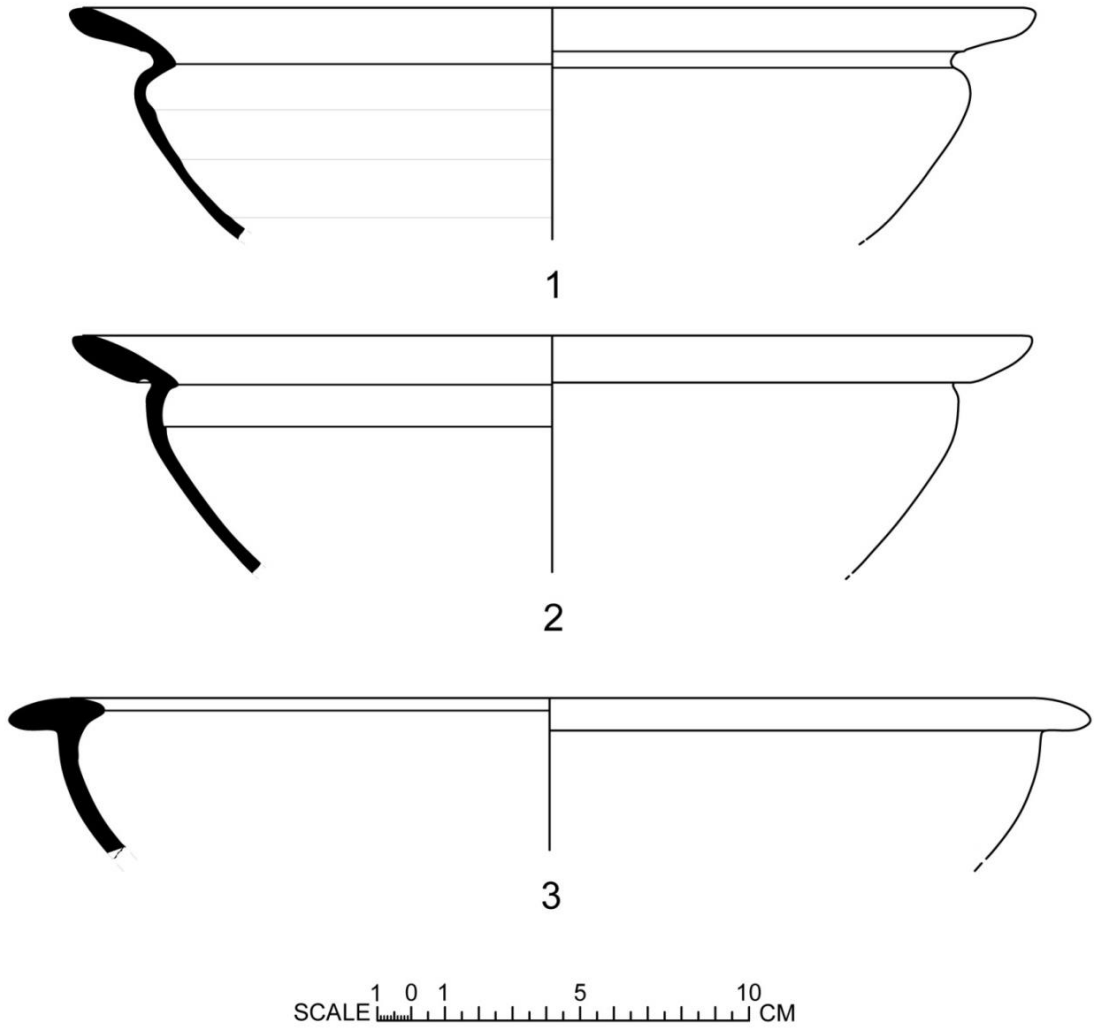


Fig. 34- comprise vase of gray ware represented with out-curved, thickened, flattened depressed grooved or ridged or flaring mouth everted undercut rim, high neck with series of horizontal lines, devoid of any surface treatment and well fired.

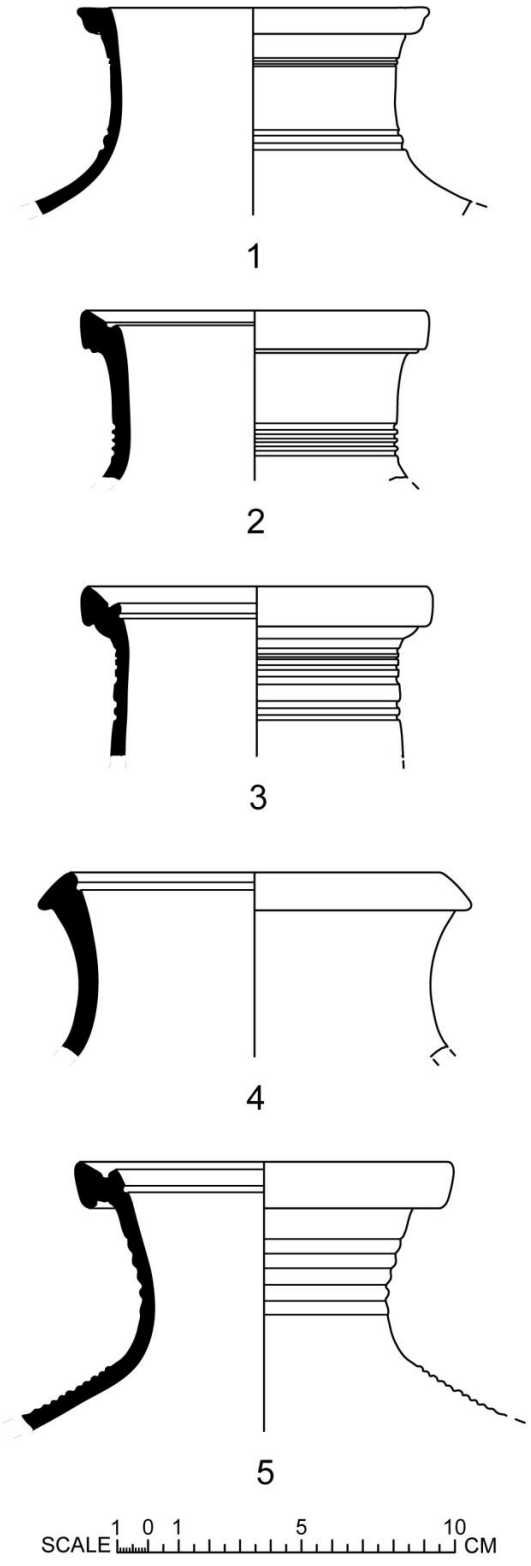


Fig. 35- comprise storage jar, bowl of red ware and gray ware, storage jars are represented with flat-topped nail headed rim; splayed out sharpened depressed rim; thickened everted rim. Bowl is of flattened nail headed rim, treated with red slip and black slip, medium fabric and well fired.

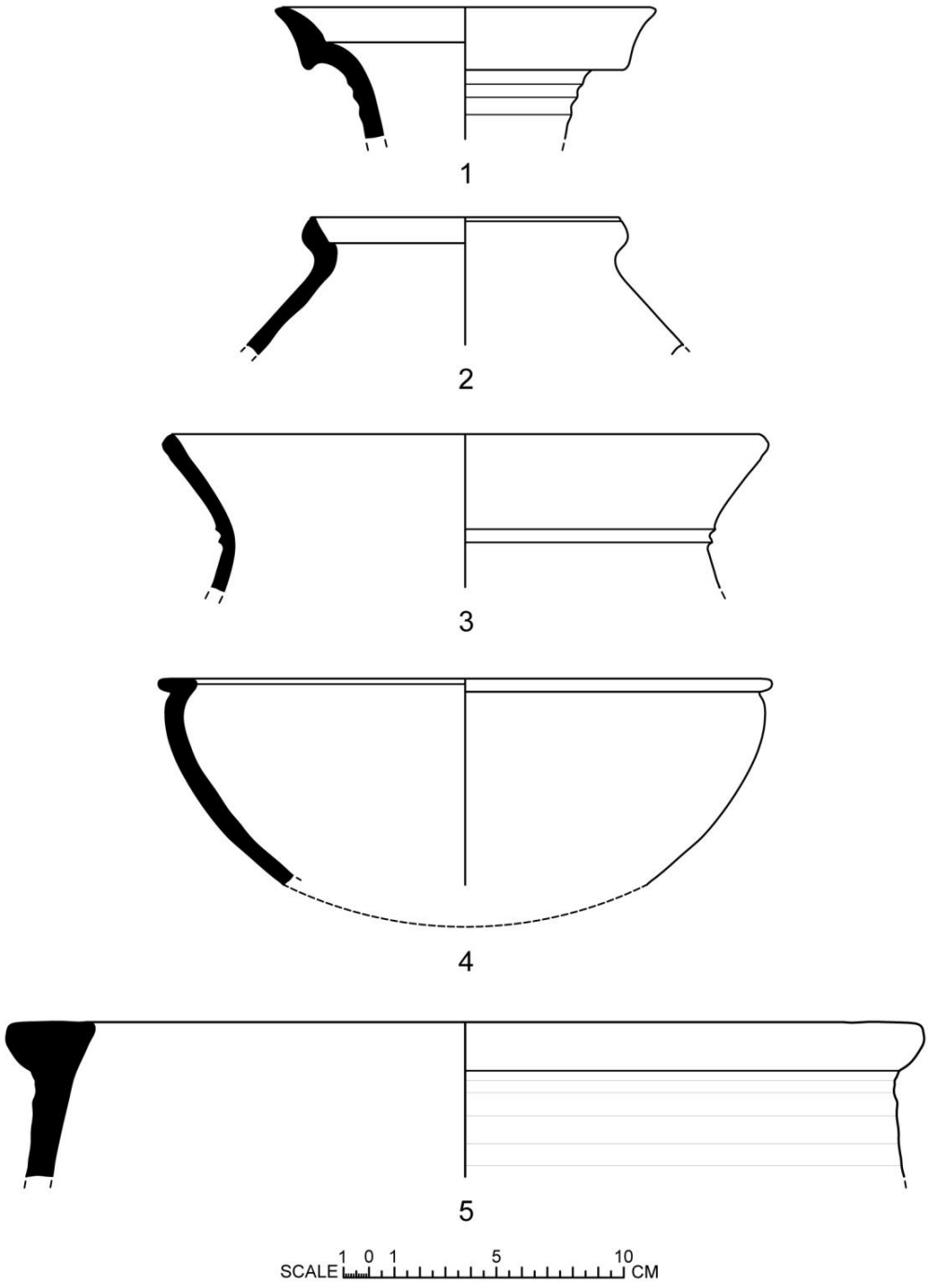


Fig. 36- comprise vase of gray ware with splayed out flat topped, internal and external horizontal single grooved rim. Sometimes incised vertical thick prominent lines on outer side of rim are present, oblique body with series of horizontal lines on shoulder part. Medium fabric, devoid of any surface treatment and well fired.

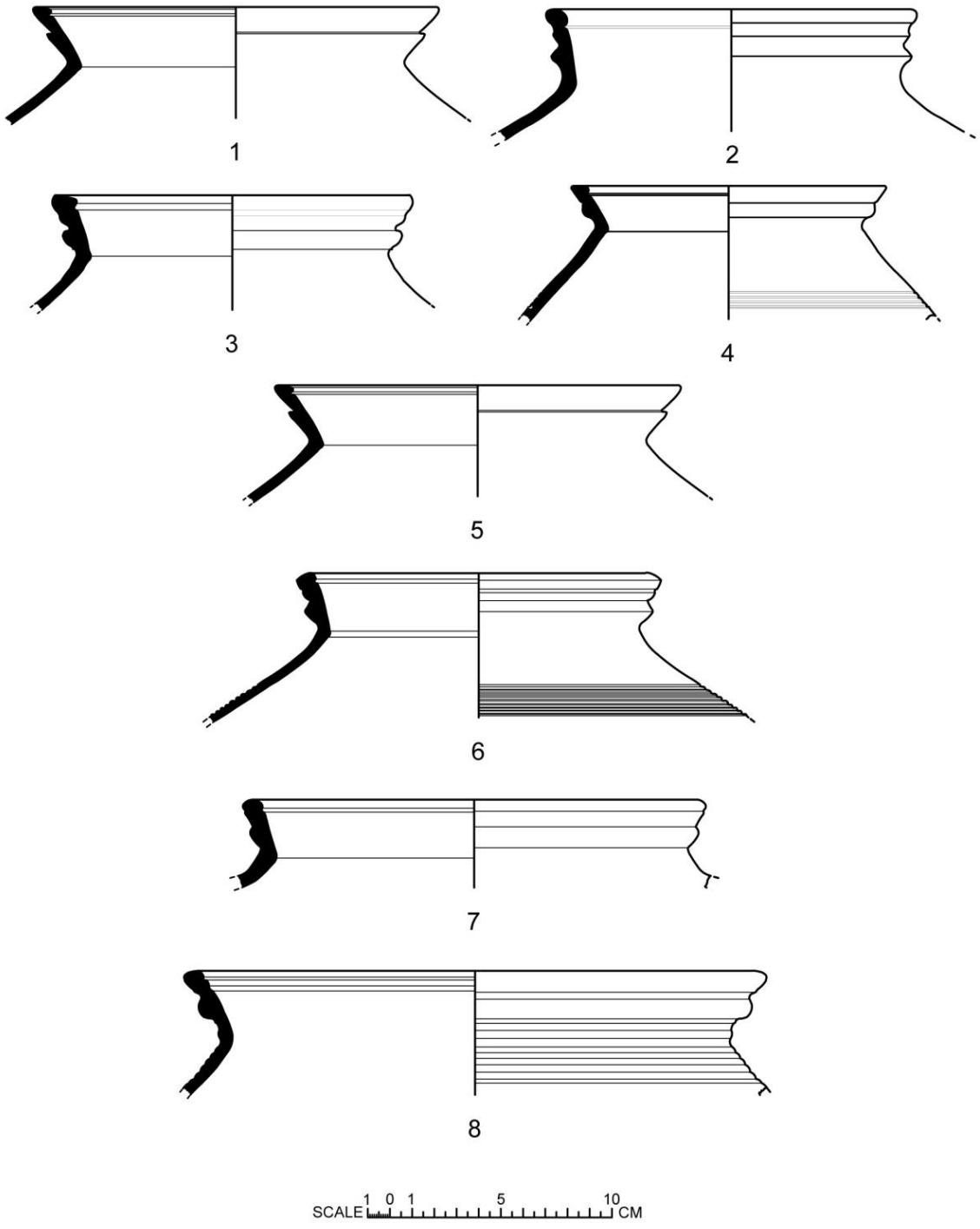


Fig. 37- comprise vase of gray ware with short everted, internal ridged rim, concave neck and oblique shoulder, ridged carination; short thickened externally and internally ridged and corrugation on both outer and inner side of rim, thickened or sharpened, splayed out rim, short concave externally groove neck; devoid of any surface treatment and well fired.

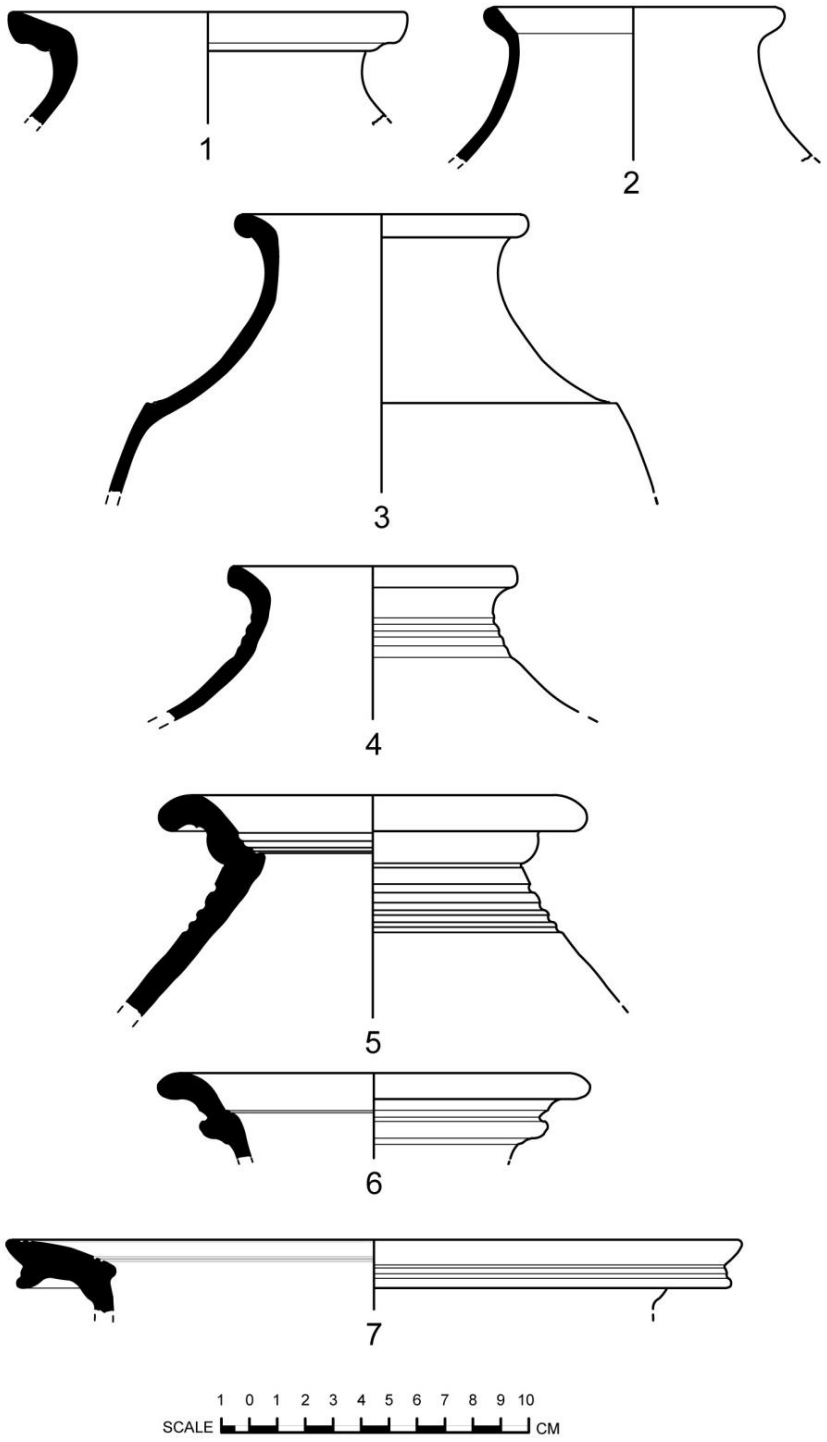


Fig. 38- comprise basin of gray ware thickened oblong, internal horizontal undercut rim neck externally distinguished with a deep groove and internally deep depression below the rim, having an incurved tapered body; splayed out and internally bevelled rim neck externally distinguished with a deep groove, having an incurved tapered body, devoid of any surface treatment and well fired, locally known as *Khurpi*.

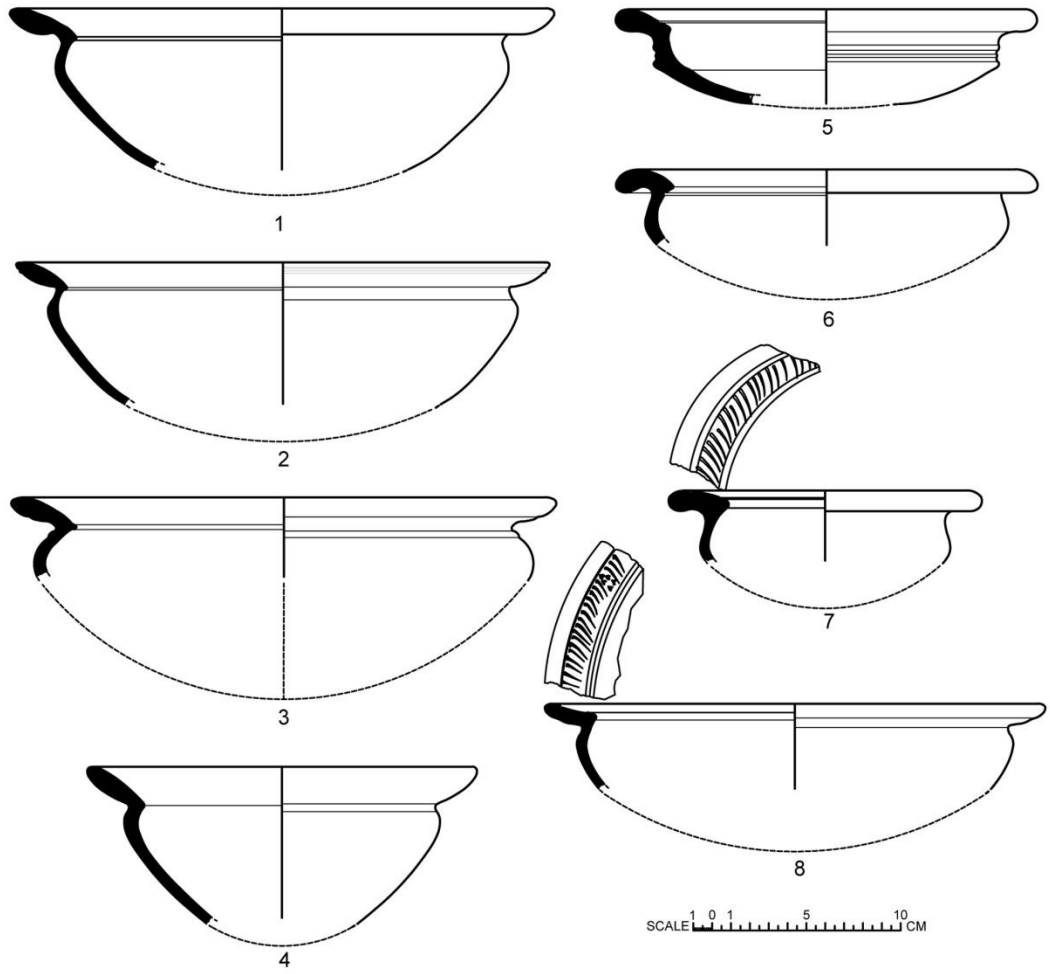


Fig. 39 –comprise handi of gray ware, the variant of storage jars are represented with sharpened splayed out flat rim, squat rim and oblique body with mild carination; sharpened splayed out externally and internally depressed rim with undercut, squat rim and oblique body with mild carination. In some cases Exterior is decorated with series of horizontal grooved lines. Medium fabric, devoid of any surface treatment and well fired.

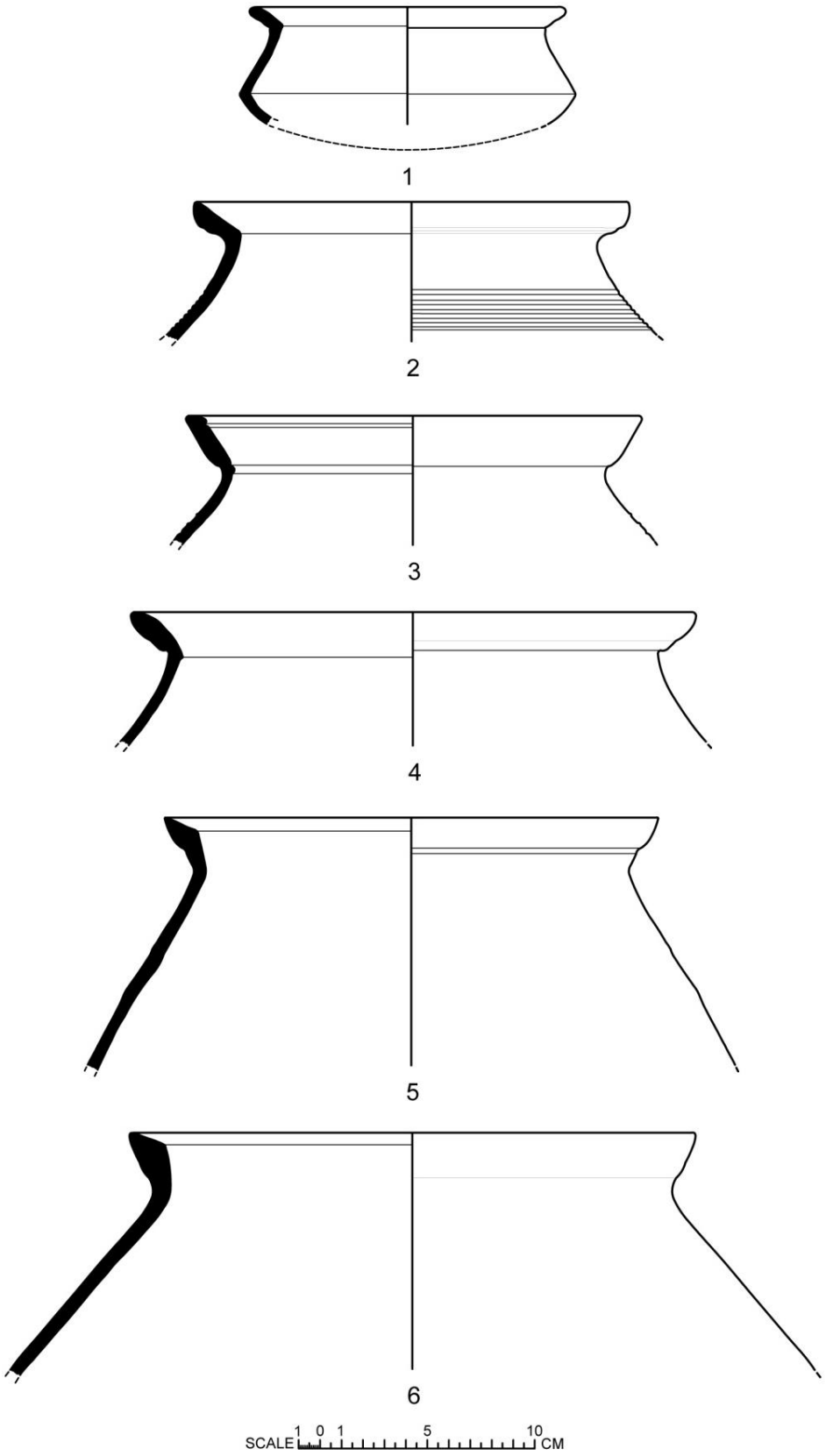

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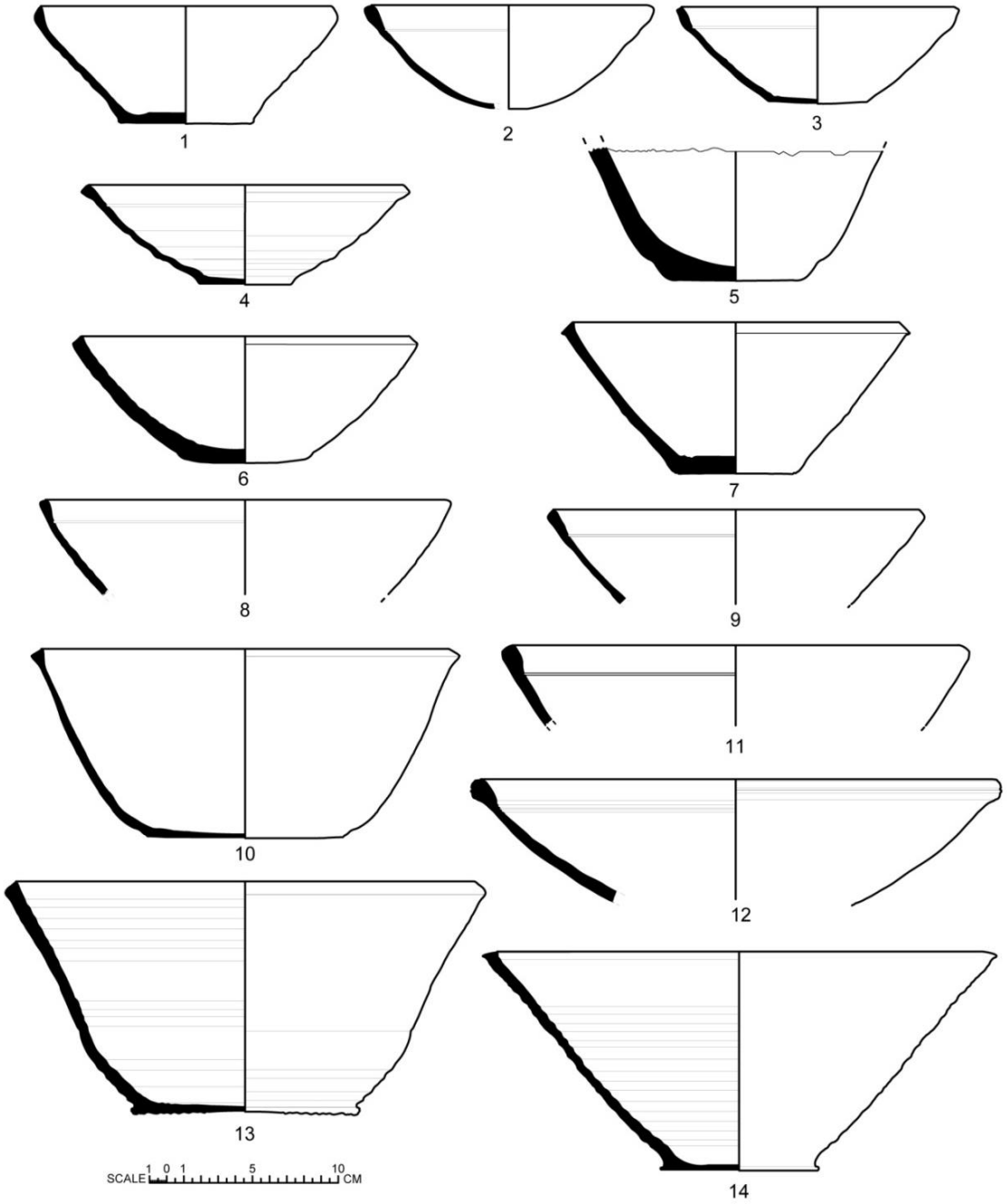
Fig. 40– comprise different types of bowl of grey ware represented with featureless inverted, featureless thickend with horizontal bottom lined rims; tapered to slight rounded body and flat base and pedestal base. Medium fabric, devoid of any surface treatment and well fired.

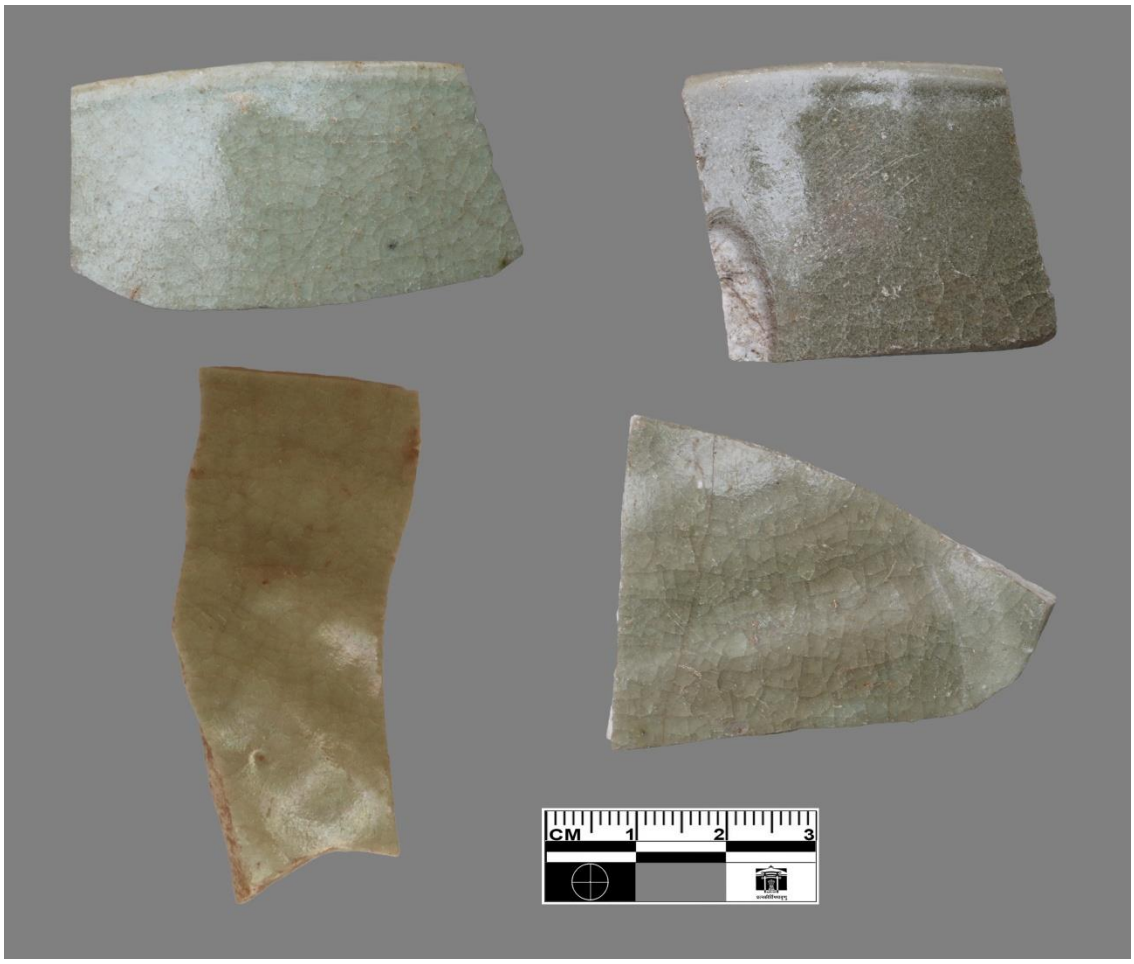


Red ware pottery

Grey ware pottery

Celadon

The scanty finding of celadon which resemblance with the Burmese ceramic/ celadon from the site paved the way to find out more material culture of south-east Asian origin or beyond and to understand the cultural interaction between India and these countries. It acts like a beacon light to understand the movement of people across the region. Though its scanty in the present scenario but it's like a drop of hope which gives aroma to the universal connectivity among the human society due to trade relationship or connection of cross regional society under the umbrella of Buddhism, believes in non-violence and brotherhood and faith on Budddha (Lord Gautam) Dharma (right principles and action) and Sangha (based on the concept of unity among the human community).



Pottery in the sculptural Art

Earthen vessels are lifeline of the society and for the monks those are living on the top of hill away from water resources. It's an essential object in the material culture

of the civilized human society. The evidence of large number of potsherds of different fabric, texture and shapes show about their utility in different stage for survival, these are may be of secular/domestic in nature or for ritualistic use. As observed during the excavation not only the potteries are found in the surface and stratigraphical context but has been seen as an attributes of Gods and Goddesses, such as Jambhala, Akshobhya, Jata-mukuta-Lokesvara, Padmpani, Vajrapani, Chunda, Bhrukuti, Naga-purusha, devotee holding bowl and offering pot, spouted pot, offering pot in the pedestal, kumbha (pitcher) as a decorative motif in the sculptural art.

Naga Purusha holding a Kalasa

Vajrapani holding sprinkler

Akshanidhi (pot with jewels) below the Jambhala

Attendants of Vasudhara holding sprinkler and bowl

Padmapani Avalokitesvara holding a sprinkler

Purna Ghata below the figurine

Gana foot resting over a pot

Purnaghata design on Vedi

Spouted Vase on the pedestal of Bodhisattva

Bodhisattva holding a sprinkler

Spouted Vase on the pedestal and bowl on stand

Chapter- VI

Conclusion

Ratnagiri is one of the significant Buddhist sites in India, uniquely representing the evolution and transformation of Buddhist practices over centuries. It is believed that Ratnagiri is one of the epicentres of the Vajrayana Buddhism, which incorporates esoteric rituals and tantric elements, and gradually spread throughout India. The site reached its zenith during 7th century CE, particularly during the Bhaumakara dynasty. The site has cultural continuity starting from early 4th century CE to the recent past having ups and downs. The material remains prior to 4th century CE have been noticed in secondary context in the form of finger-impressed brick and red- slipped pottery which are generally noticed or come across during 1st-2nd century CE in most of the sites in Odisha. A stable cultural period has been encountered in 4th -5th century CE which has changed its course during 7th century and attended grandeur monumental architectural phase. In this phase, the culmination of Vajrayana Buddhism, which combined the faith of people with complex ritualistic practices began. The chaitya complex and the monastic establishment at the site prove that the site had enjoyed its cultural and religious zenith from 7th century to 12th century CE.

In essence, the archaeological environment around Ratnagiri is not just a backdrop to the main site – it is an active part of its history. The surrounding landscape, with its scattered ruins & explored mounds, holds immense potential for further discoveries, offering a broader & more nuanced understanding of Buddhist life, architecture, and society in early medieval Odisha.

Besides, the site has quite potential in the region and is surrounded by archaeological heritage in the form of water reservoirs, rock cut caves, rock cut sculptures, brick and stone masonry structures, etc., affiliated to Buddhism. Several notable temples & religious sites affiliated to Brahmanism are also noticed in the region, which reflects the region's rich spiritual heritage. It suggests a syncretic religious environment where Buddhist & Brahmanical practices coexisted for a long time.

The landmark event of the region witnesses during *8th century CE by the initiation of Bhauma King Shivakaradeva-I (756 to 795 CE) of ancient*

Odisha, having his capital city Guhadebapatak modern Jajpur 30km away from the site under discussion was credited for sending an envoy to the court of the Chinese emperor Te-Tsong to present a Buddhist manuscript Gandavyuha to the Emperor.

Features of Site

- Ratnagiri was one of the epicentres of Vajrayana establishment in India.
- It is one of the well-set-up Buddhist establishments in Odisha, having Stupa complex, Chaitya complex and a Monastic complex.
- The significant in situ findings from the Chaitya are of three colossal heads of Buddha, measuring 1.27m 1.25m, 1.28m and four neck portions and fragments of body parts.
- Vajra-Gandha-Tara image has not been documented at any sites of Odisha and adds to its significance in the present excavation.
- For the first time in Odisha, two armed Chunda have been documented from the present excavation, along with four and six armed Chunda.
- The evolution of script from Brahmi (4th century CE), Siddhamatrika/ Kutila (8th-9th century CE), and Nagari (10^{th}-12th century CE) has been witnessed in the site.
- The records of inscriptions of different scripts indicate the continuity of the Buddhist monastic site Ratnagiri from the 4^{th} century CE to the 12^{th} century CE.
- One of the significant stone inscriptions found from the site describes the conservation of the stupa, construction of a pavilion, and digging of a Baoli (well) for the benefit of the settlers, as well as mentioning the generosity of a person in the Kaliyuga.
- The scope of excavation has been widened after findings of this stone inscription at the site. **This reference/evidence shows that the site needs to be explored and excavated completely to find out the *Baoli* (rock-cut step well) and a Pavillion which has not been find out in the previous as well as present excavation**.
- The site witnessed hundreds of votive stupas in comparison to other Buddhist sites in and around Odisha.

- During the spade work, 4 construction phase activities were identified on the site, which reflect different phases and the evolution and growth of Buddhism.
- Most of the structures noticed on the site were recreated one after another by demolishing and using the earlier material.
- The findings of finger-impressed bricks indicate the beginning of establishment around 2nd -3rd cent CE, but due to the paucity of material associated with the present context and the available material remains retrieved from the site, the beginning of the establishment is placed prior to the 4th century CE.
- The tangible evidence of fusion of art between Brahmanism and Buddhism are findings of two interesting monolithic stupa, one designed as a temple of Kalingan order consists of a Rekha vimana with a pyramidal Jagmohan housed goddesses Tara within, and another stupa is designed as a *panchayatan* temple, adorned with miniature stupa in its four corner and housed 2 male and 2 female divinities in four directions.
- The significant model of a votive stupa carved from a single stone, designed as a combination of a stupa and a temple illustrates the fusion of art and an amalgamation of Buddhist and Brahmanical art traditions.
- The present excavation retrieved three colossal heads of Buddha measuring from 1.27 m to 1.23 m in height and also four neck portions, broken hands and leg portions, suggesting the remains of huge seated Buddha images dated back to 7th-8th century CE based on the palaeographical evidence. These findings have changed the previous theories and notions of scholars and locals, who believed that the colossal heads were produced individually, but rather as part of a huge image of Lord Buddha in a seated posture in different mudra has been changed as noticed from the remains of hands, body portion, and lower body parts, i.e., in *varada mudra, abhaya mudra* and *bhumisparsha mudra*.
- The expression of art and sculptures are carved from khondalite stones of Asia Hill Range. This region provided suitable raw materials to artisans to express the ideology of the master, devotee, or self in the rock based on Buddhist ideas, and philosophical concepts of the time.

- The finding of celadon, which resembles the Burmese ceramic/ celadon (Martaban), paved the way to find out more material culture of South-East Asian origin or beyond at the site and to understand the cultural interaction between India and these countries. It acts like a beacon light to understand the movement of people across the region. Though its scanty in the present scenario but it's like a drop of hope which gives aroma to the universal connectivity among the human society due to trade relationship or connection of cross regional society under the umbrella of Buddhism, believes in non-violence and brotherhood and faith on Budddha (Lord Gautam) Dharma (right principles and action) and Sangha (based on the concept of unity among the human community).

The temple dedicated to Mahakal on the site marks a shift from Buddhist to Brahmanical dominance in the region. It was constructed during late 12th to 13th century CE, by demolishing an earlier Buddhist structure. This act of replacement was not merely architectural but also ideological, reflecting the decline of Buddhism and the growing influence of Saivism, especially after the philosophical advocacy of Adi Shankaracharya. By the 14th century CE, the remnants of Buddhist philosophy, especially during the time of Chaitanya, were further diminished.

Ratnagiri site witnesses the development of art and architecture on Buddhism from Mahayana to Vajrayana. The findings of hundreds of monolithic to ornamented votive stupas made out of brick and stone of different sizes and dimensions, also housing divinities within the niches, are significant. The syncretism of Brahmanical and Buddhist art and architecture is clearly noticed in the site. The facial features of the sculptures noticed in the sites resemble with the local inhabitants, indicating that the production of these art by local artisans made out of khondalite stone retrieved from Assia hill range or can clearly say that **Assia School of Art** has contributed a significant role in the evolution and development of Buddhist art in the region. The contribution of Bhaumakaras for the growth of art and culture and the role of women queens for the sake of Buddhism in this site, is reflected in each piece of art. The site is a perfect place to study the fusion of Buddhist and Brahmanical art and

architecture and to know the beginning and growth of Vajrayana Buddhism in this part of the **Bharatvarsha**.

Acknowledgements

I am grateful to Director General and Director (Exploration and Excavation) of Archaeological Survey of India, New Delhi for granting the permission to excavate Ratnagiri Buddhist Site, Jajpur, Odisha. I am also thankful to P.P. Pradhan- Assistant Superintending Archaeologist, Rohit Kumar - Senior Conservation Assistant, Debasish Mishra - Photographer Gr. II, R.N. Sahu former Photographer, Pabitra Mohan Barik - Draughtsman Gr. I, S. K. Khuntia- Surveyor Gr. I, and other staffs of Puri Circle and Ratnagiri Museum for their involvement. I am thankful to the Research Scholars – Saumya Singh, Bhagyalaxmi Rana, Mitali Kaushik, Kajol, Rajat, Mitrabhanu Sahu, Tusharkanta Behera, Asad, Jitendra and Indrani Mondal for their dedication and involvement and Paryatan Vikas Parisad, Bari for their cooperation and support during the excavation work.

Glimpses of Ratnagiri Excavation 2024 -2025

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09 April 2025 5:46 pm

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Archaeological Survey of India, Puri Circle
&
Tourism Bikas Parishad, Barh, Jajpur, Odisha
Celebrate
WORLD HERITAGE DAY
18th APRIL 2025
DISASTER AND CONFLICT RESILIENT HERITAGE
Photo Exhibition
Venue :
Ancient Buddhist Site - Ratnagiri

सत्यमेव जयते

प्रत्नकीर्त्तिमपावृणु

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